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To my family.

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This victory belongs to you as much as it belongs to me.
But this isn't the end. This is just the start.
The best is yet to come.

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Dedications

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my cheerful companion in moments of doubt,
I hope to continue to grow, to make you proud today and always.

Sofia AIT SIDHOUM.

Abstract

Power grids are undergoing a profound transformation, marked by growing demand, an urgent transition to more sustainable sources and the imminent disappearance of certain fossil fuels, which is encouraging the acceleration of development in the energy sector.

In response of these challenges, the use of decentralised renewable energy sources in electricity systems appears to offer a way of reducing CO_2 emissions and ensuring energy security. However, the intermittent nature of these resources poses reliability issues that require sophisticated electrical system management concepts. This thesis presents a comprehensive analysis of smart grids and microgrids, focusing on the design of a residential microgrid connected to the grid and the development of an algorithm that manages the energy flow in bidirectional way between production and consumption. The management system is based on a rule-based decision tree applied in the EMS (RB-EMS), which controls the distribution of energy between decentralised production, storage systems, and flexible consumption. The goal is to improve the use of renewable energy and the operational efficiency while reducing dependence on the grid. Additionally, a demand response strategy is supported by a pricing and incentive system as well as interaction with electric vehicles (V2G). The objective of this integration is to limit demand peaks, encourage consumption during off-peak hours, balance local energy flows, and engage participating consumers in supporting the grid.

The proposed approach has been evaluated and simulated on MATLAB/Simulink under various operating conditions. The results demonstrate the feasibility of improving energy efficiency, self-consumption and promoting a transition to smarter, cleaner, and more robust energy systems.

Keywords : Microgrid, Rule-Based Algorithm, Energy Management System, Demand Response, V2G.

ملخص

في السنوات الأخيرة، تسارعت وتيرة تطوير الشبكات الذكية بسبب تزايد الطلب على الطاقة، واستنزاف الموارد الأحفورية، وزيادة الاهتمام بالبيئة.

في ظل هذه التحديات، تبرز إمكانية الاعتماد على مصادر الطاقة المتجددة الموزعة كحل بديل يساهم في الحد من انبعاثات ثاني أكسيد الكربون وضمان أمن التزويد بالطاقة. لكن الطبيعة المتقطعة لهذه المصادر تخلق مشكلات تتعلق بموثوقية النظام الكهربائي، مما يتطلب اعتماد مفاهيم متقدمة في إدارة الشبكات الكهربائية. يتناول هذا العمل تحليل شامل لبنية الشبكات الذكية (Smart Grids) والشبكة المصغّرة(Microgrid) بالتحديد شبكة سكنية متصلة بالشبكة العمومية، وتطوير خوارزميات لإدارة الطاقة ثنائية الاتجاه بين الإنتاج والاستهلاك. تم تطبيق نظام إدارة طاقة

(Energy Management System - EMS)يعتمد على القواعد (Rule-Based) للتحكم في توزيع الطاقة بين الإنتاج الموزّع، نظام التخزين، والأحمال المرنة. يهدف هذا النظام إلى تعزيز استغلال الطاقة المتجددة، تحسين الكفاءة التشغيلية، وتقليل الاعتماد على الشبكة العمومية. كما تم إدماج نظام استجابة للطلب (Demand Response) مدعوم بآلية تسعير وتكامل مع المركبات الكهربائية(Vehicle-to-Grid - V2G) . يهدف هذا الدمج إلى تقليل الطلب الأقصى، موازنة تدفقات الطاقة المحلية، وتحفيز المستهلكين النشطين للمساهمة في دعم الشبكة.

تم تقييم السيناريوهات المقترحة من خلال محاكاة في MATLAB/Simulink تحت ظروف تشغيل متعددة. وقد أظهرت النتائج مدى فاعلية الاستراتيجية المقترحة في تحسين استهلاك الطاقة، رفع نسبة الاستهلاك الذاتي، والمساهمة في الانتقال نحو أنظمة طاقة سكنية أكثر ذكاءً ومرونة.

الكلمات المفتاحية: شبكة مصغرة، أنظمة إدارة الطاقة، خوارزمية المبنية على القواعد، نظام استجابة الطلب، مركبات كهربائية.

Résumé

Les réseaux électriques connaissent actuellement une profonde transformation, marquée par une demande croissante, une transition urgente vers des sources plus durables et la disparition imminente de certains combustibles fossiles, ce qui encourage l'accélération du développement dans le secteur énergétique.

En réponse à ces challenges, la favorisation de sources d'énergie distribuée dans les réseaux electriques permettrait de réduire les émissions de CO_2 et d'assurer la sécurité energetique. Cependant, la nature intermittente de ces ressources pose un problème majeur de fiabilité qui requière un système sophistiqué pour leur gestion. Cette thèse présente une étude generale des réseaux intelligents et micro réseaux, en se concentrant sur la conception d'un micro-réseau résidentiel connecté au réseau et le développement d'algorithmes de gestion du flux d'énergie bidirectionnel entre la production et la consommation. Le système de gestion est basé sur un arbre de décisions, appliqué sur un système basé sur les règles (RB-EMS), qui controle la distribution de l'énergie entre les sources distribuées, les systèmes de stockage, et les charges flexibles. Le but est de favoriser l'utilisation des énergies renouvelables, améliorer l'efficacité tout en réduisant la dépendance au réseau electrique. Une stratégie de réponse à la demande basée sur l'incitation et les tarifs est en plus implémentée en plus de la technologie (V2G) des vehicules electriques. L'objectif est de limiter les pics de consommation, encourager la consommation durant les heures creuses, et les impliquer dans le soutien du réseau electrique.

L'approche proposée est évaluée et simulée dans MATLAB/Simulink pour différentes conditions de fonctionnement. Les résultats valident la faisabilité de l'approche proposée pour améliorer l'efficacité énergétique, l'autoconsommation et promouvoir une transition vers des systèmes énergétiques domestiques plus intelligents et plus robustes.

Mots-clés : micro-réseau, algorithme basé sur les règles, système de gestion de l'énergie, réponse à la demande, V2G.

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Nomenclature

GHG	Greenhouse Gases
BESS	Battery Energy Storage Systems
DG	Decentralized Generation
RES	Renewable Energy Sources
FACTS	Flexible Ac Transmission Systems
HVDC	High Voltage Direct Current
ICT	Information And Communication Technologies
DERs	Distributed Energy Source
CHP	Combined Heat And Power Units
MG	Micro Grid
DERMS	Distributed Energy Source Management System
EMS	Energy Management System
MC	Microsource Controller
LC	Load Controller
MGCC	Microgrid Central Controller
DMS	Distribution Management System
RB	Rule-Based Method
MILP	Mixed-Integer Linear Programming
PSO	Particle Swarm Optimization
ANN	Artificial Neural Network
DSM	Demand Side Management
DR	Demand Response
ToU	Time Of Use
RTP	Real Time Pricing
ED	Economic Dispatch
UC	Unit Commitment
CPR	Critical Peak Rate
V2G	Vehicle To Grid
EV	Electric Vehicle
WT	Wind Turbine
PV	Photovoltaic
SOC	State Of Charge
I_{PV}	Photovoltaic Current
I_{ph}	Photon Current
V_{pv}	Photovoltaic Voltage
R_S	Series Resistance
P_{PV}	Photovoltaic Power
P_{WT}	Wind Turbine Power
SOC_{BESS}	State Of Charge of the BESS
P_{EV}	Electric Vehicles Power
P_{grid}	Grid Power

Nomenclature

P_L	Load Power
SOC_{EV}	State Of Charge of the Electric Vehicles
P_{BESS}	BESS Power
P_{SUR}	Surplus Power
P_{DEF}	Deficit Power

General Introduction

General Introduction

The world energy sector is facing an unprecedented structural challenge. The rapid growth in the population, and the increasing power demand, imposes a serious strain on modern power systems. In addition, the environmental challenges and climate regulations represent a drastic reduction in greenhouse gas emissions. Despite this, several countries still rely heavily on fossil fuels, leading to devastating consequences on their environment, economy and geopolitical implications. Electrical grids, which were initially built for centralized and unidirectional distribution of electricity, are now facing major constrains. The transition from traditional energy sources, to renewable resources such as solar and wind power, adds variability that the existing grid infrastructure can't handle. These changes call for the establishment of more flexible, efficient and robust infrastructures.

Smart grids or intelligent electric grids are a prominent solution in this system. The implementation of new information and communication technologies has made electric distribution networks more controllable and transparent. Prosumers, demand response systems, electric vehicles, and distributed generation are some of the new options for generating and consuming energy through these grids. The smart grids are likely to be the most innovative idea in this work. Of all the new infrastructure, microgrids are now a key element of optimized power systems. Microgrids are local energy system that can operate in island or connected with the main grid. They integrate renewable sources, energy storage devices, and flexible loads to optimize local energy utilization and add resiliency to the system. Among the defining features of this new model is the level of active consumer or "prosumer" who are economically rewarded for reducing or delaying their use through demand response programs and dynamic pricing arrangements on the basis of grid conditions. This behaviour is incentivized and supports enhanced grid balance, reduced peak demand, and integration of renewable energy.

The goal of this thesis is to develop and evaluate a microgrid composed of distributed energy sources, loads and an energy storage system, controlled by an energy management system (EMS). The proposed EMS combines a rule-based control strategy with an incentive-based demand response program. The aim is to improve energy efficiency, reduce grid reliance, and engage consumers in flexible, sustainable consumption.

This report is structured as follows:

The 1st Chapter presents the context of the current electricity grid, describes its technical and organisational limitations, and introduces smart grids as possible alternatives.

The 2nd Chapter explores microgrids in detail. It outlines their structure, objective and operating modes. It examines the energy management system (EMS), including the control approaches, the strategies and mechanisms. Additionally, a comprehensive literature review is presented to analyse recent advancements and challenges in microgrid energy management systems.

The 3rd Chapter gives the case study and EMS proposal. First, it provides a detailed description of the microgrid components. Then, it offers a hybrid energy management based on rule-based decisions combined with pricing and incentive-based remand response system.

The 4th Chapter describes the simulation developed on MATLEB/Simulink. Different scenarios are tested to evaluate the ability of the EMS to respond to production, storage and consumption under various dynamic scenarios.

This research contributes to the energy system transformation towards smart, flexible, and consumer-centred systems. It demonstrates the behaviour of economic incentives can be combined with technical control to prove that microgrids in residential areas can play an active part in the energy transition.

Chapter I

Evolving and modernizing power grids

I.1. Introduction

At the heart of any energy transition, the power grid faces many challenges, including the rising demand, the integration of renewable resources into the grid, and the need for a more resilient and flexible infrastructure.

However, governments are still heavily reliant on fossil fuels, with their environmental, economic and geopolitical impacts. The emergence of structures such as smart grids in the context of new evolving systems is crucial in this regard, as it allows for better control of energy flows and adaptability consumption changes.

In this chapter, we review the architecture of the electricity grid and examine the issues surrounding it, including limitations and dysfunctional elements related to its structure and operations, with a focus on microgrids. Finally, we will look at recent developments in smart grids, and the solutions and technologies being considered for optimising the electricity grid.

I.2. Energy and environmental context

I.2.1. environmental context

Since the start of the industrial age at the end of the 18th century, the global average temperature has risen by around $1.1°C$ compared with pre-industrial levels. Whereas it was rising at an average rate of $0.08°C$ per decade before 1980, this rate has accelerated to $0.18°C$ per decade since then (Figure I.1). This rise is directly linked to the growing accumulation of greenhouse gases (GHGs) in the atmosphere, which is composed mainly of carbon dioxide (CO_2), and other gases such as methane (CH_4) and nitrous oxide (N_2O).

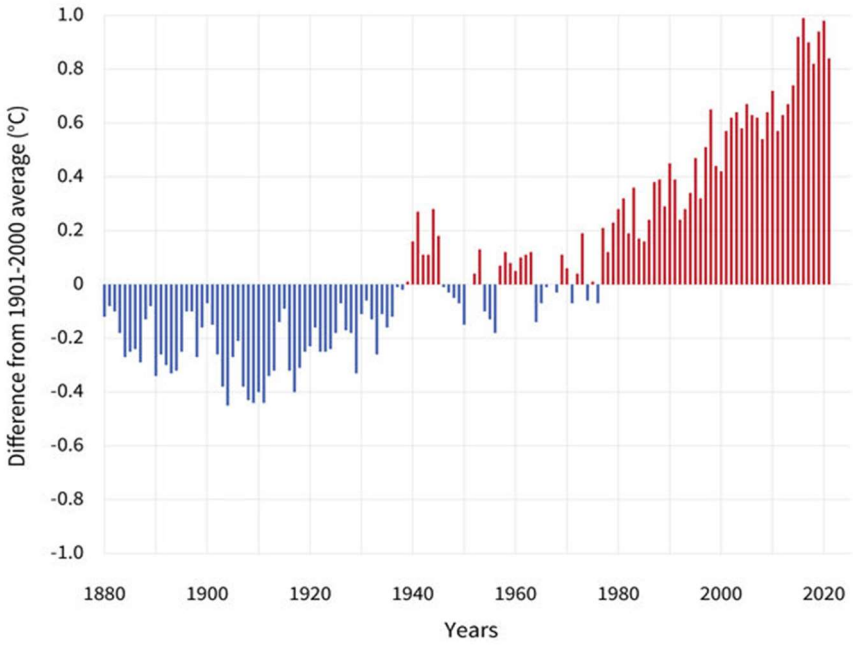


Figure 0.1: Global temperature variation from the 1901-2000 average (1880-2020) (source: Climate.gov).

Future climate change depends on the quantities of GHGs emitted in the coming decades. Currently, human activities, notably the combustion of fossil resources, inject around 40 billion tonnes of CO_2 into the atmosphere every year. This gradual accumulation is the main driver of observed global warming.

However, the transition to a more sustainable energy system faces structural challenges, especially where grid infrastructure is concerned. Many renewable energy projects are queuing up to be connected. Under-investment in grids would slow the growth of renewables, leading to higher consumption of fossil fuels (gas and coal) and contributing to massive amounts of CO_2 emissions worldwide, 58 gigatons by 2050, equivalent to four years of global emissions from the electricity sector, and making it more difficult to limit global warming to 1.5°C and risk exceeding 2°C [1].

I.2.2. Energy context

Today, our energy consumption is based on several sources: fossil fuels (petrol, gas, coal) and renewable energies (solar, wind, hydropower). Fossil fuels still represent a significant proportion of production, despite their harmful effects on the environment.

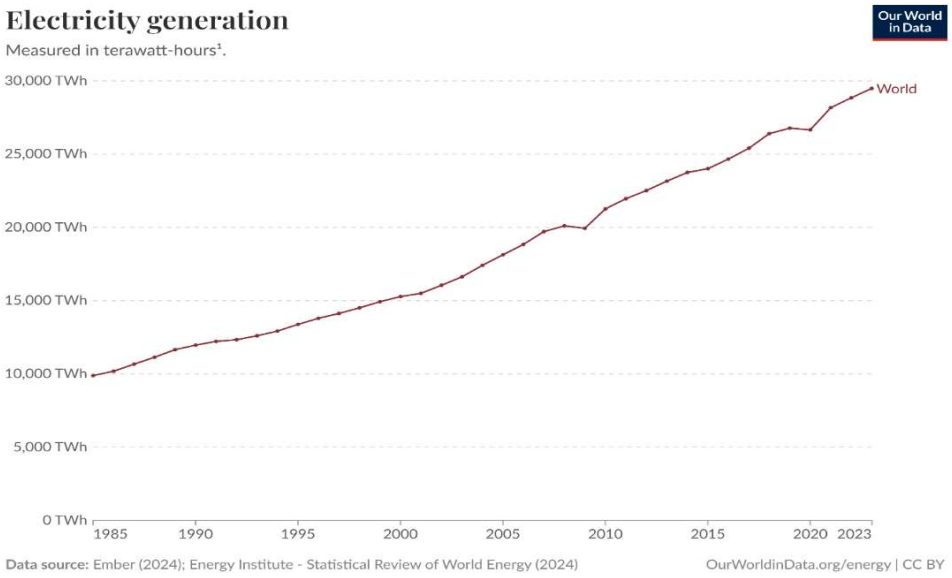


Figure 0.2: World electricity production (1985-2023) in TWh [2].

Global electricity production grew exponentially between 1985 and 2023, reaching a record 29,925 TWh (Figure I.3). This growth rate, which is 25% higher than that of global primary energy consumption, which stands at 167,778.1 TWh (Figure I.4), indicates that the global energy system continues to expand, especially with recent technological advances.

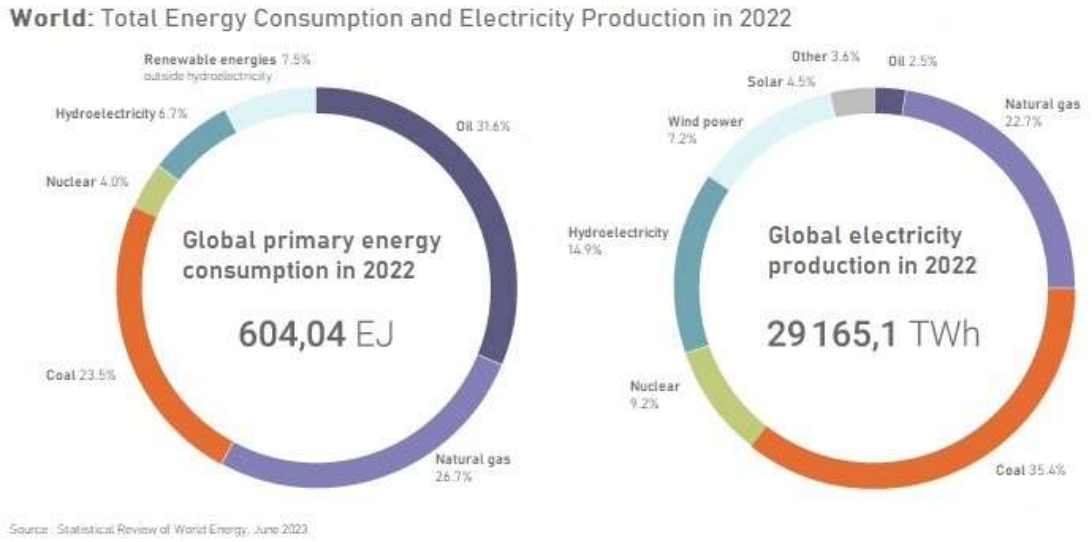


Figure 0.3: World energy and electricity mix in 2022: Breakdown of primary energy sources and electricity generation [3].

Coal has maintained its position as the dominant fuel for electricity generation despite environmental issues (Figure I.3, Figure I.4), raising the share of fossil fuels to 60.6% of total production. In contrast, electricity production from oil is gradually declining, unlike gas, which continues to increase, highlighting the global economy's dependence on these fuels.

The strong growth of renewable energies since the 2000s, particularly for wind and solar. Hydropower and bioenergy are also following an upward trend and asserting their place in renewable energy production.

Nuclear power's share of global electricity production has remained stable at around 9% for some years now. The opening of new power plants in several countries around the world has increased nuclear capacity [4].

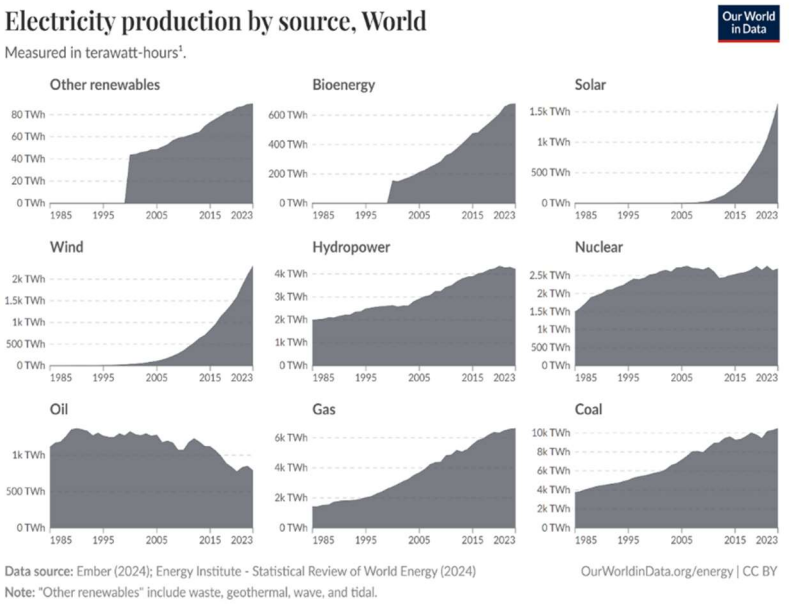


Figure 0.4: World electricity production by energy source (1985-2023) in TWh [2].

I.2.3. Geopolitical and societal context

The COVID-19 pandemic was a major event in the slowdown of progress towards universal access to energy (and therefore the energy transition), exacerbating inequalities in access to electricity, particularly in developing countries. Millions of people have seen their access to electricity jeopardized by falling incomes and lack of investment in energy infrastructure, thus multiplying energy poverty. The repercussions of this crisis are still apparent: in 2022, 733 million people still have no access to electricity, and 2.4 billion people are still cooking with fuels that are harmful to their health and the environment [5].

Furthermore, the major issue in the geopolitics of energy remains the need for governments to secure their resources and reduce their energy dependency. This dependence is the way in which international relations are shaped, from diplomatic negotiation to armed conflict. A recent example is the energy crisis in Europe, hit by the war in Ukraine, which is forcing European countries to turn to new partners and accelerate the energy transition.

Globally, geopolitical tensions and the decisions of major producers have a direct impact on the price of energy resources, as illustrated in Figure I.5. OPEC's production cuts and the conflicts in the Middle East have caused major price fluctuations, reinforcing global energy uncertainty.



Figure 0.5: Oil price trends and major geopolitical events (2022-2023) [6].

I.2.4. Economic context

The evolution of the power grid is closely linked to global economic dynamics, notably through swings in electricity prices, which have risen significantly over the last few decades. This rise is affected by changes in production costs, needs for investment and renovation of production infrastructures, and new energy transition initiatives.

By way of example, the electricity tariff for residential customers in Europe rose from €0.1256/kWh in 2012 to €0.2016/kWh in 2015, marking an increase of 60.5%, largely due to higher costs associated with grid modernization and the integration of renewable energies [7].

The graph below (Figure I.6) illustrates the evolution of global investment in fossil and renewable energies between 2015 and 2023 in billions of dollars (USD):

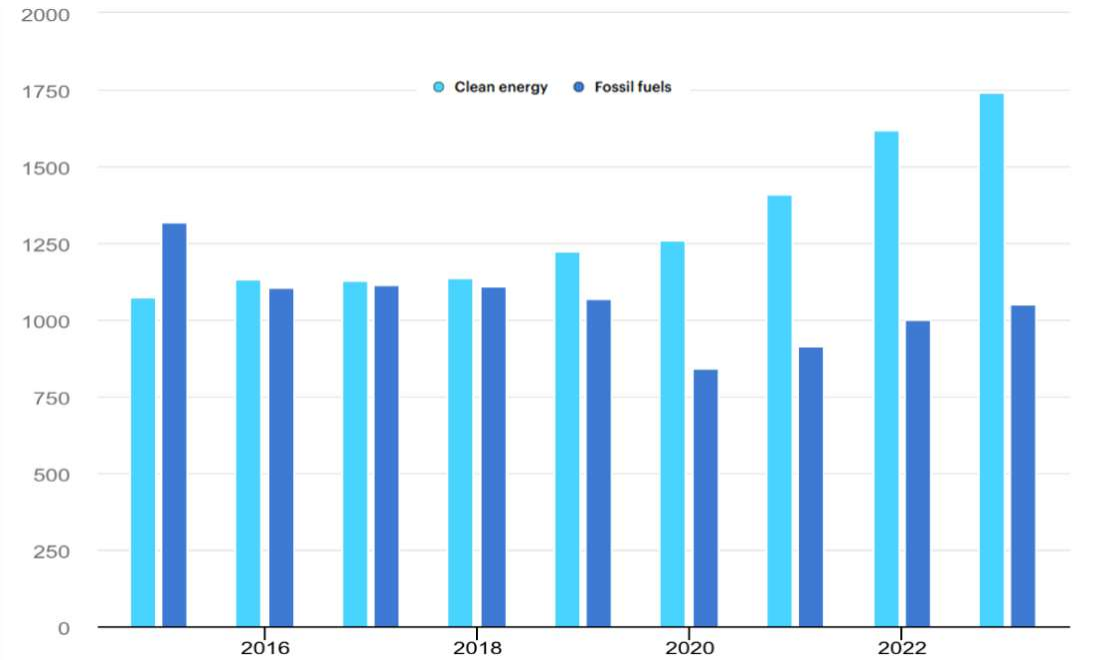


Figure 0.6: Global investment in clean energy and fossil fuels (2015-2023) in billions of USD [8].

A gradual shift in investment priorities has been observed. In 2015, the amounts devoted to fossil energies remained higher than those invested in clean energies. However, since 2019, there is a reversal in this pattern, and since then, investment in clean energies has risen steadily, while that in fossil fuels has fallen or stagnated. In 2023, the gap reaches a considerable level, with renewable energies benefiting from an amount exceeding 1,700 billion USD, i.e. around 700 billion USD more than the amount for fossil fuels. This shift reflects a structural reorientation of energy strategies worldwide.

In the 1990s, many countries began to liberalize their electricity markets, which were considered inefficient and motivated by the growing trend towards privatization of state-owned companies. This opening up to competition enabled electricity markets to reduce production costs and offer the most competitive prices to consumers, who at the same time expanded their role, encouraging them to become active players in the energy system.

I.3. Traditional power grids

The topology of the conventional power system has not changed since its creation in the 20th century. Since then, a centralized architecture has been implemented and has remained ever since [11,12] :

- Electricity is generated in high-power plants using fossil fuels (coal, gas, oil) or renewable resources (solar, wind, hydro).

- Transmission from generation plants to distribution stations via high-voltage lines to minimize energy losses.
- Distribution of electricity through substations. Industrial customers are supplied directly with HV electricity, while residential and utility customers are supplied with LV electricity.

The conventional electricity network operates in a unidirectional way, from production to consumption, and the management of supply and demand must be maintained to keep the system in balance. However, the injection of renewable energies, the increase in demand and the evolution of electricity uses (air conditioning, electric vehicles) make grid management more complex, for a variety of reasons:

- Electricity consumption variates hourly, daily and seasonally, with winter demand exceeding in summer. marked by peaks and troughs throughout the day.
- Production instability due to the intermittency of renewable energies.
- The rise of decentralized generation (DG), which means more points of injection into the grid, making it more complex to manage.

The figure below gives an overview of the power subsystems:

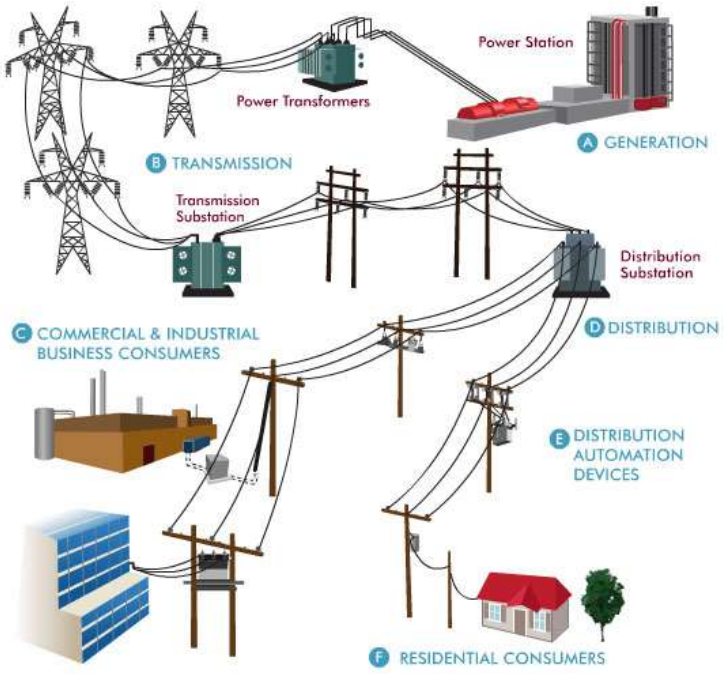


Figure 0.1: Limitations of conventional power grid [10].

I.4. Problems Associated With Conventional Grids

The traditional power grid faces a number of limitations. These issues, including the challenges of energy transition, grid reliability and modernization, energy storage and management, are examined in the following section:

I.4.1. Greenhouse gas emission

Persistent dependence on fossil fuels is a serious environmental problem. Indeed, global greenhouse gas (GHG) emissions rose by 51% between 1993 and 2023 (Figure 1.8). These emissions are an obstacle in the energy transition, because of their direct impact on global warming, and their consequences are catastrophic, as they cause extreme phenomena such as the increasingly devastating storms, floods, fires and heat waves that the world is currently facing.

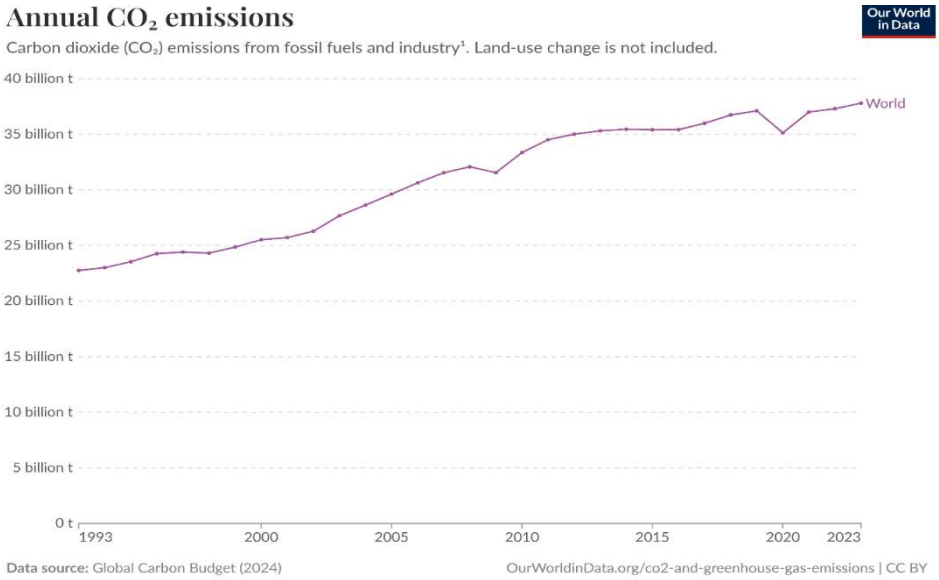


Figure 0.1: Global greenhouse gas emissions (CO_2 eq. over 30 years) [2].

Currently , the energy sector is the biggest contributor to GHG emissions, with 37.4 billion tons (Gt) in 2023 (Figure 1.9), representing 75.7% of global emissions. These are mainly due to electricity and heating, but also to the production and transport of energy from fossil fuels.

Hesitations about modernizing and extending the grid are creating a vicious circle:

- Make the integration of renewable energies more difficult.
- Increasing dependence on fossil fuels.
- Increase greenhouse gas emissions.

- Fuel global warming.

Overcoming this procrastination is essential to achieving climate goals, reducing emissions and ensuring a sustainable, low-cost future [1].

I.4.2. Fossil fuel depletion

Global energy consumption is based on the exploitation of fossil fuels (petrol, gas, coal) and fissile fuels (uranium), which continue to dominate the world's energy mix. However, the intensive use of these resources, combined with their non-renewable nature, raises a major challenge in terms of their long-term extinction.

Fuel reproduction times are generally estimated at between 100 and 300 million years. However, at the current rate of exploitation, the situation of petrol and natural gas reserves is extremely worrying, with estimated lifetimes of 42 and 65 years respectively (Figure 1.9), making them the resources most at risk of depletion in the short term. Coal, on the other hand, which has an estimated lifetime of 169 years, has vast reserves around the world, but problems of GHG emissions and mining difficulties limit its use on a large scale [3,13].

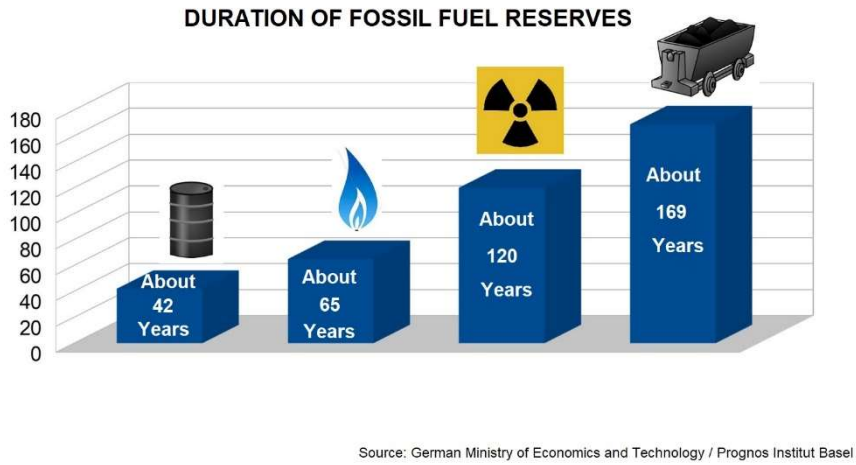


Figure 0.2: Estimated duration of the world's fossil fuel reserves [14].

I.4.3. The ageing of conventional power grid

Aging power grid infrastructure, including generation plants, transmission lines and distribution stations, as well as transformer substations and equipment, remains the most significant problem in terms of grid reliability, with the most frequent reasons being technical factors, namely obsolete and substandard equipment, poor management, lack of investment, and natural disasters. This leads to ever-increasing repair costs, higher failure rates, increased risk of power cuts, accelerated infrastructure degradation and cascading failures. In addition,

ageing power grids complicate the integration of renewable energies, limiting the management of these new sources, and consequently impeding the development of the energy transition [15,16].

Some networks use outdated equipment to compensate for reactive energy, such as synchronous generators, coils or capacitors. These solutions, although widely used, are not recommended, as they are harmful to the environment, expensive and inefficient in the face of current network and energy transition requirements.

I.4.4. Rising energy demand and peaks demand

The world's increasing population is driving a commensurate rise in energy demand, due to growing needs in both the residential and industrial sectors:

- **Residential:** Multiplication of energy-consuming connected equipment (washing machines, computers, air conditioners), home charging of electric vehicles, etc.
- **Industrial:** electrification of processes (electric ovens, thermal drying), expansion of industrial data centers, as well as the emergence of Industry 4.0 based on energy-intensive IT systems, sensors and actuators.

This demand is reflected in the appearance of peak demand that are increasingly marked over time, which raises a problem of stability.

Energy demand varies significantly over the course of a year, with consumption levels changing according to seasons. In some countries, where electric heating is widespread, the “seasonal peak” occurs in winter. In other warmer countries, where air-conditioning is widespread, the seasonal peak tends to occur in summer. Electricity consumption also varies significantly on a daily basis: in winter, for example, the “daily peak” occurs at around 7 p.m. on a typical day, a time that corresponds to the return home and the simultaneous activation of many appliances (Figure 1.10) [19].

To respond to variations in consumption, grid operators use peaking power plants, capable of starting up quickly to supply surplus energy. However, these plants are expensive and harmful to the environment.

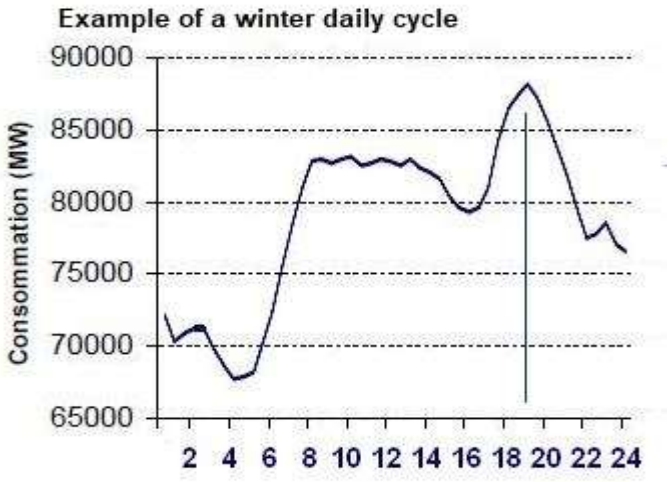


Figure 0.3: Typical daily profile of electricity consumption.

I.4.5. Intermittency of renewable energy sources

Wind and solar power are defined as intermittent renewable energies due to their fluctuating, weather-dependent nature. This renders the stability of the electricity grid difficult to maintain. For example, between 2010 and 2011, the power output from wind turbines fluctuated between a maximum of 60% and a minimum of 4% of total capacity, this fluctuation leading to too much or too little electricity being generated, impacts the grid frequency stability and voltage stability [20].

This variability can be seen in the figure I.11 below, which shows the instability of solar production over the course of a day:

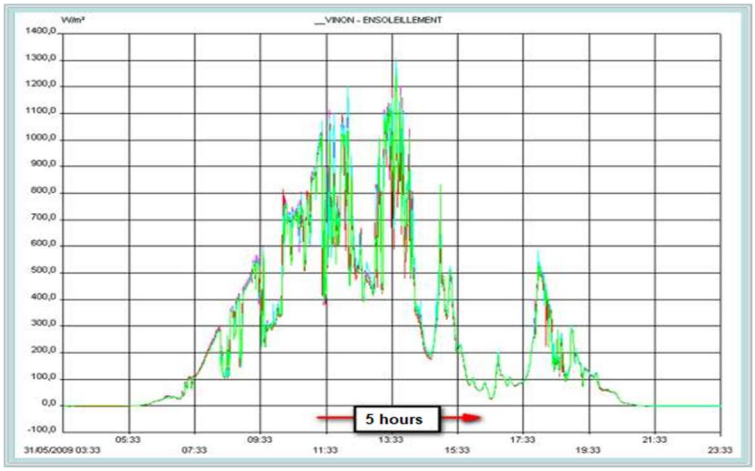


Figure 0.4: Illustration of the intermittence of solar production [18].

Unstable frequency can disturb sensitive machinery, interrupt electronic circuits, and cause industrial processes to shut down abruptly. Supply voltage variations can destroy electric motors, cause them to overheat and fail, and shorten the equipment life [21,22].

I.4.6. Large-scale energy storage

The main method of using renewable energy sources while overcoming the problem of intermittency is the use of battery energy storage systems (BESS). Such systems can temporarily store excess energy to use it in the future and thereby supply a steady power flow. But numerous issues arise with mass use, such as :

- **Costly batteries:** Although the cost of lithium-ion batteries decreased over the years, the economic impact is observed in a context where storage would be massive and enable the integration of renewable energies.
- **Poor performance:** The energy storage and restitution operation generate considerable losses due to ageing components and subsequent conversions (AC-DC-AC).
- **Environmental footprint:** The production of BESS requires necessary materials such as lithium, cobalt, or nickel, and their extraction has a serious environmental footprint with soil and water pollution, as well as GHG emissions.

Problems with storage persist in microgrids since they are small and any alteration impacts significantly. Investment and maintenance costs can account up to 40% of the overall cost of a microgrid, which has a direct influence on the size of the storage system [23].

I.4.7. Grid management and planning issues

The primary objective of system management is to match supply and demand. Failure to do so would lead to an imbalance that would compromise grid stability. Because of their unpredictable nature, renewable energy sources (RES) such as wind and solar energy can exacerbate grid stability.

This challenge comes on top of the multiplication of dispatching centers, and the control points added by energy decentralization, such as distributed sources, storage devices, and consumer participation, make real-time monitoring even harder, particularly in the absence of advanced visualization and management tools, which results in ineffective management [17].

The major consequence of this lack of management is an absence of flexibility between production and consumption, which results in significant losses as well as massive waste due to excess production over demand, lowering the system's energy efficiency and output [24].

To overcome these challenges, it is important to implement supervision and monitoring systems that study the grid and its fluctuations.

I.5. Opening a competitive electricity market

Europe has witnessed an inflexion point with the liberalization and restructuring of the energy sector in the late 1990s. This institutional transformation marked a break with the historical model of public monopolies. The new organizational model is based on the following elements:

- Targeting both the upstream and downstream sides of the value chain, including energy production and marketing, to enhance performance through competitive market structures, and encouraging technological and organizational innovation.
- Transmission and distribution are defined as natural monopolies with high fixed costs and rigid regulatory regimes.
- This transformation is accompanied by the creation of independent regulatory bodies responsible for setting transmission tariffs and monitoring the smooth operation of markets, in the general interest.

In these conditions, market liberalization attracts new competitors, whether start-ups or firms from different industries, who bring their talents, novel technology, and business strategies. These innovative business models are likely to invest in new manufacturing methods, integrate operations in novel ways, and give appealing, diverse solutions to consumers. For the end consumers, the most obvious result of these reforms is the freedom to choose their energy provider, putting the former operator (the old monopoly) in competition with new competitors [25].

I.6. Power grid evolution: towards intelligent and decentralized systems

I.6.1. Energy transition

The energy industry is undergoing fundamental changes, primarily driven by the ongoing energy transition. Reflected in ambitious decarbonization targets, greener energy mix and energy sobriety by 2030. This transition is quantified as follows :

- To minimize global warming in accordance with the Paris Agreement, greenhouse gas emissions must be reduced by at least 40% from 1990 levels.
- At least 32% of total European Union energy consumption comes from renewable sources of energy.
- Improving energy efficiency by at least 32.5% as a way to reduce energy consumption without affecting the quality of services being delivered.

These political ambitions radically change energy consumption patterns and electricity system operations. This represents an entirely new approach to grid design and management. Along with the development of alternative sources of power and policies designed to reduce energy consumption, development of new applications is also emerging such as self-consumption, electric transport, and energy storage while other traditional applications are intensifying, including electric heating and air conditioning [26].

I.6.2. Energy efficiency

Energy efficiency refers to the ratio between the useful energy delivered by a system and the energy consumed in its operation. In other words, it's the ability to obtain the same service or energy performance while consuming less energy.

This can mean, for example, using LED light bulbs, which consume less energy than conventional bulbs, encouraging electric vehicles, or choosing A+++-rated household appliances. The new system benefits consumers by enabling them to make intelligent decisions and promoting comfort and efficiency [27, 28].

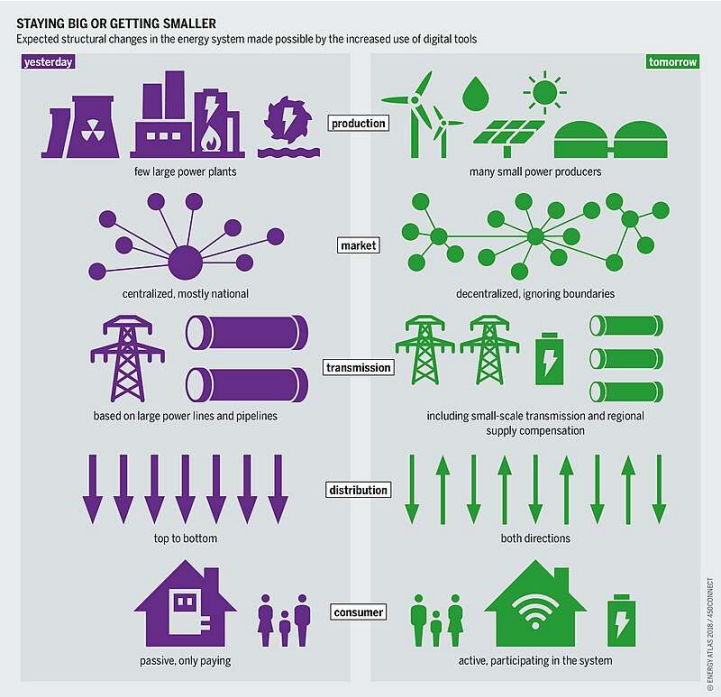


Figure 0.1: Structural evolution of the power grid [10].

I.6.3. Large-scale integration of renewable energies

The large-scale integration of renewable energies into the power grid has caused significant disruption since the 2010s with the aim of achieving an energy transition. This transformation has affected the entire grid (generation, transmission and distribution).

The installed capacity of RES has grown exponentially. This development can largely be credited to advances in decentralised production systems like wind and solar energy, which have revolutionized the traditional system of centralised production that initially made up the base of the grid system [26,29].

In particular, decentralized generation makes it possible to:

- Reduce transmission-related line losses, as energy is generated close to the consumer.
- Improve resilience by reducing dependence on large-scale generation plants.
- Promote energy autonomy for local communities and utilities.
- Compensate for the intermittent nature of renewable energies through storage and intelligent management.
- Minimize environmental impact while improving the adaptability, sustainability, and resilience of the grid.

However, this transition brings with it a number of technical challenges. Decentralization has added considerable complexity to the distribution network, due to the multiplication of production and injection points, and the transition from a unidirectional flow system towards a bidirectional flow. Therefore, the integration of smart grids that enable RES integration and real-time management of production and consumption is required.

I.6.4. Power grid flexibility and reliability

Electricity generated in high-power plants is then transmitted to distribution through the transmission grid, identified as the weak link of the power grid since it's challenging to manage and poses numerous problems. To this extent, advanced technical solutions have emerged to enhance the stability, flexibility and adaptability of the grid.

I.6.4.1 Flexible ac transmission systems (FACTS)

Flexible AC transmission systems comprise power electronic-based solutions to improve real-time control and power transfer. The main aim of these devices is to make the grid more flexible by ensuring, among other things, voltage stability, damping power oscillations and optimizing line transmission capacity. For example, systems such as SVC and STATCOM dynamically regulate voltage by injecting or consuming reactive power, thus contributing to grid stability. Other devices such as TCSC and UPFC work on power flow by regulating line impedance, preventing transmission line congestion [31].

I.6.4.2. High voltage direct current (HVDC)

HVDC is a power electronics technology that uses thyristors and IGBTs for transporting electricity as high-voltage direct current. The use of this technology has enabled [31]:

- DC power transmission over long distances, with fewer losses than AC power.
- Interconnection of asynchronous grids, where HVDC makes it possible to synchronize two systems operating at different frequencies or with incompatible stability conditions.
- Active and reactive power management through flow control through real-time balancing and voltage support.

FACTS and HVDC technologies provide solutions to meet the challenges of conventional power grids [32]:

- Control impedance and phase angle to better manage line saturation, and ensure good flow distribution.
- Reduce grid oscillations by using FACTS fast compensators and the DC switching of HVDCs for stabilizing dynamics and minimizing losses.
- Integrate renewable energies, with precise control of active and reactive power, facilitating the connection of wind and solar farms while compensating for their intermittency.

I.6.5. The emergence of smart grids

Smart grids represent an improvement on conventional power grids, integrating advanced communication, information and management technologies to manage the efficiency and sustainability of the electricity system. These advances systems enable dynamic, real-time management of electricity generation, transmission, distribution and consumption, and facilitate the integration of renewable energies and the active participation of consumers. The Table I.1 below compares smart grids with traditional power grids.

Table 0.1: Comparison between the traditional power grid and the smart grid [33].

Characteristics	**Traditional Grid**	**Smart Grid**
Control	Electromechanical	Digital
Communication	Unidirectional	Bidirectional
Generation	centralized	Centralized and distributed
Sensing	Limited Sensors	Widespread use of Sensors
Monitoring	Manual	Remote and autonomous
Restoration	Manual	Self-healing
Costumer Options	Limited	Diverse
Consumer Role	Passive	Active (prosumers)

Guelzim et al. (2016) have explained smart grids as *"a modern electricity infrastructure based on ICT and digital networks to collect information from the grid, i.e., consumed and produced electricity, and use this information to generate real-time consumption patterns in order to optimize the state of the grid in real time."* [34].

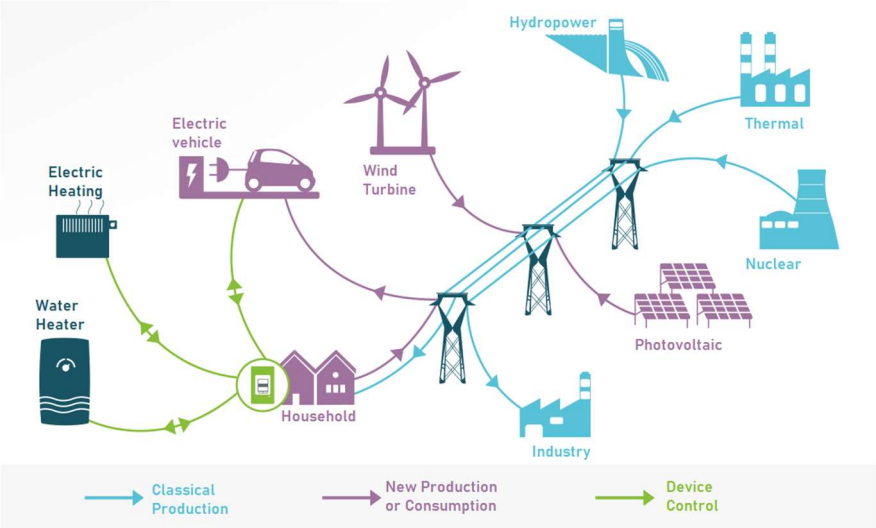


Figure 0.2: General structure of a smart grid [10].

The fusion of these two information and communication sectors in smart grids provides power system operators better optimized services in responsiveness, flexibility, and security. These grids aim to optimize the equilibrium between supply and demand by maximizing the use of data across the entire energy value chain, from generation to consumption, by taking advantage of management of transmission and distribution networks [10,33].

I.6.6. Information and communication technologies (ICT)

Information and communication technologies (ICT) include all of the equipment, services, and methods used to process, store, and record data. ICTs in the energy sector are based on the integration of communicative metering, management instruments and devices to the grid [18]. These include:

- Communicative and smart meters.
- Grid automation actuators.
- Fast and smart disconnection devices
- Communicating sensors.
- Wireless (Wi-Fi) and wired (ADSL, fiber optic) communication modules.

Already common in transmission and distribution networks, the application of ICTs in a global way permits the development towards smart grids, increasing the observability and credibility of the power grid. The main interests of ICTs in smart grids are to :

- Enable real-time exchange of information between producers, consumers and grid operators through a bidirectional communication.
- Real-time fault detection and system anomaly identification.
- Ensure preventive maintenance based on collecting and processing data from smart meters, sensors, and other devices.
- Guarantee the security of data exchanges and protect against cyber-attacks.
- Enable consumers to track their consumption, sell their own energy, and participate in grid management, thus turning them into prosumers.

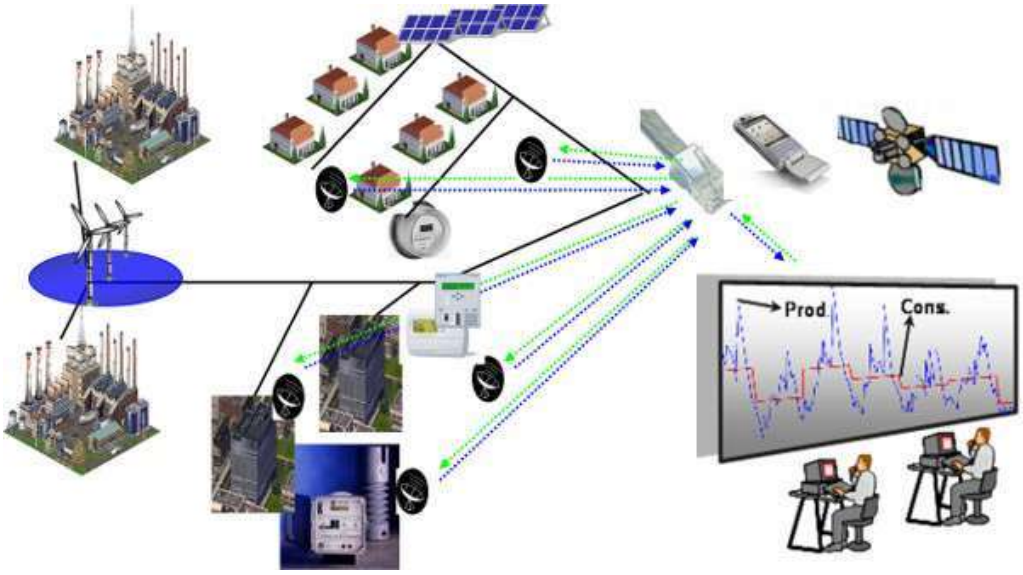


Figure 0.3: Communication diagram in an ICT-based smart grid [18].

I.6.7. Goals and challenges of smart grids

In any smart grid, challenges and objectives are identified to guide its design and architecture, define its level of complexity, the components and solutions to be used [27].

- **Extending energy coverage :** Smart grids facilitate the electrification of isolated areas not connected to power grids, such as rural areas and small towns in developing countries. They reduce dependence on fossil fuels to the population.
- **Supply reliability assurance :** The islanded mode secures the energy supply of smart grids, by providing electricity during power outages.

- **Optimize energy bills :** Consumers in residential areas can optimize their electricity consumption by choosing to draw power from the main grid, or from renewable energy infrastructures such as wind turbines.
- **Green energy production and consumption :** smart grids mark a major step forward in terms of energy transition, with the integration of renewable energy sources. Storage units help to offset the intermittent nature of renewable energies, thanks to their ability to store energy for long periods.
- **Maintaining grid balance :** Smart grids rarely experience transient disturbances, as the controllers installed at the heart of the systems serve to maintain a stable frequency and voltage.

I.6.8. Strengths of smart grids

Smart grids represent an innovative solution for modernising energy systems. They combine efficiency, sustainability and resilience. Their main advantages include:

- **Demand management and consumer flexibility**

Smart grids aim to transform energy demand, such as load control or demand management. The objective is to limit consumption during peak periods and achieve a balance between production and demand [10,27].

Demand response refers to the flexibility in electricity use, i.e. the ability of consumers to adjust their consumption:

- **Load reduction (load shedding) :** when demand on the power system is high (peak demand), consumers are encouraged to reduce their consumption.
- **Load increase :** when there is a shortfall in consumption, customers are encouraged to use available resources and therefore benefit from the excess generated by renewable sources of energy.

Consumers (previously passive) then become active, or “prosumers”, involved in managing their own energy demand.

- **Real-time control and communication**

Smart grids enable the optimization of energy systems through advanced control centers, operating as control towers managing real-time transmission and distribution flows.

These facilities rely on sensors, meters and different ICT devices to ensure :

- Supervision of grid status.
- Optimal flow management.
- Reactivity to defaults.
- Two-way communication to both operators and consumers.

- **Distributed energy production and storage**

In parallel with demand response, distributed generation involves the local production of energy via small-scale facilities connected to low-voltage grids. This approach promotes security of supply, price stability and energy autonomy.

However, the intermittent nature of renewable energy sources necessitates the use of storage technologies to achieve balance in the electrical grid system. The progress in electric vehicle technologies have spurred improvements in battery technologies in terms of their durability and economic viability and driven their use in the substations and power plants [44].

I.7. Conclusion

The evolution of the power grid is a multifaceted problem needing a critical understanding of the matters involved. In this chapter, we've examined the issues facing the energy industry, the functioning of the traditional grid and its limitations. We have also explored the recent concepts of smart grids, as well as their challenges and objectives. Changes need to be made to accommodate the growing electricity demands while fitting within the present environment of the electricity infrastructure.

The following chapter addresses intelligent microgrids, including their flexible and decentralized nature, design, classification, management, and optimization.

Chapter II

State of the art of microgrids

II.1. Introduction

Currently, the decline of fossil fuels, the deregulation of electricity public services, the advancements in the field of renewable energies as well as public awareness of environmental policies are the main factors driving the energy transition. At the same time, the heavy integration of DERs into the grid presents an energy dilemma that impacts the quality of energy supplied to consumers. In this context, microgrids appear to be an appropriate solution.

The emergence of bidirectional energy flows introduces a new terminology of "prosumers", who are both consumers and producers of energy. However, the extreme deployment of renewable energy sources, combined with the integration of new technologies such as electric vehicles, is turning the energy chain upside down, requiring changes at all levels.

In this chapter, we will review the state of the art of microgrids, from their objectives to their basic components and classifications. Specifically, we will analyse microgrid controllers and study the operating modes and structures designed to achieve optimal management.

In the second part, we will focus on energy management systems, their objectives, architecture, techniques and strategies within these frameworks.

Finally, a review of recent work is presented, providing an overview of the technological advances and current challenges facing these systems.

II.2. State of the art of microgrids

II.2.1. Definition and characteristics of microgrids

The US Department of Energy's definition of a microgrid is as follows: "*A microgrid is a group of interconnected loads and distributed energy resources within clearly defined electrical boundaries that acts as a single controllable entity with respect to the grid. A microgrid can connect and disconnect from the grid to enable it to operate in both grid-connected or island-mode"*. The ability of a microgrid to isolate itself and control the energy exchanged with the grid opens up a wide range of possible uses, from increasing grid resilience, providing backup power, delaying grid investment and increasing grid hosting capacity for distributed generation [35].

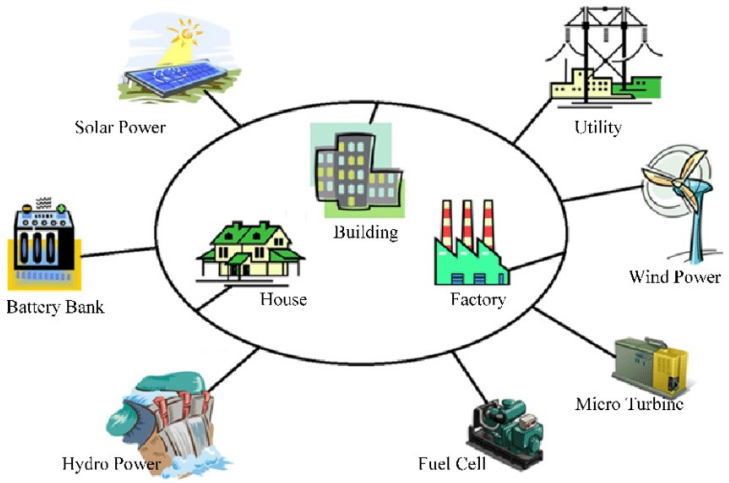


Figure 0.1: Basic structure of a microgrid [36].

A microgrid is identified by characteristics, such as autonomy, flexibility and diversity, with its composition of numerous micro sources, combining fossil and renewable energy sources, along with energy storage devices that ensure stability (Figure II.1). The interactivity and controllability enable mutual support between the microgrid and the main grid, allowing connected and island operation modes. Independence allows the microgrid to operate autonomously to meet local self-sufficiency demands [37].

II.2.2. Microgrid objectives

The emergence of microgrids is part of the energy and ecological transition initiative. The aims of microgrids are to :

- **Price control for extended energy coverage**

The installation of microgrids not only reduces dependence on the main power grid, but also offers an alternative for extending the grid to supply isolated areas.

- **Intelligent energy management**

By using energy optimisation tools, microgrids can efficiently energy from local production. This software analyses production capacity and identifies consumer needs.

- **Grid reliability**

In the case of an incident on the power grid, some local systems can take over to supply at least some of the equipment designated as high priority.

- **Contributing to the energy transition**

Since microgrids are part of the energy transition initiative, they prioritise the use of renewable energies to reduce GHG emissions.

II.2.3. Main components of the microgrid

The microgrid is based on a modular architecture that include a number of fundamental components that ensure autonomous and stable operation. Figure II.2 illustrates the overall architecture of a microgrid based on three main layers: infrastructures (loads), distributed energy sources, information processing and management systems.

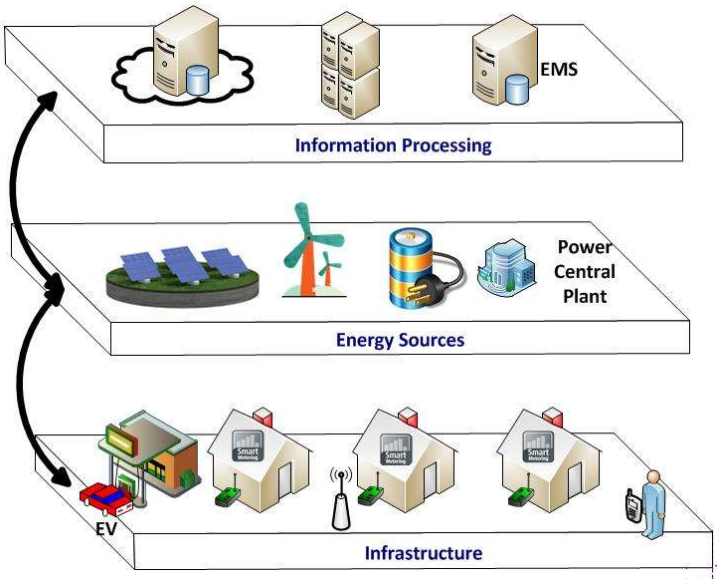


Figure 0.2: Main components of a microgrid.

II.2.3.1. Loads

Loads can be classified according to different factors [38]. These include :

- **Power:** Alternative current (AC) loads, such as household appliances (refrigerator, oven), pumps, air conditioners. Direct current (DC) loads, such as electronic equipment (computers, LED televisions), communication systems, DC charging points for electric vehicles.
- **Load classification :** these loads are divided into three levels.
    - **Level 1 :** The most critical loads, hospitals for example, which should never be unloaded.
    - **Level 2 :** Loads including air conditioning/heating of non-essential buildings or street lighting can be shed in the short term to reduce peak loads.

- **Level 3 :** Back-up generators for commercial installations and residential customers, which can be off-loaded to protect the microgrid and avoid a power failure.

II.2.3.2. Energy sources

Microgrids rely on distributed generation (DG) units in addition to the main grid, either at the point of use or nearby. The IEEE defines distributed generation as the production of electricity by facilities that are small enough relative to generating plants to allow interconnection at almost any point on a grid [39].

In microgrids, two types of generation technology can be applied: renewable sources such as solar photovoltaics, wind power, small hydro and hydroelectric power, among others, and non-renewable sources comprising internal combustion engines, gas turbines or modern combined heat and power units (CHP) [11].

II.2.3.3. Energy storage

Energy storage devices can be divided into three categories: electrochemical systems (batteries, electric vehicles and accumulators), kinetic energy storage systems (flywheels) and potential energy storage (water pumping and compressed air storage). In microgrid applications, they can be used to absorb excess power and discharge to cover the power deficit, thus serving to improve power imbalance and quality, reliability and stability between loads and the production of distributed resources [40].

II.2.3.4. Communication, supervision and control

Microgrids involve numerous levels of distributed intelligence in the grid, including all the components needed to build the command-and-control system. It is a mutually connected system that collects data from sensors, alarm systems and electrical measurements (ICT) that enables energy management through communication channels with microgrid components and parameters. This microgrid energy management system can be central or distributed depending on the size of the microgrids to satisfy control requirements according to their complexity [36].

II.2.4. Advantages of the micro grid

- **Efficiency**

Reduced power losses in transmission and distribution lines result in significant efficiency due to the proximity of micro-sources within the MG.

- **Better integration of renewable energies**

Microgrids facilitate the integration of clean energy, helping to reduce greenhouse gas (GHG) emissions.

- **Main grid support**

MGs can support the local grid and facilitate increasing production, improving system reliability, power quality, as well as grid stability due to the unique characteristics of MGs.

- **Continuous supply of loads in island mode**

Microgrids have the ability to independently and continuously supply power from microsources In the event of any failure. Also known as 'plug-and-play' ability, MGs can switch between grid-connected and islanded modes and act as the main power source, while maintaining voltage and frequency stability for all local loads.

- **Bi-directional power path**

By integrating MGs into a distribution grid, the concept of unidirectional power flow (substation to load designed for conventional distribution systems) can be transformed into a bidirectional structure [38].

II.3. Technical and economic limits of microgrids

- **Flexibility limits**

MGs are capable of operating in grid-connected as well as islanded mode. However, significant imbalances can cause severe frequency and voltage issues between generation and loads. If connections are made to a large number of generators at the same time, plug-and-play capability can be a serious problem [41,42].

- **Integration of renewable energy production**

The variability, unpredictability and weather dependence of renewable energy sources are some of the main challenges for integrating (RES) generation into the main grid. As a result, electricity production from these resources can vary suddenly and frequently, disturbing the stability of the microgrid. In addition, one of the problems encountered is that the increasing share of renewable energy can lead to congestion (saturation) of the distribution grids. Other problems may include the intermittency of renewable energy production and the lack of dispatching capacity.

- **Protection**

Protection of the microgrid is the biggest challenge, as it is not easy to design an appropriate protection system that must respond to faults in both the main grid and the microgrid. In fact, the magnitude of the fault current in the system depends on the operating mode of the microgrid and can vary considerably between grid-connected and stand-alone operation. Traditional power grids have been designed with unidirectional current flow for distribution systems. However, the integration of distributed energy sources into the main grid with microgrids changes the current flow from unidirectional to bidirectional. The microgrid is connected to the main grid by a fast static switch to protect a microgrid in both operating modes against all types of faults.

- **Profitability of microgrids**

Overcoming these technical and regulatory obstacles won't resolve the economic problems. High investment costs impede microgrid commercialization. Technologies such as photovoltaics, fuel cells, and storage struggle to become profitable without financial support [42].

In the mid-to long-term, additional costs such as operation and maintenance add to the economic burden. In addition to regular upgrades and developments to remain competitive [43].

Variations in electricity price impact expenses, an increase of 50% in purchase price can raise operating expense by 20-40%, affecting profitability. Dynamic pricing is applied by some projects to respond to this. studies indicates that user empowerment can reduce expenses but is complex to manage and increases upfront cost [44].

II.4. Classification of microgrids

Microgrids can be classified according to several parameters, including phase, power type, operating mode, supervisory control and application, as shown in Figure II.3.

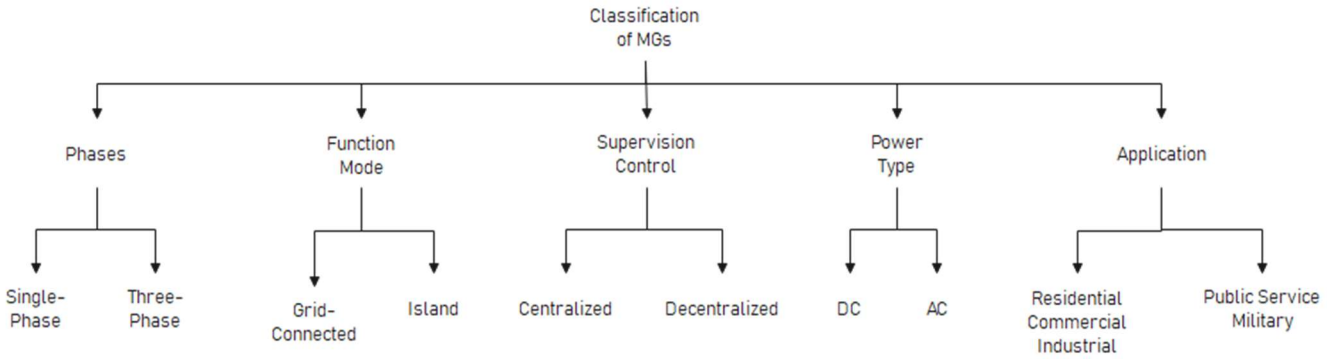


Figure 0.3: Classification of microgrids.

II.4.1. Classification by phases

- **Single-phase:** used for small loads, particularly residential loads.
- **Three-phase:** required for large loads and heavy equipment such as industrial loads.

II.4.2. Operating mode

- **Grid-connected mode:** the microgrid is connected to the main grid, that enables energy exchanges (purchase/sale) and greater stability.
- **Isolated mode (island):** operates independently of the main grid. It must ensure a balance between production and consumption.
- **Autonomous mode:** the microgrid manages its resources without external support. This mode encourages intelligent self-governance through advanced control systems.

II.4.3. Supervisory control

Control structures can be classified as follows:

- **Centralised control:** Relies on a single controller that manages, communicates with and controls the entire microgrid.
- **Decentralised control:** there are several individual local controllers. Decentralised control uses algorithms to enable components to make decisions.

II.4.4. Types of power supply

Microgrids can be categorised according to the type of current and the way their buses are connected. There are three groups based on the type of current:

II.4.4.1. Alternating Microgrids AC

AC microgrids are the most common type of microgrids, as they can be easily integrated into conventional AC power systems (figure II.4), giving them more control and flexibility than other types of microgrids [10,38].

AC microgrids offer a number of advantages:

- Can be integrated into the main grid or operate in island mode.
- Compatibility with AC equipment and loads such as motors.
- Cost-effective optimisation of protection systems
- No need for an inverter for AC loads.
- Improved energy availability for loads.

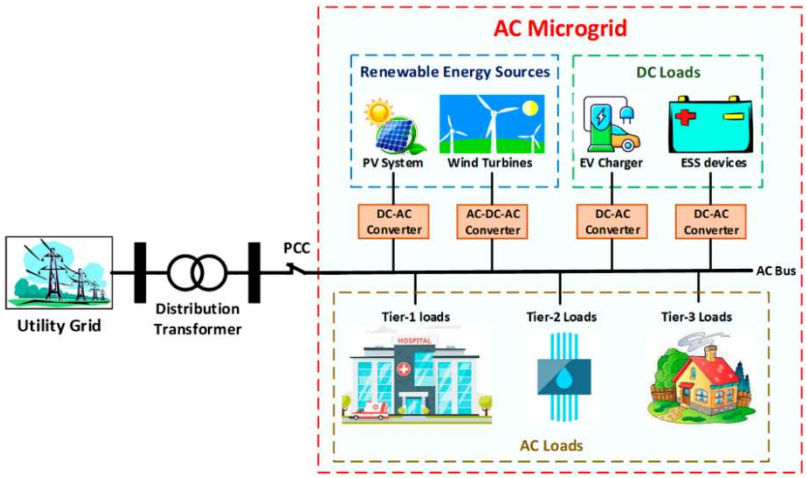


Figure 0.4: Structure of an AC microgrid [38].

II.4.4.2. DC Microgrids

The operating principle of a DC microgrid is similar to that of AC, with a common DC bus used to connect to the grid via an AC-DC converter and link the various components of the microgrid.

DC microgrids offer greater stability because they do not generate reactive power, they simplify voltage control, and facilitate the integration of renewable energies. In addition, the major advantage of these installations lies in the energy and economic benefits they offer due to the reduction in the number of converters, since these microgrids reduce conversion losses, requiring only a single conversion to connect the DC bus (Figure II.5).

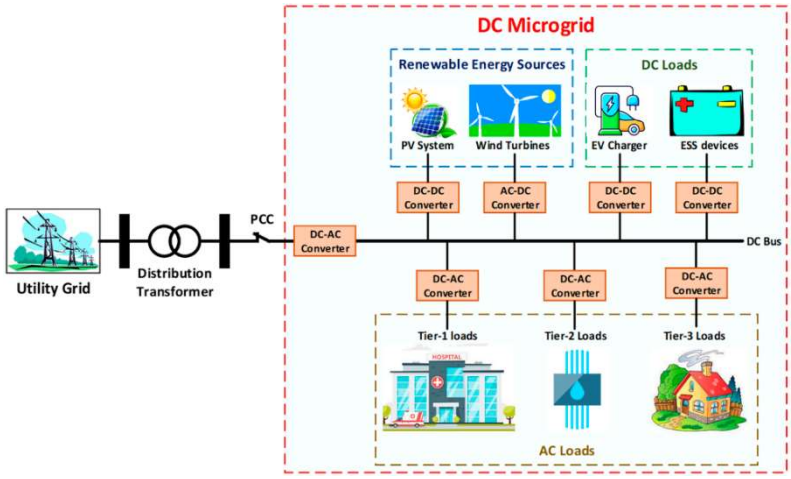


Figure 0.5: Structure of a DC microgrid [38].

However, the increase in DC loads in the system and the desire to integrate more renewables into microgrids have drawn attention to hybrid microgrids.

II.4.4.3. Hybrid microgrids (AC-DC)

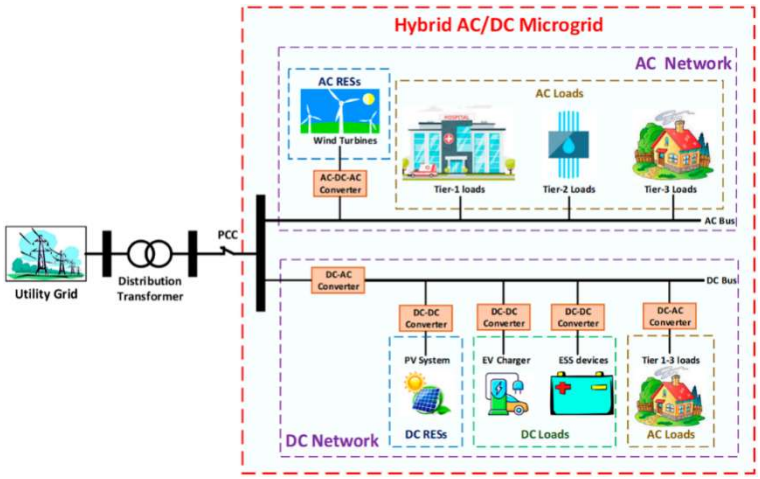


Figure 0.6: Structure of a hybrid microgrid (AC-DC) [28].

Hybrid microgrids are obtained by combining (DC) and (AC) microgrids (Figure II.6). They combine all the advantages of both structures, since they enable the integration of decentralised production, renewable energies and energy storage systems. They also make it possible to reduce the number of interfacing devices, resulting in fewer conversion stages, lower power losses, lower overall costs and greater reliability.

Hybrid microgrids offer the possibility of integrating (AC) and (DC) loads without the need for synchronisation, so that consumers can use electricity according to their needs.

II.4.5.Classification by applications

There are five types of microgrid applications:

- Commercial or industrial microgrids are typically connected to the grid to reduce demand and costs. These systems also provide a back-up power source against operational disruptions.
- Residential microgrids are designed to reinforce the stability of the public grid due to the high consumption of residential and utility customers.
- Public service microgrids are installed by institutions such as universities, hospitals and industrial facilities, because of the requirements for uninterrupted power supply.
- Military microgrids are implemented where both physical and cyber security are required. These microgrids are installed where the power grid is not available.
- Remote microgrids generally operate in island mode and are powered by diesel generators in addition to renewable energy sources.

II.5. Operating modes of microgrids

II.5.1. Grid-connected mode

The microgrid is connected to the main grid according to energy needs, and can receive partial or total energy depending on demand and power sharing. It can also inject (or sell) surplus energy when local production exceeds consumption. This mode of operation provides optimum energy and economic distribution, through autonomous voltage and reactive power regulation, which improves overall stability [46].

II.5.2. Island mode

In the event of failure of the main grid or of planned actions upstream (e.g. maintenance operations), the microgrid can easily disconnect from the grid, and switch to island mode to operate as a stand-alone system, while ensuring system stability.

This islanded mode of operation must consider load magnitude, load type, load reliability requirements, and DER availability in order to balance load and generation in this autonomous mode. Several techniques exist, such as load-following, generation-tracking and load-shedding [46,47].

II.6. Microgrid control

II.6.1. Definition

A microgrid control system includes the control functions that define the microgrid as a self-managing system, that can operate isolated or connected to the grid, and dynamically transition between the main distribution grid for the exchange of energy and delivering gris support services. It represents the single point of interface with any distributed energy resource (DER), DER management system (DERMS) or energy management system (EMS) as illustrated in figure II.7.

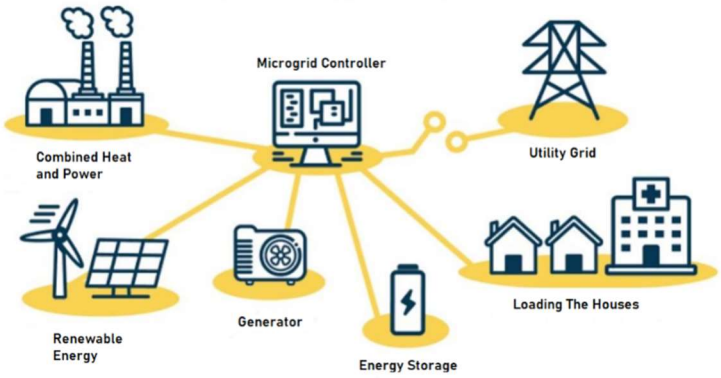


Figure 0.7: Central role of the controller in the grid architecture.

The system must incorporate real-time control and energy management functionalities that operate in the following scenarios:

a. Operation in grid-connected and island mode.
b. Automatic transition to island mode, ensuring a seamless transition for microgrid loads in the event of failure or maintenance.
c. Resynchronisation and reconnection from island mode to grid-connected mode.
d. Energy management to optimise the production and consumption of active and reactive power.
e. Ancillary services provision, support of the grid, and participation in the energy market and/or utility system operation, as applicable.

The microgrid control system consists of software, hardware or a combination of both, and can be physically implemented in a variety of ways, including centralised or distributed, as well as Energy Management System (EMS) capabilities needed to dispatch microgrid resources. These functions are often used in commercial and industrial installations [48].

II.6.2. Standards for microgrid controllers

The design of a microgrid requires high-performance controllers, paired with the distributed energy sources that form its infrastructure. These controllers must comply with various international standards, amongst which IEEE Std 2030.7 (IEEE Standard for the Specification of Microgrid Controllers) is one of the most widely recognized.

This standard establishes the technical criteria for microgrid controllers and sets out the basic specifications for energy management systems. It guides the two main control functions, which are the management of transition zones and the distribution of energy. Another essential standard is IEEE Std 2030.8 (Standard for the Testing of Microgrid Controllers), which defines the protocols for testing controllers, their functions in connected and islanded modes, power flow management, local control and load management [48,49]. Table II.1 summarises the essential standards for microgrid control.

Standard	Description
IEEE 2030.7	Specifies the architecture, functions and interfaces of a microgrid controller.
IEEE 2030.8	Defines methods for performance testing and validation of microgrid controllers.
IEEE 1547-2018	Standard for the interconnection of distributed energy resources to the grid.
IEC 61850	Communication standard for the automation of power systems (substations, microgrids, etc.).
IEC 62257	Technical recommendations for hybrid or renewable energy systems in rural areas.

Table 0.1: Summary of the main standards applied to microgrids.

II.6.3. Control strategies for microgrids

To classify the control architectures developed for microgrids, two factors can be used: the complexity of the model (used for dynamic control of the grid) and the degree of communication (between controllers in different units or hierarchical levels). Based on these factors, control structures can be classified as follows:

II.6.3.1. Centralized control

The centralized architecture involves gathering all the measured data from all the DERs in the microgrid, which is used to adjust the setting for each piece of control equipment and sending it to the central system (Figure II.8). This control is particularly suited to small-scale microgrids. Moreover, from an economic perspective, centralized hierarchical control offers an efficient solution. [40,51].

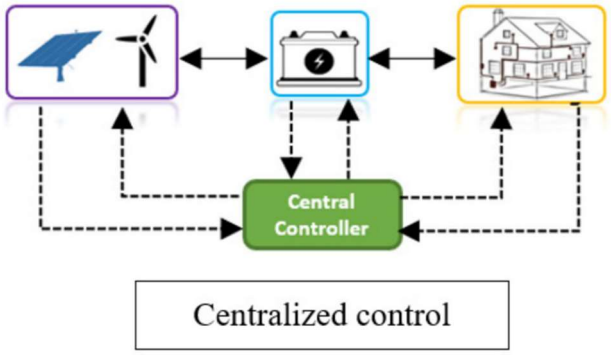


Figure 0.8: Structure of Centralized control [50].

However, this type of control is unreliable and lacks redundancy. Other disadvantages of this type of control include potential communication problems and the requirement to shut down the entire system during maintenance or failure.

II.6.3.2. Decentralized control

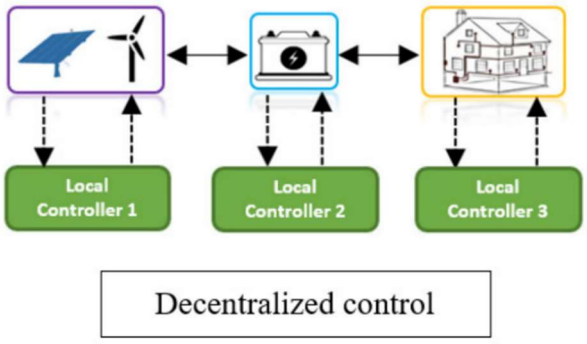


Figure 0.9: Structure of Decentralized control [50].

Decentralized control is characterised by a monitoring system made up of several local controllers, each operating autonomously, with no exchange of information with the others. As there is no communication system between units as shown in figure II.9, each manages its voltage and frequency regulation independently, based on local measurements. In this way,

each DG contributes to the sharing of active and reactive power according to local strategies (such as droop control). This type of control is more suitable if the users of the microgrid have different objectives or a different operating environment. As a result, the decentralised structure significantly reduces computational requirements and removes dependency on a communication grid, which improves the robustness and reliability of the microgrid.

Nevertheless, this approach is not perfect because of statism, which causes uncontrolled variations in voltage and frequency, and can make synchronisation between sources difficult, especially if they are far apart [40,46].

II.6.3.3. Distributed control

This architecture is based on a compromise between centralisation and decentralisation. Unlike decentralised control, distributed control is made up of local controllers that are connected to a bidirectional communication bus used to exchange data between loads and power generation sources such as frequency and voltage (Figure II.10) [52].

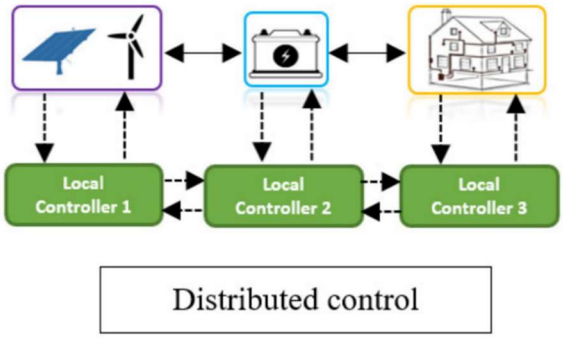


Figure 0.10: Structure of Distributed control [50].

The following table provides a summary comparison of the different microgrid control architectures, presenting the advantages and disadvantages of each:

Table 0.2: Comparative table of control architectures for Microgrids.

Method	Advantages	Disadvantages
Centralized	– Globally optimal solutions	– Calculation complexity – Communication infrastructure required – Stability affected by the communication grid – Reduced scalability

Decentralized	– Greater reliability – No communication required – No computational burden – Improved scalability	– No globally optimal solution
Distributed	– Better reliability – Reduced computational demands – Improves scalability	– Communications infrastructure required – Stability affected by communication grid – Sub-optimal solutions

II.6.4. Control architecture

In order to ensure optimal operation of a microgrid, the control architecture is built in a hierarchical way and is based on several levels of controllers as shown in Figure II.11 below and described as follows :

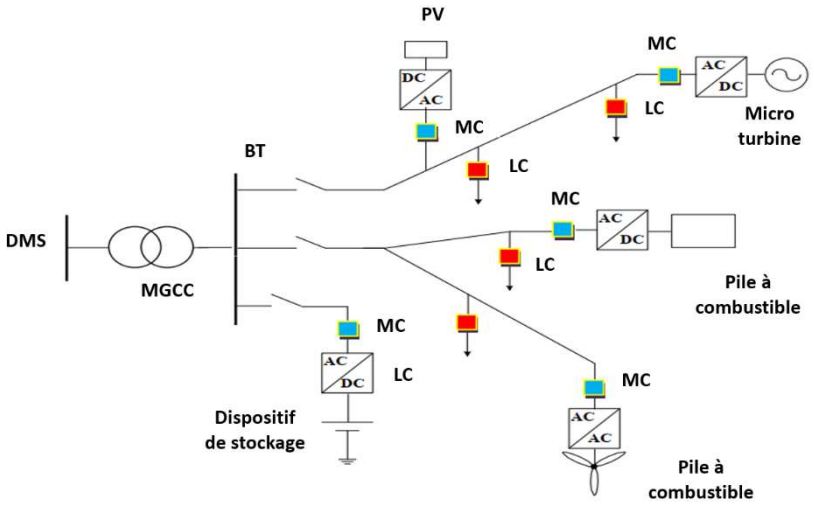


Figure 0.11: Microgrid control architecture [47].

II.6.4.1. Microsource controller (MC) and load controller (LC)

These two controllers are located at the local sources and loads level. The MCs are directly connected to distributed energy generation (DG) units such as photovoltaic panels, wind turbines or generator sets. They ensure local regulation of primary parameters (voltage, current and frequency), as well as managing the transition from grid-connected to islanded mode (and vice versa).

LC controllers are dedicated to managing and optimizing electrical load consumption. They guarantee local supervision of the power consumption of each load, and enable automatic load shedding where necessary (e.g. switching off non-priority loads). In addition to load shedding, LCs also help to stabilize the voltage/frequency of the microgrid through coordinated load reduction [46,52].

II.6.4.2. Microgrid central controller (MGCC)

The central microgrid controller (MGCC) acts as the primary control unit coordinating the microgrid's all energy sources and loads. It decides generation, storage, and consumption in real time and issues commanding signals to the LC and MC controllers. The MGCC's tasks can be listed as follows [47,56] :

- control and monitoring of active and reactive power of distributed resources.
- Supervision of grid connection and disconnection operations.
- Regulation of energy flow.
- Fault level management in case of an irregularity.
- Execution of cold start actions and load shedding operation.
- Supervision of energy exchanges on the market.

II.6.4.3. Distribution management system (DMS)

The Distribution Management System (DMS) operate at the largest scale of microgrid control and is an aggregator monitoring production, storage, and consumption from multiple microgrids at once. Its aim is to make the optimized choices to ensure reliable, cost-effective operation of the microgrid, and to maintain quality of service even when operating in island mode.

In a context of interconnected microgrids, the DMS must in particular:

- Manage and coordinate connection and disconnection operations performed by MGCCs;
- Coordinate active and reactive energy flows between microgrids;
- Optimize information exchange between local units (LC, MC, MGCC) to ensure overall stability;

- Evaluate and manage the disconnection and resynchronization of microgrids during or after faults;

The DMS thus enables a flexible approach to be implemented, which can be centralized with supervision provided by the MGCC, or decentralized, by distributing specific operational responsibilities to local LC and MC controllers. It also contributes to the appearance of microgrids as simple concentrated loads or small power plants that are flexible from the grid's perspective [10,46].

II.6.5. Management control

In order to control the various systems in the grid, it is important to define a set of control objectives that will ensure the smooth operation of the system. The hierarchical control structure consists of three levels, with primary, secondary and tertiary control, as shown in Figure II.12:

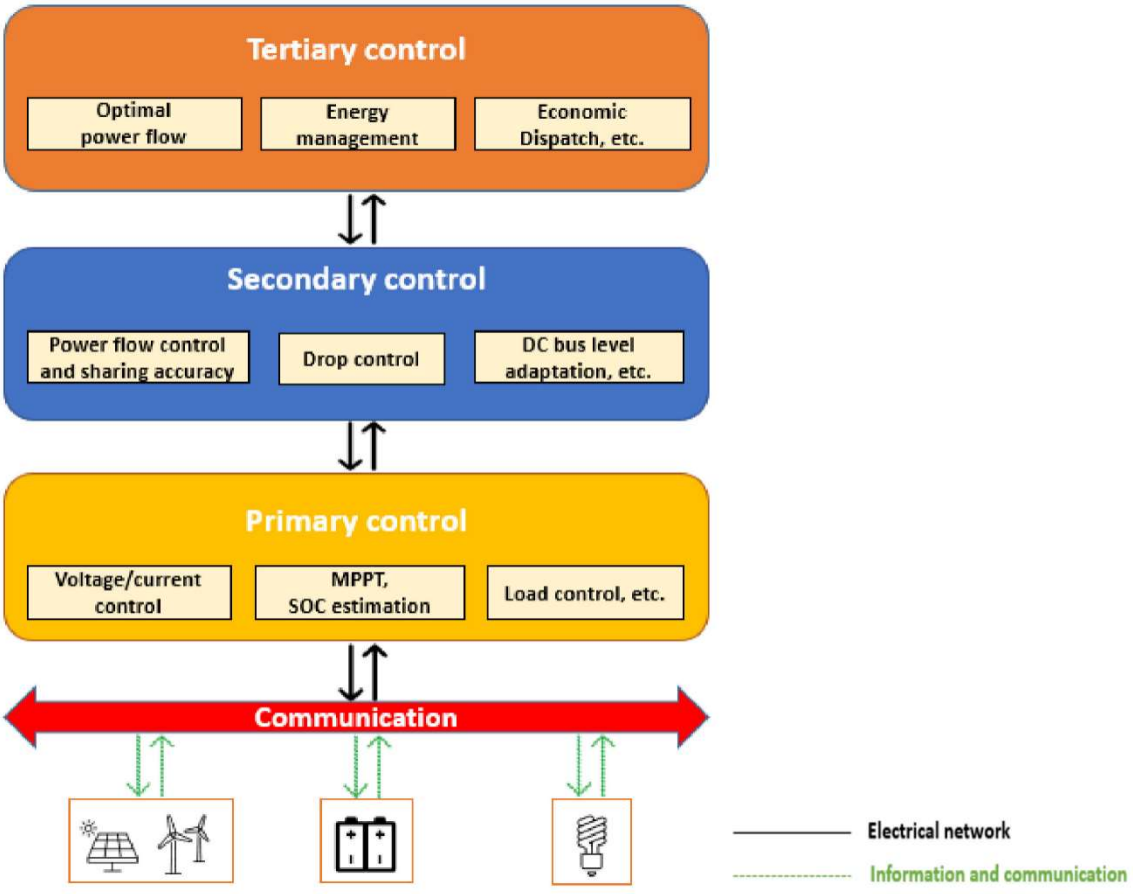


Figure 0.12: Goals and typical control functions in a microgrid [52].

II.6.5.1. Primary control

Primary control is the first level of the control hierarchy, it constitutes the fastest-response layer operating at the level of each source, storage and load unit in the microgrid. The aim is

essentially to regulate local voltages, frequencies and currents in the presence of disturbances. This enables better control of circuit breakers/switches and load [50,52].

II.6.5.2. Secondary control

When connecting or disconnecting large loads in a microgrid, the system may show unstable dynamic behaviour, sometimes triggering overvoltage or undervoltage protection, due to deviations induced by primary control.

In this case, secondary control (coordinated by the MGCC) takes over at a higher level, to attenuate frequency and voltage (DC bus) deviations by dynamically adjusting converter setpoints using control methods such ats proportional-integral (PI) control, adaptive controllers, or predictive approaches. These techniques enable the droop control characteristic curves to be shifted in order to restore the setpoints.

The secondary control level is responsible for improving the precision of power sharing between distributed sources (DER), as well as facilitating microgrid synchronization with the main grid. It also ensures optimal economic management by adjusting power flows according to grid conditions and MGCC operational strategies [50,52,53].

II.6.5.3. Tertiary control

Tertiary control, the highest level of the microgrid control hierarchy, is often integrated into the energy management system (EMS). It coordinates interactions with the main distribution grid and microgrids. This level incorporates external economic, environmental and technological criteria (electricity prices, weather data, etc.) and the dynamic MG system (energy storage, distributed sources) to guide energy management over an extended planning horizon [54]. Tertiary-level control is responsible for determining the energy flow (to be injected or consumed) that secondary control must satisfy. It is also at this level that decisions are taken regarding the sale of surplus energy, the microgrid's participation in the market, and the optimization of exchanges with other grids, in order to enhance the microgrid's profitability and integration into the overall energy system [52,55]. The following table summarizes the three control levels and their functions:

Table 0.3: Summary table of hierarchical levels of control in a microgrid.

Level	Controller	Function	Time Scale
Primary	Local Controllers (LC) and Micro-source Controllers (MC)	Very fast local control (V/f) without external communication	Microseconds to seconds
Secondary	Microgrid Central Controller (MGCC)	Coordination between units, restoration of set values	Seconds to a few minutes
Tertiary	Distribution Management System (DMS)	Economic optimisation, interaction with the main grid, management of energy flows	Minutes to hours (or even days)

II.7. Energy management systems in microgrids

II.7.1. Definition of EMS

The main feature of microgrids is the notion of controllability, which differentiates them from active distribution grids. Microgrid control architectures feature the Energy Management System (EMS) as a fundamental component.

Microgrids are defined by their controllability, which makes them different from active distribution grids. One of the control levels of microgrids is the energy management system. Through this process, microgrids systematically manage energy transactions with the main grid to meet technical, environmental, and economic constraints. Optimizing the production and consumption of different types of energy carriers (e.g., electricity) within a microgrid maximizes its benefits. The energy management system is the most important element of the microgrid and is necessary to accomplish this task. The energy management system is responsible for collecting information, controlling various entities including distributed energy resources (DER) and energy storage devices, executing demand response programs, selecting optimal microgrid operating strategies based on different circumstances, and forecasting renewable energy resource production and load consumption (Figures II.13, II.14) [57,58].

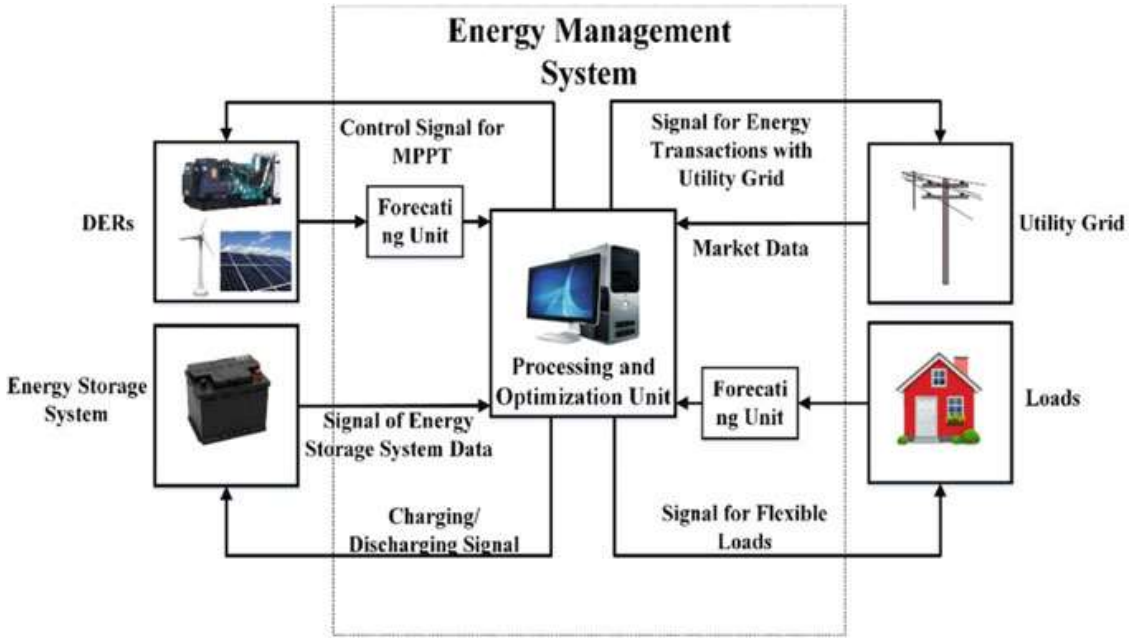


Figure 0.13: Architecture of the energy management system (EMS).

Some of the main responsibilities of the energy management system are:

- Monitor real-time generation and load profiles across production/generation units.
- Ensuring the balance between production and consumption.
- Ensuring compliance with and implementation of the rules for connecting the microgrid to the main distribution grid.
- Make optimum use of existing resources.
- Minimise overall operating costs.
- Separate the microgrid from the main grid in the event of an emergency.
- Provide a practical control strategy for reconnecting to the main grid after islanding.

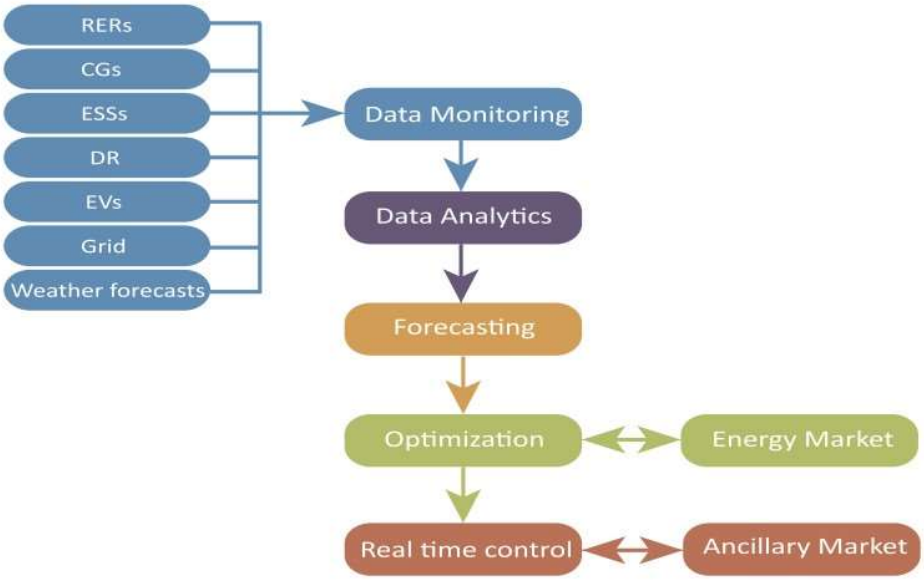


Figure 0.14: Schematic diagram of a microgrid energy management system [60].

II.7.2. Comparison of energy management strategies

- **EMS in a grid-connected microgrid**

In a microgrid connected to the main grid, economic optimisation is generally the main objective of energy management. Energy is exchanged with the main grid to minimise costs by buying energy during low periods and selling the surplus during periods of high demand. Renewable resources are exploited as a priority, with any generation-consumption imbalance managed through grid exchanges.

- **EMS in an autonomous microgrid**

The autonomous microgrid has to guarantee a balance between production and consumption at all times, using only the distributed generation available in the microgrid. As a result, it relies heavily on storage systems to achieve this balance.

As opposed to the connected microgrid, energy surpluses cannot be sold back to the grid; they are either stored or dissipated. Deficits, on the other hand, need to be compensated rapidly to avoid voltage and frequency drops, or even power outages [59].

The following table II.4 summarises and compares the fundamental objectives of the two microgrids:

Table 0.4: Comparison of energy management objectives in a connected and autonomous microgrid**.**

Criteria	Grid-connected Mode	Autonomous Mode
Main objective	Economic optimisation	Reliability and stability
Short-term management	Exchanges with the grid	Autonomous V/f control
Long-term management	Minimisation of costs	Self-consumption
Role of storage	Economic arbitration	Critical backup

II.7.3. Advantages Of EMS

Energy management systems offer the following advantages [10,46,51]:

- Energy optimisation : by integrating data such as load profiles, weather forecasts, consumer profiles, etc.
- Cost reduction : Minimise operating costs, energy losses and dependence on the grid.
- Greater reliability : EMS provide stability and continuity of energy supply, even in the event of a fault.
- Reduced GHG emissions : Prioritising energy from renewable resources to reduce GHG emissions.
- Peak shaving : Energy control and management techniques are used to reduce or offset peaks caused by high demand for electricity.
- Autonomy and stability : EMS guarantee the durability and resilience of the service both when connected to the main grid and in island mode.

II.7.4. Architecture of EMS

Microgrid EMS control architectures are classified into two types [61] :

- **Centralized EMS**

In the centralized approach, the central controller collects all the information including DER production, electricity prices, weather forecasts, consumption schedules, etc. in order to determine an optimal energy programming strategy.

Its main objective is to maintain the energy balance of the entire electricity grid. This approach focuses primarily on improving reliability and reducing costs.

This architecture is referred to as top-down, because there is little feedback from the units to the EMS, which imposes decisions in a hierarchical structure.

- **Decentralized EMS**

Decentralised EMS is a controller that integrates several local controllers, treating each entity in the microgrid as an independent system. Unlike the centralised EMS, the local controllers make the decisions, which reduces the communications and computing load.

With the uncertainties associated with RES production and consumer demand, both can complicate the energy management, local controllers need to determine the optimum power locally so that they achieve a reconciliation between local and global objectives.

However, this architecture is becoming more complex in terms of security, as it is susceptible to cyber-attacks due to the complexity of the system and the multiple sources of information.

II.7.5. Energy management techniques

Various energy management strategies have been proposed in the literature to achieve optimal and efficient operation of microgrids. In some practical cases, an EMS may combine several of the optimisation techniques.

Optimisation techniques can be classified in several ways in the literature. Figure II.15 illustrates the classification into : traditional, mathematical, metaheuristic, artificial intelligence technique or stochastic techniques.

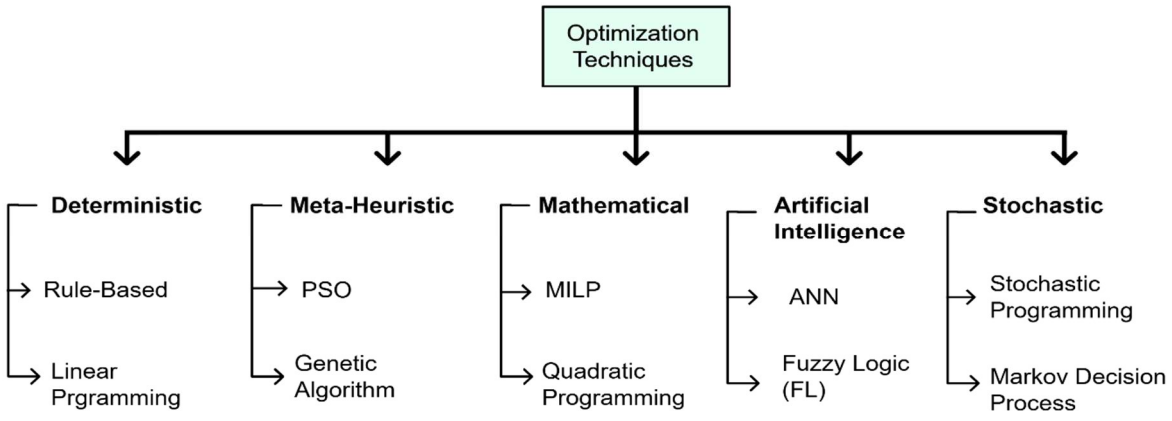


Figure 0.15: Classification of Energy management optimisation techniques.

- **Traditional technique (deterministic)**
  - **Rule-Based Method (RB)**

The rule-based method is a conventional deterministic approach that relies on a set of logical rules (if-then-else) to perform optimised energy management of the microgrid. Decisions are taken on the basis of the actual conditions. This method does not require any specific knowledge or optimisation algorithm.

The RB method requires real-time monitoring variables, such as the SOC (state of charge) of the batteries, the active power, the price of electricity, etc.

This technique stands out for its simplicity of use, low computational requirements and rapid adaptation to real-time systems. However, its robustness to adaptability remains limited [62].

The rule-based method will be developed for the energy management of the microgrid studied, because of its ability to ensure rapid decision-making in a dynamic environment.

- **Mathematical technique**
  - **Mixed-Integer Linear Programming (MILP)**

Mixed-Integer Linear Programming (MILP) is a mathematical technique used to manage energy in a microgrid. In this method, some variables take integer values (0 or 1), while others take continuous values. MILP is used to model energy planning problems that involve binary decisions (e.g. activating/deactivating a unit), and at the same time, continuous energy flows (power, tariffs, etc.). The aim of this method is to minimise cost while reducing losses.

MILP is effective for hourly or daily scheduling in complex microgrids, but it is computationally intensive and becomes less flexible in real time [63].

- **Metaheuristic technique**
  - **Particle Swarm Optimization (PSO)**

PSO (practice swarm optimization) is a metaheuristic technique inspired by the collective behaviour of swarms of birds or fish. The candidate solutions, called particles, evolve in a multidimensional space by adjusting their positions and speeds according to their best-known positions and the best position of the whole.

This memory-based technique helps to solve complex problems. It is particularly well suited to non-linear, multi-objective problems in microgrids where conventional methods fail because of discontinuities or the presence of local minima [45,63].

- **Artificial intelligence Technique**
  - **Artificial Neural Network (ANN)**

ANN is a technique inspired by the human brain, capable of simply learning from data and then modelling complex, non-linear systems. This technique is used to predict or adjust energy variables such as power load, solar/wind generation or dynamic control decisions. However, this method remains limited, since ANN requires a sufficient data set and is very often difficult to interpret (black box) [51,60].

- **Stochastic technique**
  - **Stochastic programming**

Stochastic programming is a mathematical optimization method designed to make robust decisions under variable parameters. It enables two-stage action. It performs advance planning, followed by real-time adjustment. Stochastic programming is ideal for highly uncertain environments such as intermittent RES. Except that it requires a great deal of computing time [64].

II.7.6. Energy management strategies

New management strategies are also essential to improve the overall performance of the power grid. In a microgrid, the bidirectional flow of electrical energy and data enabled by smart technologies enhances energy efficiency, reduce operational costs, ensures production-consumption balance, limits GHG emissions and optimizes the use of utility resources.

Traditional electricity companies strive to match production to demand. However, this represents a significant long-term cost, as load is unpredictable and continues to increase seasonally, requiring additional generation capacity. A mismatch between supply and demand can lead to cascading faults across the grid, or even blackouts.

In a microgrid, demand-side management (DSM) manages the consumer's use of energy in relation to supply conditions in a fairly intelligent way. The microgrid does not need to

continuously adapt generation to load; rather, it equalizes demand and supply using control technology or by actively involving consumers.

Table 0.5: Main demand management strategies.

Strategy	Description	Main objective	Reference
Demand Side Management (DSM)	Methods of managing consumption through load shifting, demand response (DR), dynamic pricing (ToU, RTP).	Peak reduction, integration of RES, grid efficiency.	[60]
Economic Dispatch (ED)	Continuous optimisation of power dispatch between generation units, minimising hourly cost.	Reduction of operating costs and better use of resources.	[65]
Unit Commitment (UC)	Planning the activation or shutdown of production units over the medium term (day, week).	Reduction in medium-term operating costs.	[65]

II.7.7. Demand side management

Demand Side Management (DSM) is an initiative implemented by utilities to encourage consumers to adopt mutually beneficial procedures and practices. These practices include any systematic interventions designed to influence consumption patterns and reshape load profiles.

The DSM strategy includes various policies such as energy efficiency, consumer load management and demand response (DR).

The six most commonly used DSM techniques are peak shaving, valley filling, load shifting, strategy conservation, strategic load growth and load shape flexibility as shown in Figure II.16 below [68,69] :

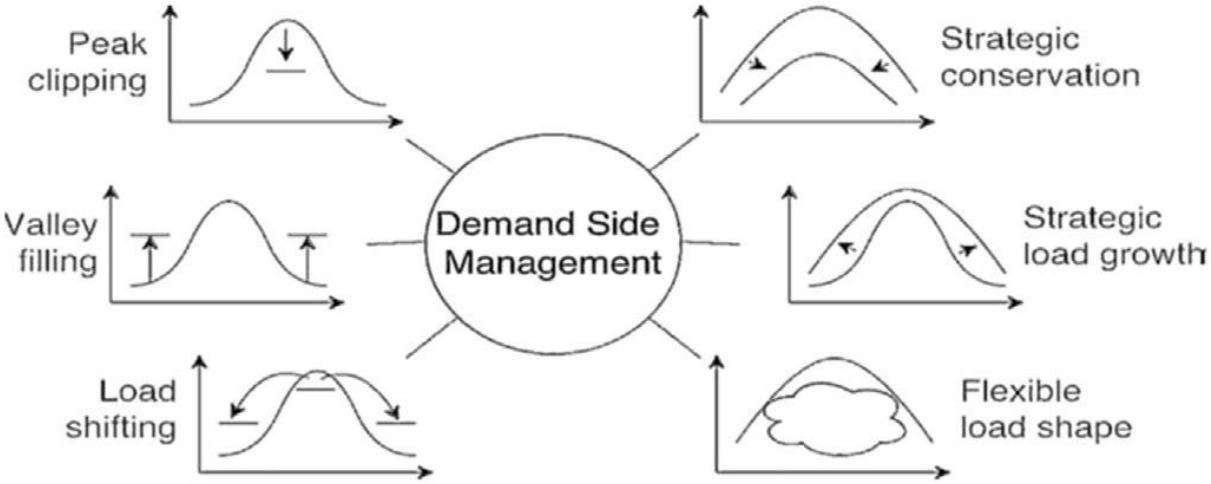


Figure 0.16: Demand management techniques [68].

- **Valley filling :** Consists of encouraging consumers to consume outside peak hours in order to benefit from low energy costs. This approach is valuable because it lowers the average cost of electricity and improves system efficiency.
- **Load shifting :** This technique enables peak workloads to be shifted to off-peak periods without altering total consumption. It also uses the time-independent characteristics of certain appliances to shift their use to off-peak hours.
- **Strategic conservation :** Consists of reducing seasonal energy consumption mainly by increasing consumption efficiency and reducing energy waste. This program is relativly comprehensive and includes incentives for technological change.
- **Strategic load growth :** This strategy controls the growth in seasonal energy consumption by using intelligent systems and processes, more efficient equipment and more competitive energy sources to achieve its objectives.
- **Flexible load shape :** This technique operates by identifying consumers with flexible loads, who are most ready to respond to the incentive.
- **Peak shaving :** Involves reducing the demand for a large load, i.e. when demand exceeds supply. The peak must be reduced to achieve the utility's desired load profile.

II.7.8. Demand response programs

Demand response (DR) is a short-term load modulation and optimisation program that aims to influence consumer behaviour through changes in electricity prices and incentives. According to the US Department of Energy, demand response is defined as *"changes in electricity use by end-use consumers from their normal consumption patterns in response to changes in the electricity prices over time, or to incentive payments designed to induce*

lower electricity use at times when wholesale prices are high or system reliability is at risk" [70].

As shown in Figure II.17, DR programs are divided into incentive and market-based programs:

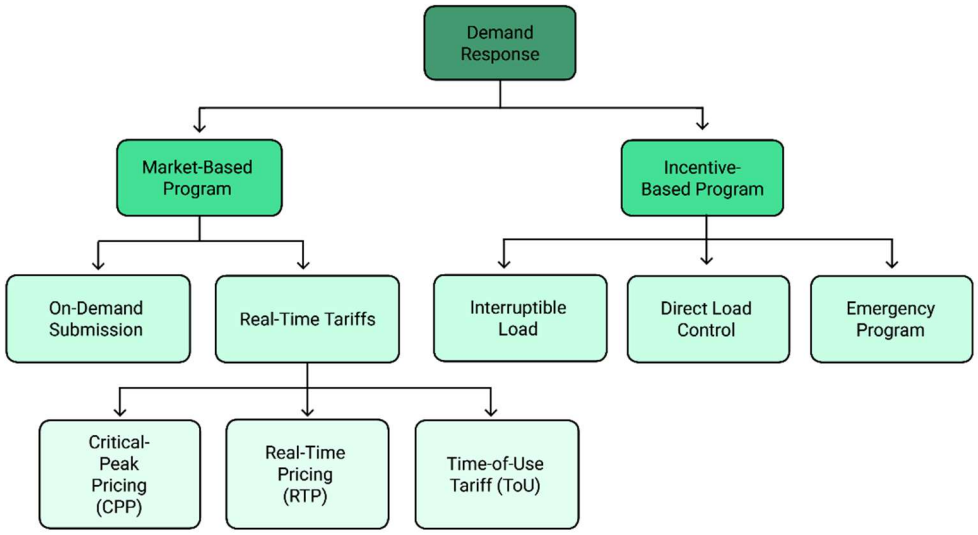


Figure 0.17: Categorization of demand response techniques [71].

Incentive-based DR programs are based on a commitment by consumers to reduce their charges in exchange for economic reductions and incentives. This program is based on the following elements :

- **Interruptible load program :** This program is aimed at large industrial and commercial consumers who have a major impact on the electricity market, with the intention of suspending their loads for a short period. In return, they benefit from a rate reduction in exchange for interrupting their load at the appropriate time. However, these consumers may incur penalties if they fail to fulfil their obligations when required [71].
- **Direct load control program :** This program consists of directly interrupting or reducing the power supply to consumers (non-critical loads, e.g. air conditioning) during periods of peak consumption after they have been informed. Participating customers receive financial compensation for their curtailed consumption [70].
- **Emergency program:** Consumers are encouraged to reduce their consumption in the event of system disturbances. Unlike the interruptible load program, this program imposes no financial consequences on consumers if they are unable to participate [72].

Another classification of demand response is the market-based program. This tariff program offers consumers the possibility of adjusting their electricity consumption, while benefiting from economic advantages. Market-based DR programs include the following elements :

- **Demand submission program:** This program allows large consumers to submit offers for specific load reductions. Consumers remain at a fixed price and receive high payments when electricity wholesale prices are high.
- **Real-time pricing programs :** In order to resolve price fluctuations over time despite undesirable average fixed costs, in particular for commercial and industrial customers, the following real-time pricing programs have been introduced [70,72] :
  - **Real-time pricing (RTP) :** Often referred to as dynamic pricing, consumers pay rates that differ every hour according to actual market rates. Using day-ahead pricing schedule to allow consumers to plan their consumption in anticipation. Many companies consider the RTP program to be the most acceptable, efficient and flexible.
  - **Time-of-use (ToU) :** This residential tariff is the most widely considered and used throughout the world. It is a predefined tariff offered on a seasonal, monthly, weekly or daily basis, where producers set prices according to peak/off-peak consumption times. As a result, the cost of electricity is high at peak times and lower at off-peak times, with the aim of encouraging consumers to shift their consumption patterns.
  - **Critical Peak Rate (CPR) :** This tariff promotes power saving as it charges market prices during peak hours. Customers are alerted to price increases when there's high demand, (peak consumption during a heatwave, for example), so they can cut down on use.

II.8. Literature review

Many research teams have designed load regulation and peak management methods using various techniques to reduce the load curve.

For example, in reference [73], Ravin S Phutane and Ashok Kumar Jhala studied the effects of management algorithms on the performance of microgrids, and proposed an intelligent EMS based on a PSO-optimised fuzzy logic algorithm applied to an electric bus charging station. This approach promotes dynamic management of solar panels, batteries and ultracapacitors while connected to the grid.

In the context of island microgrids, Al-Wesabi and colleagues [74] designed an approach based on a combined PSO-GA algorithm combined with an active disturbance rejection controller (ADRC) demonstrating robust performance in terms of voltage regulation and power quality on an autonomous DC microgrid. On the other hand, reference [75] focused on the sizing optimisation of a microgrid using a new hybrid JAYA-Grey Wolf (JGWO) configuration with satisfactory results on the reduction of the total annual cost.

In reference [76], the authors presented an energy management strategy in a DC microgrid, combining an adaptive droop control and a fuzzy state-of-charge (SOC) algorithm. This approach dynamically balances the SOC of the batteries, ensures intelligent power distribution, stabilises the bus voltage and reduces battery degradation. In a related development, Ouassima El Qouartiet and colleagues [77] have proposed an AC/DC architecture with an EMS capable of controlling active/reactive power and synchronisation interconnected microgrid segments.

Other studies focus on the integration of EVs into the grid. The authors of reference [78] present a solution based on a hierarchical EMS using predictive MPC control to enable dynamic management of photovoltaic arrays, battery and hydrogen storage, and a bidirectional charging station connected to several electric vehicles. Safitri and his colleagues [79] have proposed a model for an electric vehicle charging station connected to a photovoltaic system with intelligent bidirectional flow control using V2G. Using a fuzzy logic control algorithm, the EMS dynamically regulates the current, voltage and state of charge (SOC) of the batteries in order to optimise solar energy consumption, avoid overloads and smooth the grid demand curve.

S. Saurabh [80] takes an in-depth look at the techno-economic assessment, analysing the environmental and financial implications of integrating PV and BESS into a grid-connected system.

Given the strategies, architectures and algorithms explored in the literature, the present study focuses on the management and optimisation of a residential microgrid connected to the grid and structured around centralised control. This microgrid includes several renewable energy sources: photovoltaic (PV) arrays, wind turbines (WT), a battery energy storage system (BESS) and an EV charging station equipped with V2G technology

The management strategy is based on a rule-based EMS algorithm, chosen for its ease of implementation and ability to make real-time decisions in a dynamic residential environment. A hybrid approach combining Demand Side Management (DSM) and Demand Response (DR)

methods has also been adopted. Additionally, one demand response strategy has been implemented that combines an incentive program with time-of-use (ToU) pricing. This combination enables users' consumption profiles to be optimised, thereby encouraging self-consumption and grid stability.

The system's primary objective is to showcase the technical and economic feasibility of integrating a two-way energy flow, residential flexibility, and dynamic pricing mechanisms within an energy management system designed to facilitate the energy transition.

II.9. Conclusion

Microgrids are a promising solution to the energy and environmental challenges. This chapter has presented the concept of the microgrid, its components, its different classifications, the objectives and challenges, and the different approaches to control. We have also reviewed energy management systems, going through the different management algorithms, the resulting strategies, demand response, as well as the different management techniques proposed in the literature, focusing on the management technique proposed in this dissertation.

The following chapter explores a comprehensive analysis of the microgrid's components and infrastructure. Additionally, we will introduce the design of the energy management system implemented in this microgrid

Chapter III

Design of an energy management system for a hybrid microgrid

III.1. Introduction

In this chapter, we present an overview of the structure of the microgrid we are studying. We will review its general architecture and its main components.

The principal objective is to propose solutions for the optimal management of energy resources at the microgrid scale. In this context, a hybrid energy management algorithm has been implemented, combining a rule-based management system and a demand response strategy based on incentive and pricing.

III.2. General description of the microgrid

The system developed in this study is illustrated in figure III.1, it consists of a residential microgrid connected to the main grid. The dimensioning of our microgrid is partly inspired by the approach proposed by Bekele et al [81] in their study on the planning of microgrid clusters. The power values adopted, in particular the residential and dynamic loads, are close to those used in our study, which reinforces the consistency and realism of our dimensioning. However, some adjustments have been made to incorporate the specific features of our microgrid. For example, the estimated production of distributed energy sources has been based on the power requirements of residential, commercial and institutional loads, as well as EV charging. This approach ensures a balance between local production and energy consumption [81].

Our microgrid connected to the main grid upstream, through a $66\ kV/6.6\ kV$ step-down transformer that delivers power to the whole system at a medium voltage distribution level.

The microgrid includes two photovoltaic fields and a wind farm, as well as a battery storage system. other loads include commercial activities, hospitals, etc., supplied by a voltage of $6.6\ kV$, as well as residential areas supplied via a split-phase pole-mounted transformer, which provides a $240\ V$ phase-to-phase voltage to the houses. In addition, the microgrid includes an electric vehicle station with a bidirectional V2G recharging through a $6.6\ kV/600\ V$ step-down transformer. All managed by an EMS real-time supervision and management system.

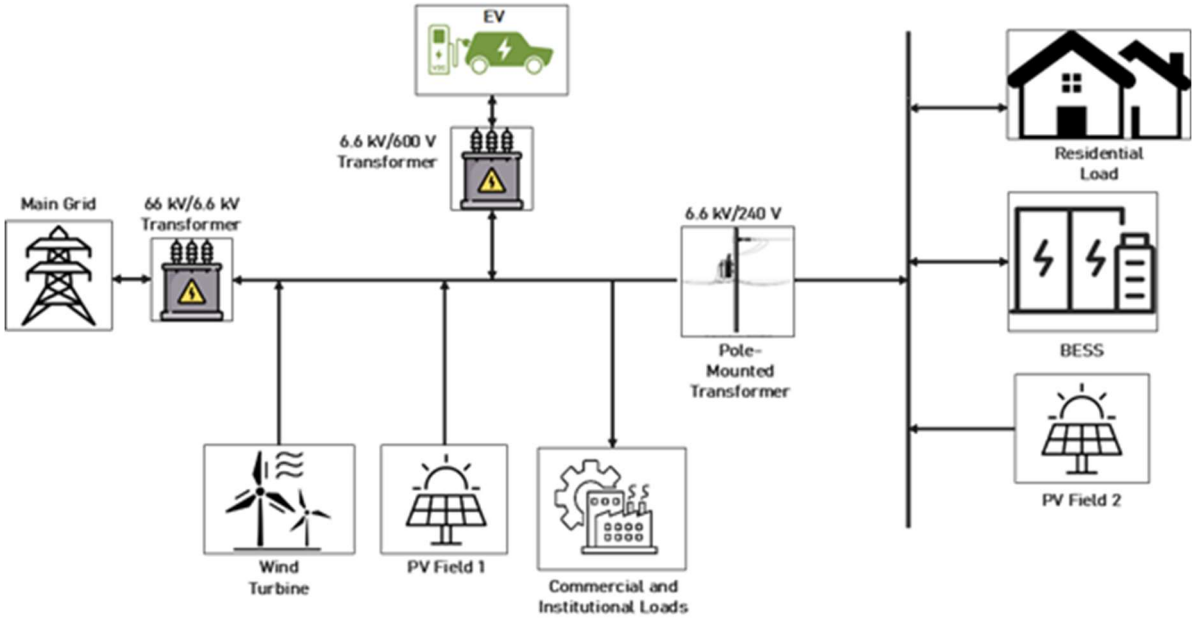


Figure 0.1: Architectural diagram of the microgrid.

III.3. Description of phasor mode

Continuous mode, discrete mode and phasor mode are the three modes used to simulate electric systems on MATLAB/Simulink. Figure III.2 illustrates the differences between the three modes of voltage representation in the time domain.

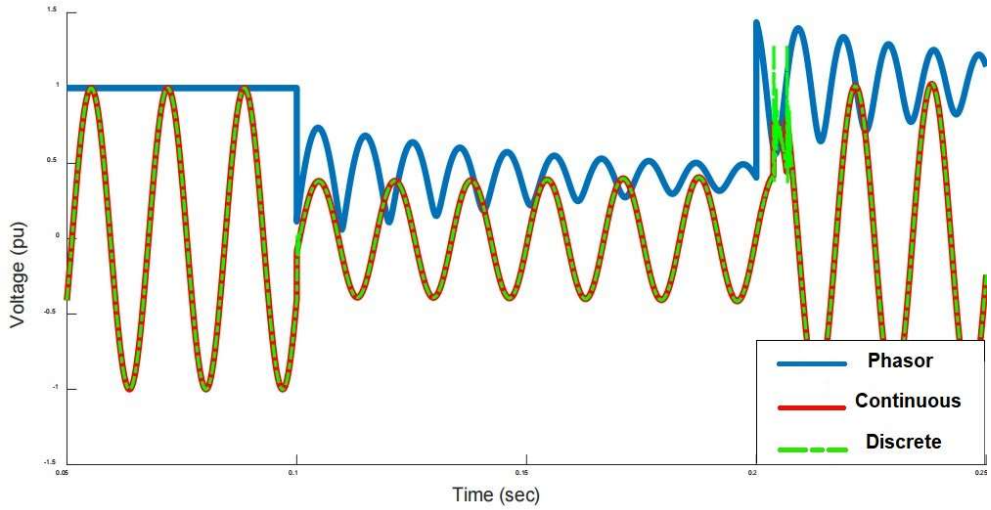


Figure 0.2: Representation of the voltage in the three simulation modes [82].

Continuous mode provides accurate calculations but it can be very slow for real-time simulations over long periods.

Discrete mode has constant sampling steps with fewer calculations than continuous mode, but results are step size dependent. A large step size may result in calculation errors and loss of accuracy [82].

Compared with the two proposed methods, the PHASOR MODE is the one chosen for this study:

- Speed of calculation, particularly for real-time simulations over long periods (several hours or days).
- Compatibility with power flow studies.
- Better management of interactions between distributed sources and storage systems thanks to the unified representation of electrical quantities.

In this simulation, the phasor mode is used. This mode is well suited to analysing power management and steady-state behaviour, particularly for studying the impact of demand response (DR) strategies and management algorithms on the overall behaviour of the grid. The aim is not to analyse fast transient phenomena, such as the details of a fault or fluctuations. This mode simplifies the calculations while maintaining a realistic representation of the voltage, current and power P/Q.

III.4. Description of microgrid components

III.4.1. The photovoltaic system (PV)

The photovoltaic (PV) system is one of the main sources of energy for our microgrid. Its use is a priority in the energy management strategy in order to make the least possible use of the main grid. The microgrid has two separate photovoltaic fields for optimum energy distribution

- A 100 kW field installed exterior and directly connected to the main grid, which ensures continuous solar production with a maximum solar radiation of 1000 W/m^2.
- A second 38 kW array with similar solar radiation to the exterior one. This is installed inside the residential perimeter. This field can be considered as the consumers' own production, conforms to local self-consumption.

The analytical model used is the most common model of a photovoltaic solar cell, treating the PV as an irradiance dependent current source connected in parallel with a diode and in series with a series resistor R_S.

The equivalent circuit diagram of the PV cell for this model is shown in Figure III.3 :

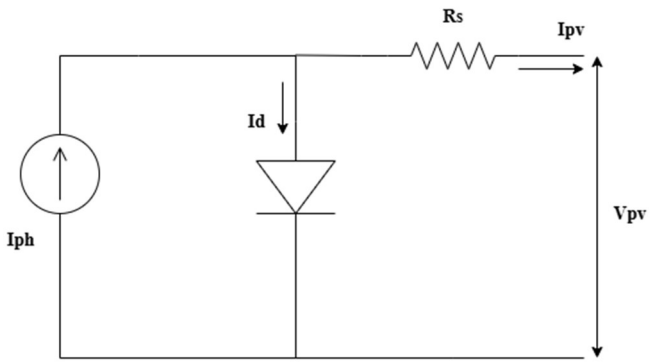


Figure 0.3: Diagram of the equivalent model of a PV cell [83].

The four parameters appearing in the equation for the $I_{PV}(V)$ characteristic are the photon current I_{ph}, the series resistance R_S, and two diode characteristics I_S, and n.

The electric current produced by the cell is then given by the following expression [83] :

$$I_{PV} = I_{ph} - I_S(exp^{\left(\frac{V_{pv}+I_{pv}\cdot R_S}{V_t \cdot n}\right)} - 1) \qquad \text{(III.1)}$$

In the MATLAB/Simulink simulation in phasor mode, the photovoltaic source is modelled in a simplified way as an irradiance-dependent active power generator. The two arrays are considered as a single equivalent PV system, which gives us the equation (III.2) for the total injected power expressed as follows [83] :

$$P_{PV} = G * A * \eta \qquad \text{(III.2)}$$

Where:

P_{PV} : Output power (W).

G : Solar irradiance (W/m^2).

A : Surface area of photovoltaic panels (m^2).

η : PV system efficiency (without unit).

The equation estimates the power available in real time according to irradiation variations, the surface area of the arrays and efficiency by integrating the profiles of both a sunny day and a cloudy day based on data from solar irradiation in Texas, USA [88] as shown in the figure III.4. The PV system injects this power into the grid and supplies the loads.

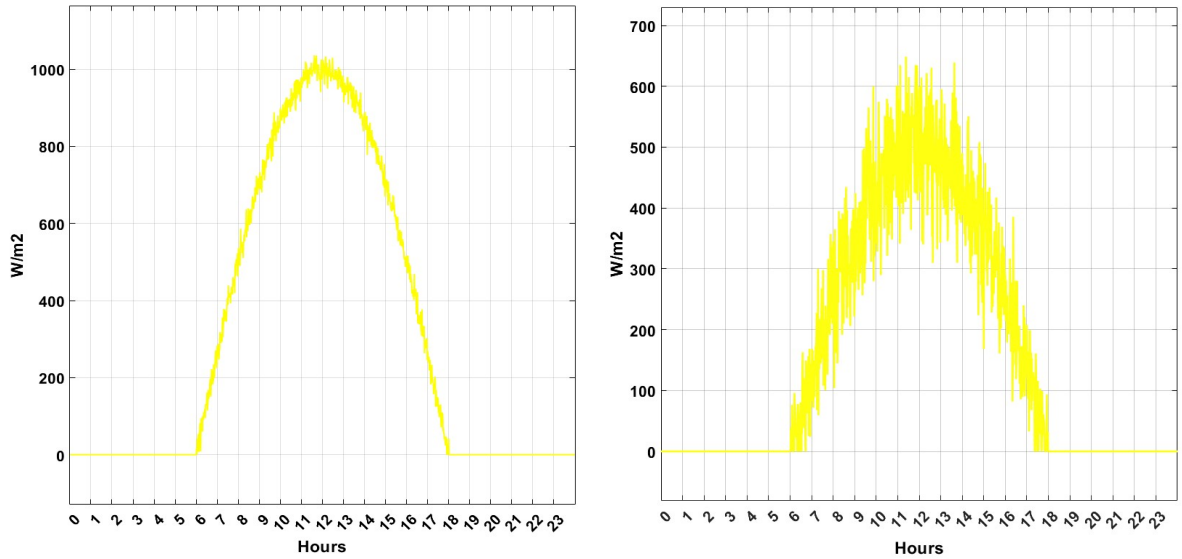


Figure 0.4: Solar irradiation during (a) sunny day. (b) cloudy day.

III.4.2. The wind turbine system (WT)

Wind energy is the main renewable source in the microgrid. It also has priority over the main grid for supplying the load. This method aims to emphasise the contribution of energy from distributed renewable sources over that from the main grid.

The wind turbine system is composed of a $150\ kW$ windfarm, representing an example of five (5) wind turbines with a power of $30\ kW$ each, directly connected to the grid. The modelling, purposely simplified for this study, concentrates on the power exchange processes, without any complex aerodynamic design or internal electronic or mechanical controls.

The maximum power of the wind turbine is expressed as a function of the air density, the area swept by the turbine blades and the wind speed in equation (III.3) [84] :

$$P_{WT} = 0{,}5 * \rho * A * \omega^3 * C_p \quad \text{(III.3)}$$

Where:

P_{WT} : Power output (W).

ρ : Air density (kg/m^3).

ω : Wind speed (m/s).

A: Swept area of the wind turbine (m^2).

C_p : Power coefficient (without unit).

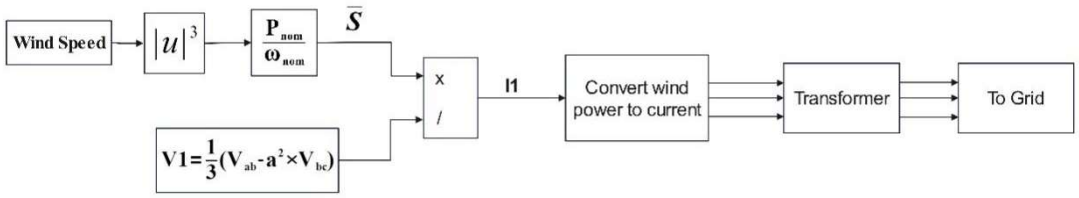


Figure 0.5: Simple modelling of wind turbine in power system in phasor mode.

The representation of wind turbine simplified modelling is illustrated in figure III.5. The cube of the wind speed determines how much power a wind turbine can capture. The active power as a function of the instantaneous wind speed is modelled using the following simple expression [85,86]:

$$P(\omega) = P_{\text{nom}} \cdot (\frac{\omega}{\omega_{\text{nom}}})^3 \qquad \text{(III.4)}$$

Where:

$P\ (\omega)$: The instantaneous active power output in (W).

P_{nom}: The nominal power of the wind turbine in (kW)

ω_{nom}: The nominal wind speed.

This real power is used to build the complicated power in order to integrate the wind turbine output with the grid in phasor-mode simulation:

$$\bar{S} = P(\omega) + jQ \qquad \text{(III.5)}$$

Where Q is the reactive power, either set to zero for unity power factor or assigned based on control strategy. The injected current is then computed as:

$$I_1 = \frac{\bar{S}^*}{3 \cdot V_1} \qquad \text{(III.6)}$$

Where:

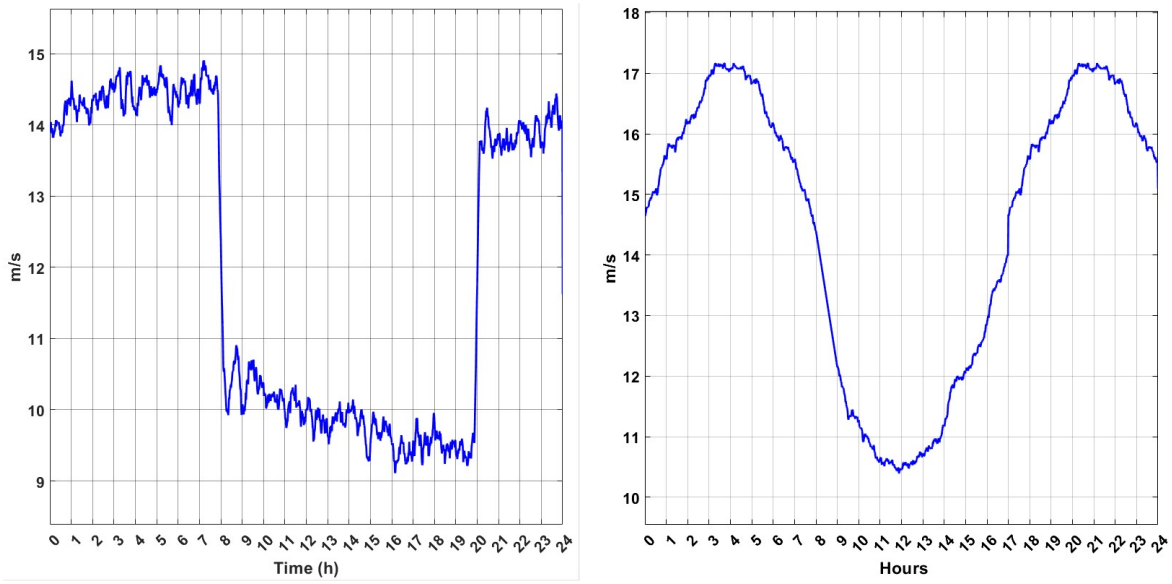
I_1: The current space vector.

V_1: The voltage space vector.

* : denotes complex conjugate.

This modelling strategy works well for power system studies that don't need full dynamic behaviour and instead employ steady-state or phasor-based simulations [87].

Curves (a) and (b) in Figure III.6 illustrate the intermittent behaviour of wind energy on a typical day and on a day with strong winds, and highlight the impact of wind on the variability of electricity.



(a) **(b)**

Figure 0.6: Wind profile during (a) normal day. (b) windy day.

This wind profile is representative of day-to-day variability. These profiles, based on real data from the state of Texas in the USA [88], have been pre-processed by smoothing (filtering) to limit abrupt fluctuations, thus retaining the real wind dynamics while ensuring numerical stability. Robustness of the system is evaluated by different scenarios of simulating realistic wind turbine power output deviations.

III.4.3. Battery energy storage system (BESS)

The Battery Energy Storage System (BESS) is an essential component of the microgrid. With the different variations in consumption and distributed generation, the batteries store the excess energy produced during periods of low demand, or at times when electricity prices are relatively low. This energy stored in the energy storage device can be released to the grid during periods of high load. [89].

In this study, the BESS is composed of lithium-ion technology batteries, and the storage system is sized for a capacity of $180\ kWh$ ($750\ Ah$). The state of charge is formulated in equation (III.7) as follows:

$$SOC_{BESS}(t) = \frac{C(t)}{C_{ref}(t)} \qquad \text{(III.7)}$$

Where;

$C(t)$ is the remaining capacity at instant t (kWh), and $C_{ref}(t)$ is the nominal capacity of the battery (kWh).

The state of charge of the batteries is dependent on the power balance of the microgrid. The relationship between the SOC and the powers is formulated in equation (III.8) :

$$SOC_{BESS}(t) = SOC_{BESS}(t-1) + \frac{P_{PV}(t) + P_{WT}(t) + P_{EV}(t) + P_{grid}(t) - P_L(t)}{C_{ref}(t)} \cdot \Delta t \qquad \text{(III.8)}$$

Where :

$SOC_{BESS}(t-1)$: State of charge at the previous time step.

$P_{PV}(t), P_{WT}(t)$: Power generated by (PV) and (WT).

$P_{EV}(t)$: Power exchanged with electric vehicles.

$P_{grid}(t)$: Power exchanged with the main grid.

$P_L(t)$: Power consumed by loads.

Δt : Unit time interval (1 hour).

The maximum (SOC_{MAX}) and minimum (SOC_{MIN}) battery limits are set to ensure the safety and longevity of the batteries, so that :

$$SOC_{MIN}(t) \leq SOC_{BESS}(t) \leq SOC_{MAX}(t)$$

Our model incorporates a system of protection against overcharging and deep discharging. We have set the minimum and maximum limits at 20% and 95% respectively in order to limit premature ageing of the batteries and guarantee their correct functioning.

The battery control system was modelled using MATLAB/Simulink in order to simulate the dynamic behaviour of the storage system (Figure III.7). This model, with integrated conditional logic, allows charge/discharge cycles to be managed without exceeding the defined minimum and maximum thresholds.

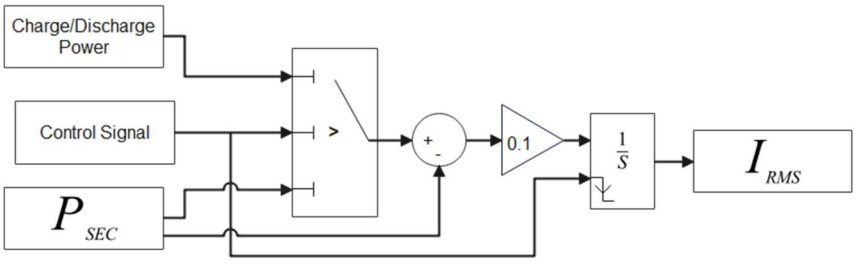


Figure 0.7: Simplified model of BESS control on MATLAB/Simulink.

III.4.4. Electric vehicles (EV)

In the age of energy transition, electric vehicles are the dominant players in the automobile sector. in addition to their low greenhouse gas emissions, these vehicles now represent a genuine revolution due to Vehicle-to-Grid (V2G) technology. This technique enables energy to be exchanged between the grid and the EVs via a bidirectional charging station [86].

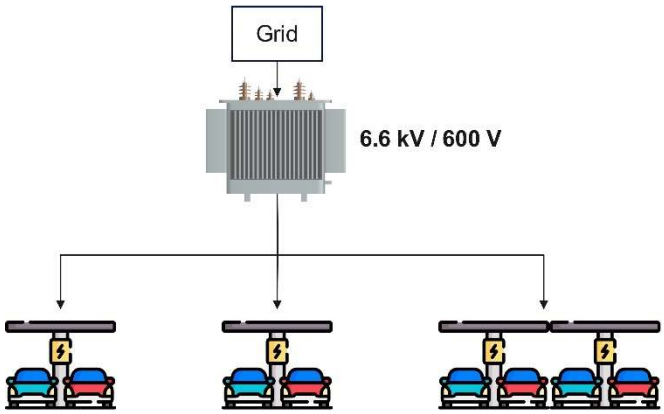


Figure 0.8: Model of electric vehicle stations in the microgrid.

Three (3) parks of electric vehicles using a bidirectional charging infrastructure were made available for the study of this microgrid (Figure III.8). These parks accommodate a total of 8 EVs, each with a power of 30 kW and a storage capacity of 44 kWh. The profiles of each park vary during the day according to users charging patterns.

Table 0.1: Distribution of electric vehicles by park.

Parks	**Number of EV**	**Power**
1	4	120 kW
2	2	60 kW
3	2	60 kW
Total	**8**	**240 kW**

Aggregators enable the distributed power to form a controllable unit, and consequently coordinate the energy flows between the EVs and the grid, as shown in figure III.9. The aggregators also ensure that a controlled injection rate of between 10% and 30% of the capacity of the vehicle batteries is maintained without exceeding critical limits [90,91], allowing the EVs to be used as a flexible resource while maintaining a certain degree of autonomy.

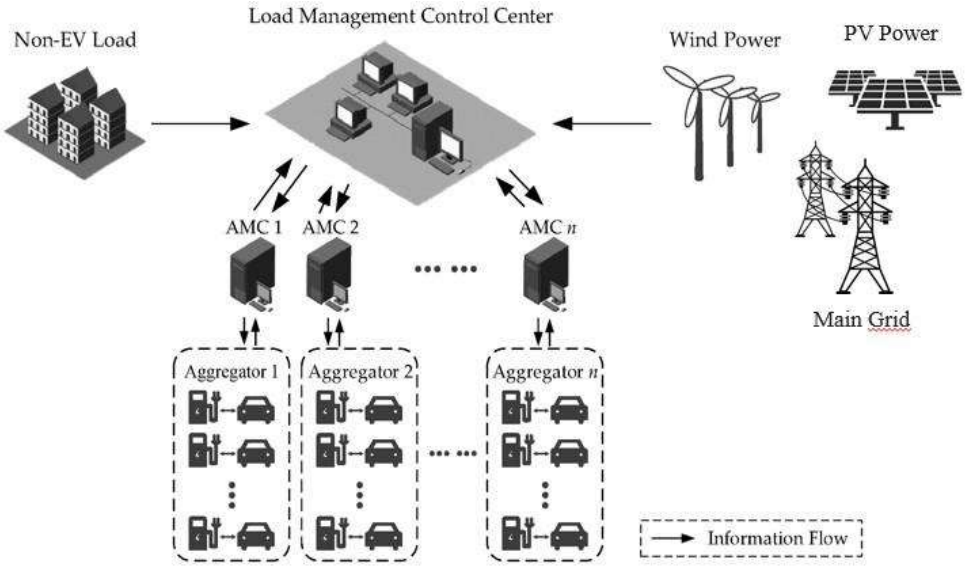


Figure 0.9: Load management architecture with EV aggregators.

- **Activation condition:** EV charging/injection depends on two parameters:
  - **Plug state:** Check whether the EV is connected to the bi-directional terminal
  - **State of charge (SOC_{EV}):** The minimum and maximum states of charge are 20% and 100% respectively:
    - If, $SOC_{EV} > SOC_{MIN}$: The user can authorise 5% to 30% of the EV's state of charge to be injected into the grid.
    - If, $SOC_{EV} < SOC_{MAX}$: The EV can be recharged.
- **Charging situations:** EVs can be charged in two situations:
  - During surplus of renewable energies production.
  - Off-peak times when electricity prices are low.
- **Injecting situations:** Electricity is injected into the grid when:
  - Renewable energies are unable to meet energy demand.
  - Peak hours, when consumers are at home (peak at 7pm, weekends, etc.).

In order to model EV behaviour realistically, three plug state profiles have been defined (Figure III.10), representing the average random distribution of the three parks; each profile is characterised by specific connection, charging and discharging habits.

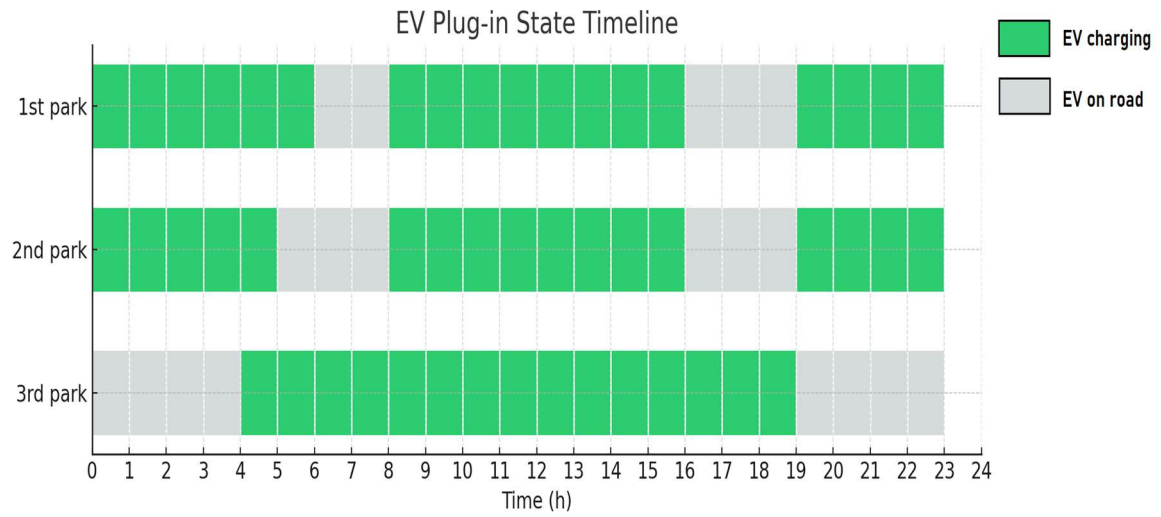


Figure 0.10: Plug state profiles of the three electric vehicle parks.

III.4.5. The main power grid

In this study, the microgrid is connected to the main grid, to maintain a bidirectional exchange of electrical energy. In fact, when the energy from the PV, WT and BESS is not sufficient to satisfy the needs of consumers, the lacking energy can be delivered by the main grid.

At the same time, if the production of DGs exceeds consumption, including the charging of BESS and VE, the excess energy can be injected into the grid, and sold at the same price as that of the purchased electricity.

This operation is based on an energy balance which ensures the balance between production and consumption, ensuring the stability of the system. This balance is formulated by the following equation (III.9) as follows:

$$P_{grid}(t) + P_{PV}(t) + P_{WT}(t) + P_{BESS}(t) + P_{EV}(t) = P_L(t) \qquad \text{(III.9)}$$

The exchange of power with the grid is presented as follows:

- $P_{grid}(t) > 0$: electricity imported from the grid.
- $P_{grid}(t) < 0$: injection into the grid (sale).

In this study, a dynamic pricing program is implemented, based on the principle of time-of-use (ToU) pricing. This type of pricing offers different prices at peak and off-peak times, as illustrated in Figure III.11:

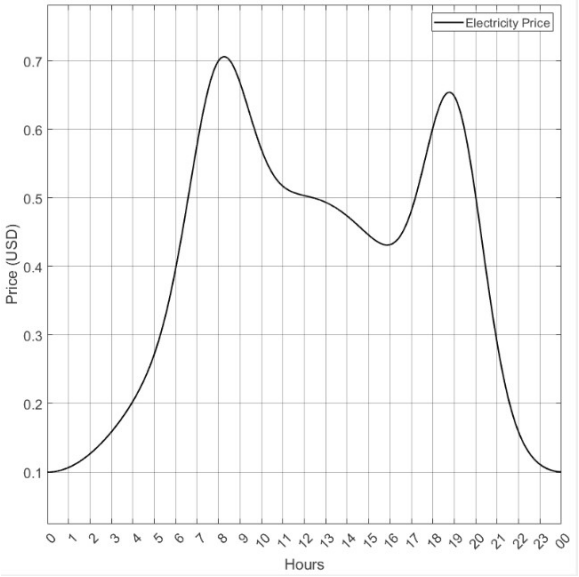


Figure 0.11: Variation in electricity prices throughout the day [93].

During peak hours, usually in the morning and evening, rates are relatively high due to high energy demand. This leads to the activation of peak power plants, which are called upon to meet growing energy demand. But these plants are often more expensive and polluting.

Conversely, during off-peak hours, usually at night and in the afternoon, rates decrease significantly. This reduction is intended to encourage users to consume during these periods.

On the other hand, consumers can benefit from their own distributed production, such as the residential photovoltaic system integrated into our microgrid. They can shift their consumption to off-peak hours and sell their surplus electricity to the grid at a higher price, and consume their own production at the same time as electricity from the grid during off-peak hours at advantageous rates [89].

In our case, the energy management system, thanks to the demand management strategy, allows consumption to be managed by prioritising distributed energy production. It also programs flexible loads during off-peak hours and feeds surplus production into the grid.

III.4.6. Description of the loads

It is difficult to estimate an exact model of load profiles due to the variability of numerous parameters, such as the number of connected devices, consumer profiles and habits, and external factors such as weather data.

Our microgrid consists of two types of loads:

III.4.6.1. Residential load

The residential load of the microgrid is represented by eight individual houses, each with an average consumption of 6.6 kW, giving a total average residential power of 53 kW. This load includes the daily use of household appliances, lighting, heating/air conditioning systems, etc., though our study does not model individual devices. The daily consumption profile, as illustrated in Figure III.12, shows two peaks, the first around 8 a.m. (start of morning activities) and the second around 7:30 p.m., which corresponds to consumers returning home.

In this configuration, demand response influences the residential loads since they are considered as flexible loads. They are responsive to incentives, meaning they can be encouraged to reduce or postpone consumption in order to reduce peaks. Through this demand response strategy, these loads help to optimize the energy balance.

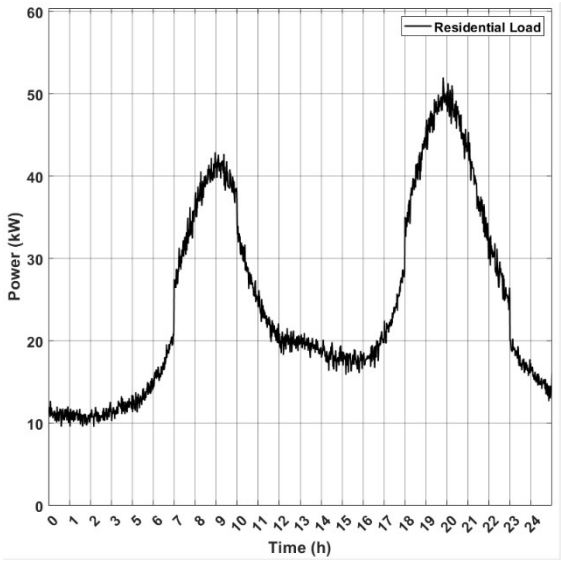


Figure 0.12: Residential load during weekday.

III.4.6.2. Industrial load

This type of load closely matches the one in Toba village documented by Bekele et al.'s study [81], it includes commercial activities, a school, hospital, and other similar buildings, with a total consumption of $150\ kW$.

These loads are modelled in MATLAB/Simulink as a three-phase dynamic load, powered by a voltage of $6.6\ kV$. The model is based on equations representing variations in active power (P) and reactive power (Q). The equations (III.10, III.11) make it possible to simulate the variable nature of the loads [91].

$$P(s) = P_0\left(\frac{V}{V_0}\right)^{n_p} \cdot \frac{1 + T_{p1}s}{1 + T_{p2}s} \quad \text{(III.10)}$$

$$Q(s) = Q_0\left(\frac{V}{V_0}\right)^{n_q} \cdot \frac{1 + T_{q1}s}{1 + T_{q2}s} \quad \text{(III.11)}$$

With :

P_0, Q_0 : Nominal active and reactive powers.

V : The positive-sequence voltage.

V_0: Nominal voltage.

n_p, n_q : Voltage exponents for active and reactive power.

T_{p1}, T_{p2}, T_{q1}, T_{q2} : Time constants representing the dynamic behaviour of the load.

The figure III.13 illustrates the industrial load profile within the microgrid during a typical weekday. The consumption profile is high and sustained during working hours, reaching consumption of over $110\ kW$ from 8 a.m. to 6 p.m., showing intense activity during these hours. Outside this period, the load decreases but remains relatively constant due to the presence of the hospital, which operates continuously.

This profile highlights the inflexible nature of these loads, including the hospital, school and commercial activities, whose energy needs are not easily modulated.

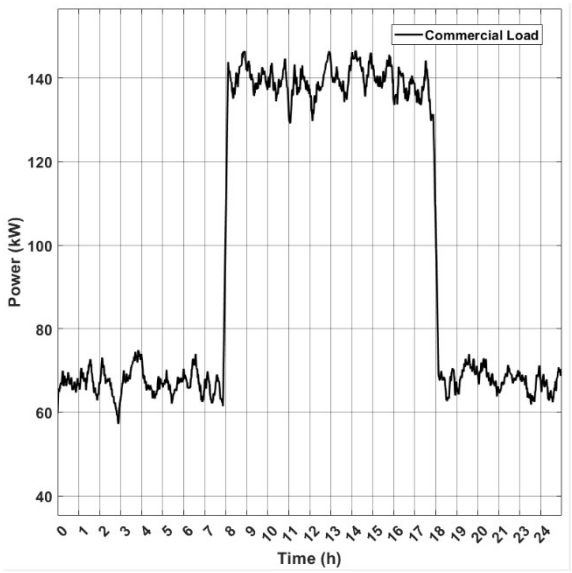


Figure 0.13: Industrial load profile during weekday.

III.5. Design of the energy management system

III.5.1. Rule-based energy management system (RB-EMS)

Our study's management system is based on a rule-based (RB) algorithm. It is highly valued for its simplicity of design and interpretation and its high responsiveness at all times.

This method is based on a set of predefined logical rules of the type (if-then-else). These rules are structured using production and consumption data in order to make the best decisions for real-time energy management.

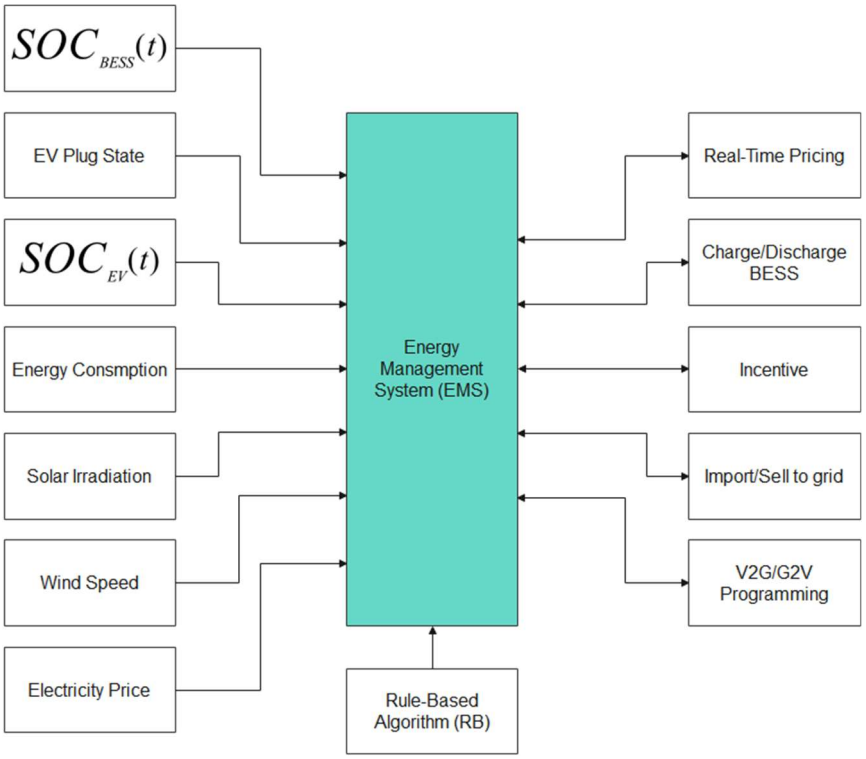


Figure 0.14: Energy Management System of the microgrid.

The energy management system receives the forecasting values in real time of load consumption, PV and WT production, the state of charge of the BESS storage system, the plug state and SOC of EVs, and electricity pricing (Figure III.14).

Based on this information, the EMS makes decisions to optimise energy flow, adjusting the price of electricity in real time, the power distributed to consumers, the power imported from or injected into the grid (sale), the power used to charge or discharge batteries, and the power used to charge and inject EVs.

The microgrid management system follows a number of predefined rules:

- Photovoltaic (PV) and wind turbine (WT) energy are the main sources of power for the microgrid.
- BESS batteries and EVs are charged from surplus production during the day.
- The BESS and EVs discharge when PV and WT production is insufficient to meet consumption, in order to eliminate consumption peaks.
- The main grid takes over and supplies the loads not covered by the BESS and EVs.

- When production exceeds consumption and the BESS and EVs are fully charged, this surplus is fed into the main grid.

III.5.2. Demand response program (DR)

To complement the rule-based energy management system, a demand response (DR) strategy was implemented for the economic management and optimisation of the microgrid:

The demand response strategy used in this study is a combination of two programs, namely incentive-based programs and pricing-based programs, more specifically the time-of-use (ToU) program.

The incentive-based program is based on the customer/producer relationship (Figure III.15). When energy consumption exceeds production, residential consumers are encouraged to reduce or postpone non-essential consumption. On the other hand, when a surplus is available, consumers are encouraged to consume more.

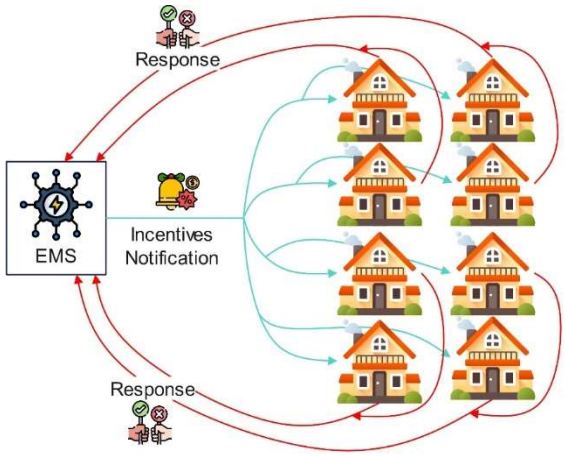


Figure 0.15: Operation of the microgrid incentive program.

Pricing-based programs rely on predefined electricity prices based on off-peak and peak consumption hours. Prices are high during peak loads and low during off-peak hours. This approach aims to encourage users to reduce consumption during peak hours or shift it to off-peak hours when rates are cheaper (Figure III.16).

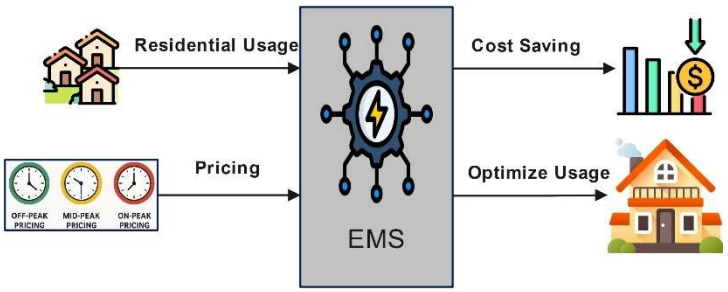


Figure 0.16: Operation of the microgrid ToU program.

The combination of these two demand response strategies is the ideal compromise for energy and economic management, where both parties (producers and consumers) win.

III.5.3. Management system chart

The balance of power between production and consumption in the microgrid is presented as follows:

$$P_L(t) = P_{PV}(t) + P_{WT}(t) \qquad \text{(III.12)}$$

III.5.3.1. Energy management in case of surplus

When $P_L(t) < P_{PV}(t) + P_{WT}(t)$, The surplus production is given by the following equation:

$$P_{SUR}(t) = P_{PV}(t) + P_{WT}(t) - P_L(t) \qquad \text{(III.13)}$$

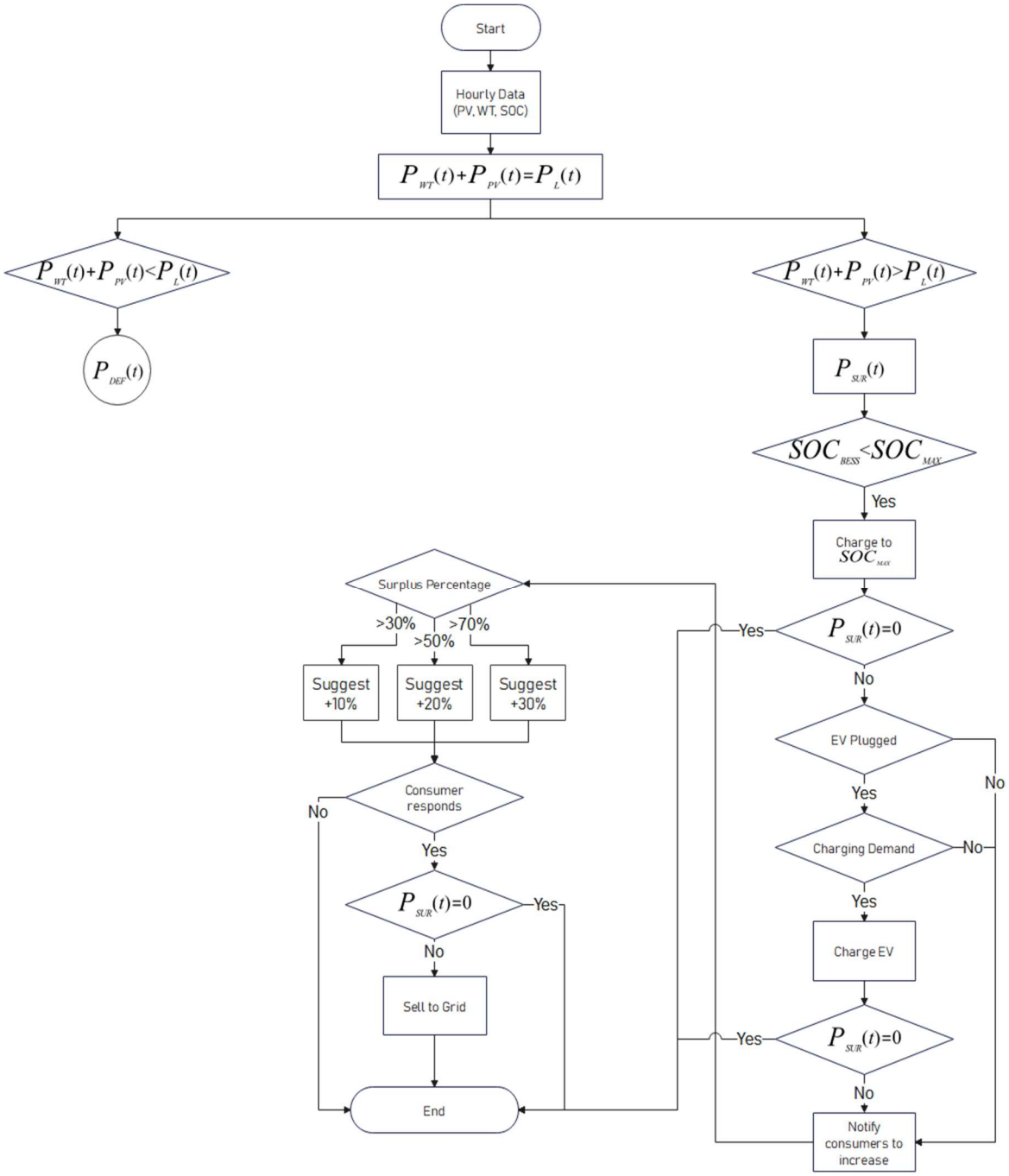


Figure 0.17: Flowchart of the RB-EMS in the case of a surplus.

The EMS follows a hierarchical surplus management strategy consisting of the following steps as shown on figure III.17:

- **Battery storage:** BESS batteries are used first.

- The EMS checks whether the SOC of the batteries is below SOC_{MAX} in order to charge them.

- The surplus charges the batteries until $SOC_{MAX} = 95\%$.

➢ **EV Recharging:** The remaining surplus energy is allocated to electric vehicles.

- The EMS checks whether EVs are connected to the charging stations.
- If so, and these EVs are requesting a charge, they are recharged with the available surplus.

➢ **Consumer incentives:** When surplus energy remains, the EMS suggests it to the consumers:

– By $+10\%$, if the surplus is $> 30\%$
– By $+20\%$, if the surplus is $> 50\%$
– By $+30\%$, if the surplus is $> 70\%$

- If the consumer responds positively, the EMS checks whether the surplus is completely absorbed.

➢ **sale of surplus to the grid:** As a last resort, the EMS injects the remaining surplus into the main grid.

III.5.3.2. Energy management in case of deficit:

When $P_L(t) > P_{PV}(t) + P_{WT}(t)$, The deficit is expressed as follows in equation (III.14)

$$P_{DEF}(t) = P_L(t) - [P_{PV}(t) + P_{WT}(t)] \qquad \text{(III.14)}$$

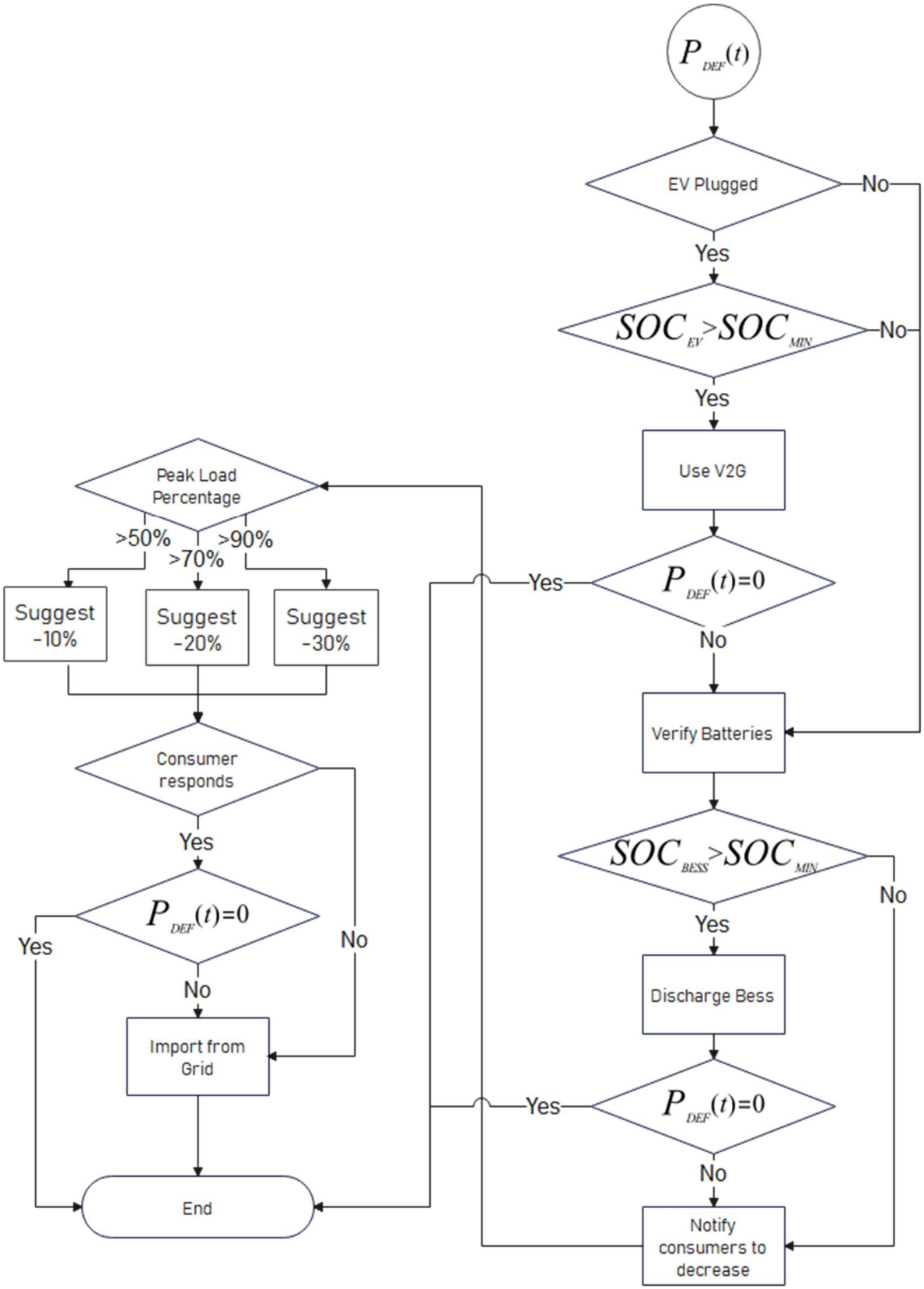


Figure 0.18: Organisational chart of the RB-EMS in case of a deficit.

When an energy deficit $P_{DEF}(t)$ occurs, the EMS follows a multi-step strategy to reduce this deficit in order of priority as illustrated in Figure III.18:

- **Use of V2G technology:** electric vehicles are called upon first

- The EMS checks whether the vehicles are connected to bidirectional charging stations.
- If so, it checks that the SOC_{EV} state of charge is above the minimum limit.
- If the conditions are met and the user authorises injection, the EMS uses 5% to 30% of the EV power through V2G to cover the deficit.

- **Battery discharge:** if the deficit persists, the EMS turns to the BESS.

- The EMS checks whether the SOC of the BESS is higher than SOC_{MIN}.
- If this is the case, the batteries are discharged until $SOC_{MIN} = 20\%$ to help make up the deficit.

- **Load reduction through incentives:** Consumers are notified to reduce their consumption to reduce the remaining deficit.

- Depending on the rate at which the charging power is exceeded, the incentive levels are:

– -10% if the deficit is $> 50\%$
– -20% if the deficit is $> 70\%$
– -30% if the deficit is $> 90\%$

- if consumers respond favourably and the deficit is absorbed, the process is complete.

- **Use of the main grid:** this is the last resort of the strategy to cover the deficit

- if none of the previous steps have completely eliminated the deficit, the EMS imports the missing energy from the main grid to restore balance.

III.6. Conclusion

In this chapter, we have described the fundamental elements of our microgrid, also designed a rule-based energy management algorithm, as well as a demand response program to optimise the flow of energy from production to consumption. In the next chapter, we will study the behaviour of the implemented EMS in the MATLAB/Simulink environment through a series of scenarios illustrating different operational situations. The results obtained will be analysed to determine the technical and economic impact of the energy management system within the microgrid.

Chapter IV

Simulation and discussions

IV.1. Introduction

In this chapter, we present the MATLAB/Simulink simulation of the residential grid-connected microgrid. Different scenarios are analysed to evaluate the energy management system performances, considering the weather changes that influence distributed energy sources, the state of charge of storage systems, and different consumption profiles. Each scenario is studied to highlight the flexibility and adaptability of the energy management system developed in order to cope with the dynamic conditions of the microgrid.

IV.2. Simulation Parameters

The general architecture of the elements of the microgrid studied includes a main photovoltaic park, a wind farm, a battery energy storage system (BESS), and a bidirectional electric vehicle charging infrastructure (V2G station) as illustrated in figure IV.1. Loads are divided into commercial and residential. The grid provides support in the event of a local production deficit.

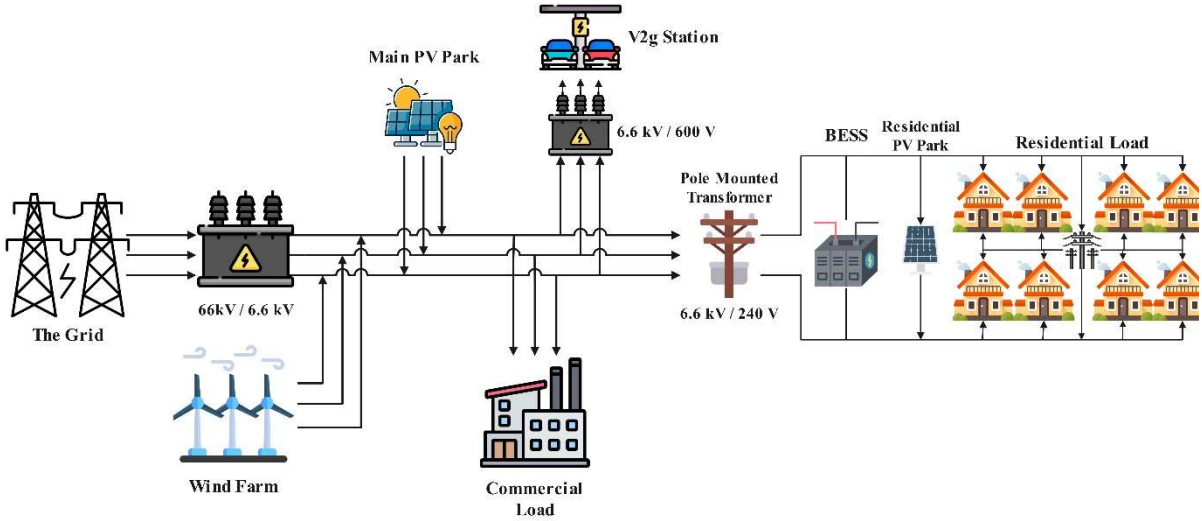


Figure 0.1: Structure of the developed microgrid.

The simulation was performed using MATLAB/Simulink in phasor mode in order to model the dynamic interactions between the components of the electrical system in real time, while maintaining the fidelity of the microgrid's energy behaviour. Table IV.1 shows the simulation parameters selected for this study.

The BESS operates through a smart inverter, which automatically charges when solar production reaches a determined limit.

Table 0.1: Parameters of the microgrid simulation.

Parameter	Value
Wind Farm	$150\ kW$
Main PV Park	$100\ kW$
Residential PV Park	$38\ kW$
V2G Station Power (for one vehicle)	$240\ kW\ (30\ kW)$
EV Capacity (for one vehicle)	$352kWh\ (44\ kWh)$
Commercial Load	$150\ kW$
Residential Load	$53\ kW$
BESS	$180\ kWh$
$\boldsymbol{SOC_{BESS}max}$	95%
$\boldsymbol{SOC_{BESS}min}$	20%
$\boldsymbol{SOC_{EV}max}$	100%
Simulation Time	$86400\ s\ (24\ h)$
Frequency	$60\ Hz$

The simulations were carried out over a 24 hours period for all the proposed scenarios. These scenarios were designed to reflect realistic operating condition, including weather conditions (sunshine, wind), consumption habits (working days, weekends, EV consumption, etc.), and the initial state of charge of the batteries (SOC_i).

The scenarios tested are as follows:

- **Scenario 1:** Weekday - Sunny - Normal wind - $SOC_i = 50\%$
  - **Mode 1:** Without EMS and V2G
  - **Mode 2:** With EMS and V2G
- **Scenario 2:** Weekend - Cloudy - Windy - $SOC_i = 20\%$
  - **Mode 1:** Without EMS and V2G
  - **Mode 2:** With EMS and V2G
- **Scenario 3:** Holiday - Sunny - Normal wind - $SOC_i = 80\%$
  - **Mode :** With EMS

IV.3. Results and discussions

IV.3.1. Scenario 1

Weekday - Sunny - Normal wind - $SOC_i = 50\%$

- **Mode 1:** Without EMS and V2G

This scenario is based on a typical weekday, characterized by sunny weather and normal wind speeds. The initial state of charge of the BESS is 50%, and the V2G technology is disabled. The specific feature of this mode is that the EMS is deactivated in order to study the random behaviour of the microgrid without supervision.

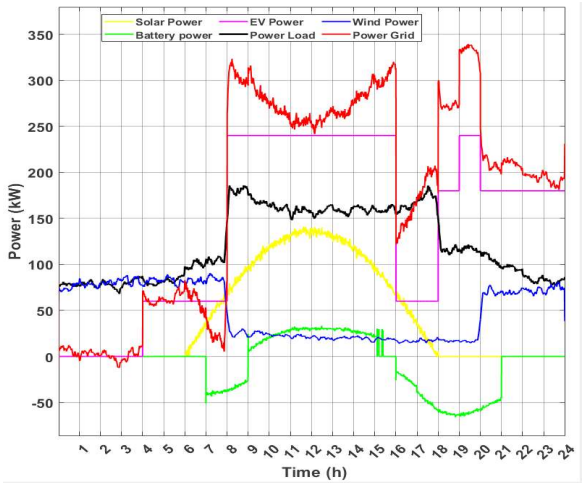


Figure 0.2: Power variation of the microgrid on a sunny weekday without EMS.

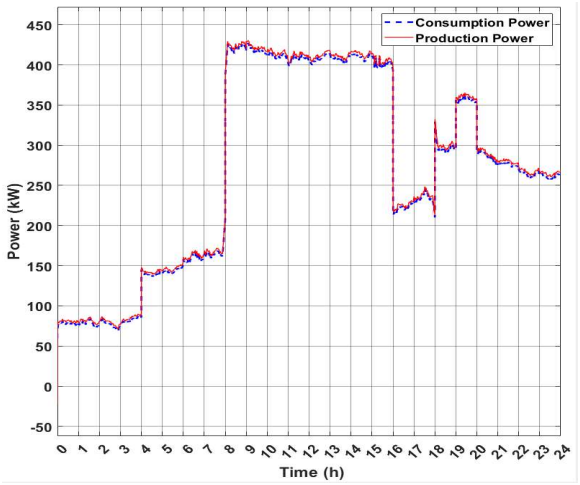


Figure 0.3: Production-consumption comparison in the microgrid without EMS.

A near-perfect alignment between production and consumption is illustrated in Figure IV.3. The slight difference between the two curves can be explained by losses associated with the grid lines and the phasor simulation parameter. This balance is what guarantees the stability of the grid frequency.

The supply of residential and industrial loads is approximately 90 kW (black curve). These loads are supplied by the main grid in conjunction with the WT system (between 80 and 90 kW). As for the BESS and EVs, it remains inactive during these hours, with its state of charge remaining at 50% for the batteries (Figure IV.4, IV5).

At 7 a.m., sunrise allows solar production to begin, reaching its peak power of approximately 140 kW at 1 p.m. Despite high solar and wind energy production, the system continues to draw heavily on the main grid, which reaches its peak of over 300 kW (Figure IV.2). This is because the three parks are being charged simultaneously, which alone represents a demand of 240 kW, in addition, the charging of the BESS that increases the SOC from 40% to 71% at 4 p.m. (Figure IV.5). This situation highlights the major inefficiency caused by the absence of EMS and, consequently, the lack of management and planning.

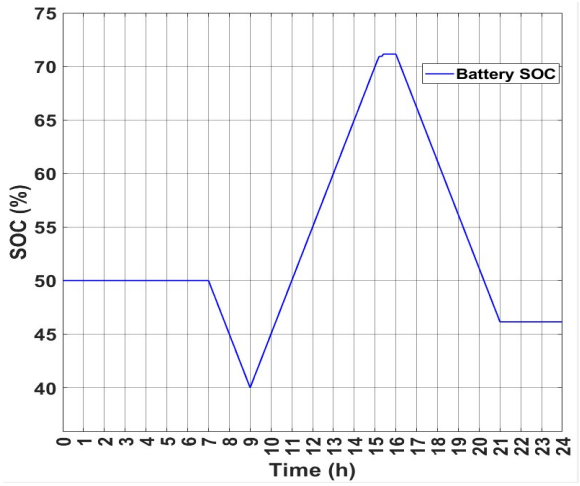


Figure 0.4: Battery state of charge during the day without EMS.

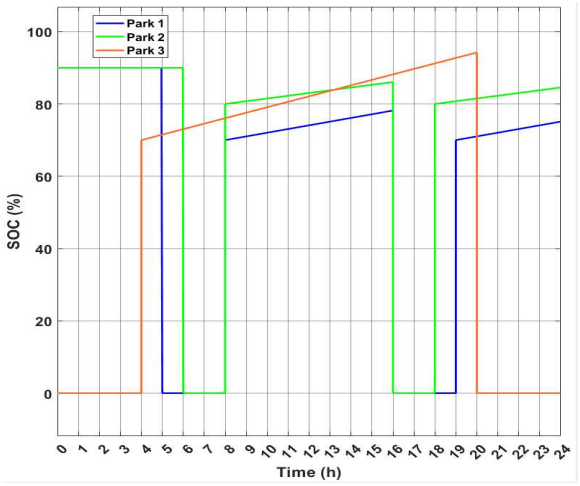


Figure 0.5: EV state of charge during the day without EMS.

From 6 p.m. forward, commercial load gradually decreases, and solar energy is no longer available as the sun has already set. Only wind turbines remain active but low. The BESS starts to discharge to a SOC of 46% in order to sustain the microgrid. A pronounced evening demand appears, requiring a significant reliance on the main grid for covering the remaining deficit. During the night, the plugged EVs are being charged with a power of approximately 180 kW (Figure IV.5).

The random behaviour clearly demonstrates the inefficiency of a microgrid without an energy management system. The lack of decision-making and coordination leads to paradoxical situations where the microgrid imports energy when local production is at its maximum. Furthermore, the irregular charging/discharging of EV and BESS batteries can lead to overcharging or deep discharging of the devices, which does not effectively serve autonomy and energy optimisation.

- **Mode 2:** With EMS and V2G

This mode represents the same scenario of a sunny weekday, with normal wind speed and an initial SOC of 50%, but this time with the energy management system activated in order to compare the two modes.

In order to observe the balance between production and consumption, Figure IV.7 highlights the performance of the microgrid in the presence of the energy management system, which ensures the frequency stability.

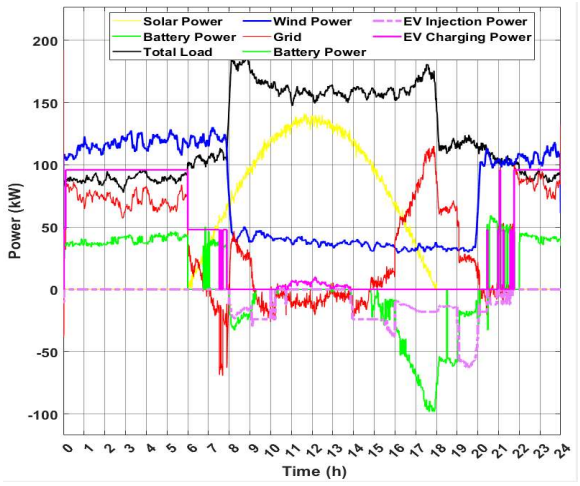


Figure 0.6: Power variation of the microgrid on a sunny weekday with EMS.

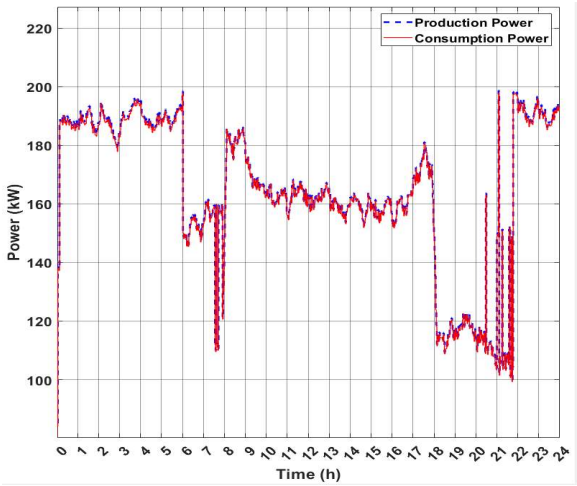


Figure 0.7: Power variation of the microgrid on a sunny weekday with EMS.

During the night (0-7 a.m. and 9 p.m.-midnight), energy demand is low and the price of electricity is at its lowest according to the blue curve of Figure IV.14 ($0.25\ \$US/kWh$). The wind system and the grid therefore work together to recharge the BESS and EVs and supply the active loads at that time (e.g. hospital). Figures IV.8 and IV.9 show respectively that at dawn (6-8 a.m.), the BESS reaches an SOC of around 95% and two of the three EV parks are charged to start the day.

At 8 a.m., commercial activities begin, leading to a significant increase in load. The EMS responds by calling on the energy stored in the connected EVs, which reduce the deficit by $20\ kW$ (Figure IV.6), and the BESS, which contributes 8% of its SOC until 9 a.m. (Figure IV.8). Consumers are then asked to reduce their non-essential consumption; each participating household reduces its load by between 0,5 and 2 kW. The residual deficit is covered by the grid to ensure stability.

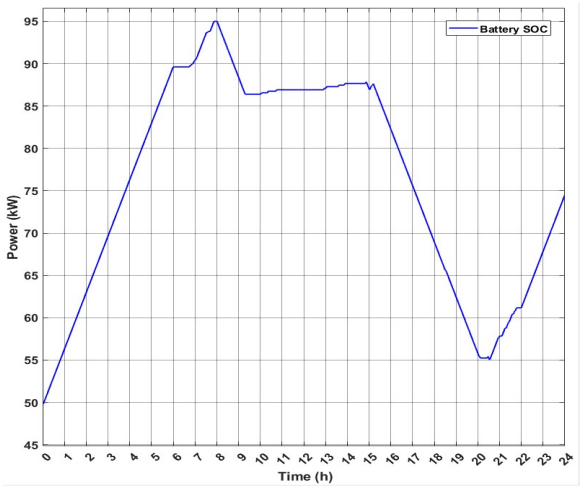


Figure 0.8 : Battery state of charge during the day with EMS.

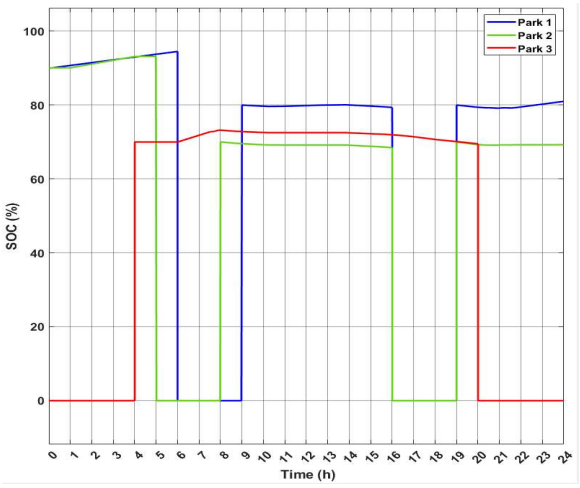


Figure 0.9: EV (V2G) state of charge during the day with EMS.

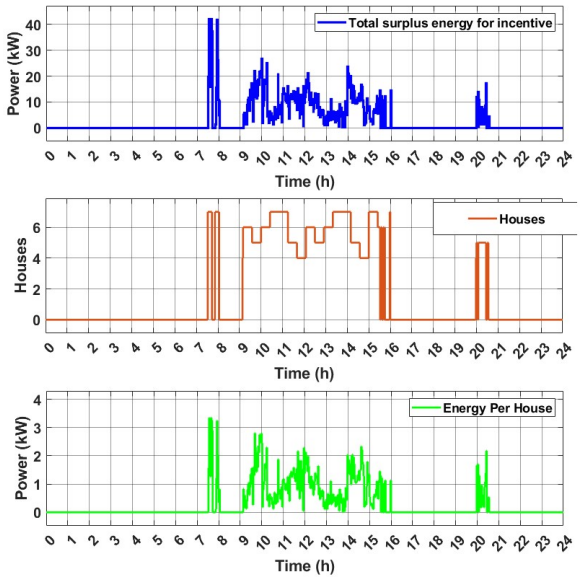


Figure 0.10: Energy Increase Incentives: Total, Household Count, and Average per Time Slot

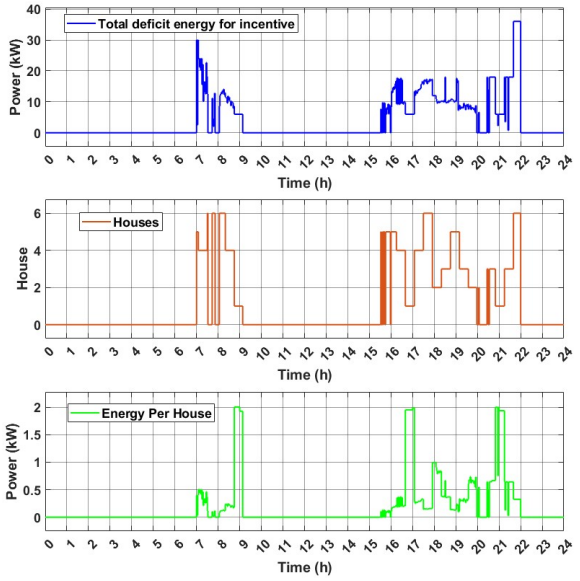


Figure 0.11: Energy Reduction Incentives: Total, Household Count, and Average per Time Slot

We also note surplus situations at 7 a.m. and between 9 a.m. and 4 p.m. (Figure IV.6). The EMS distributes it to consumers. Figure IV.10 shows that between 4 and 6 homes benefited from the incentive notification between 7 a.m. and 3 p.m., at a cheap energy pricing of 0.3 $\$US/kWh$ (Figure IV.14). The remaining surplus is sold to the grid (Figure IV.15). Figure IV.12 shows a peak in profits reaching 7 $\$US$ at 7 a.m. This sale perfectly coincides with the price peak, maximising revenue.

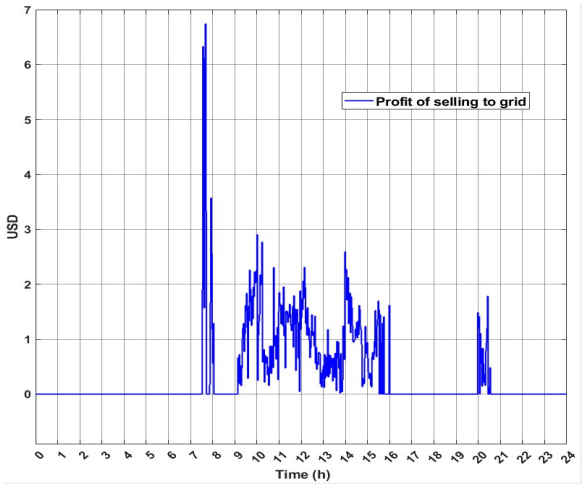


Figure 0.12: Profit from Grid Energy Sales – Weekday Scenario 1 with EMS, V2G, and Incentives.

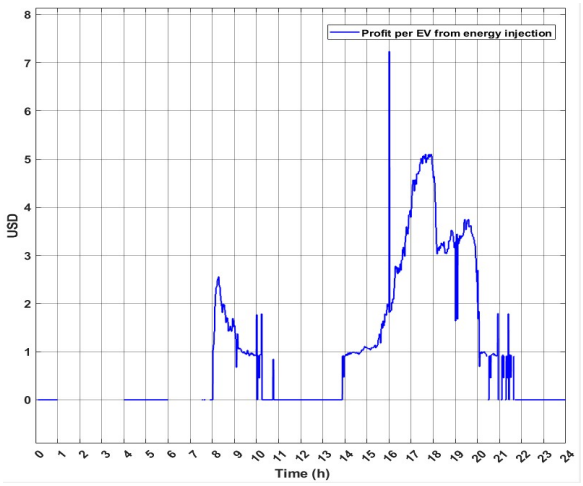


Figure 0.13: Revenue from V2G Export to the Grid – Weekday Scenario 1 with EMS, V2G, and Incentives.

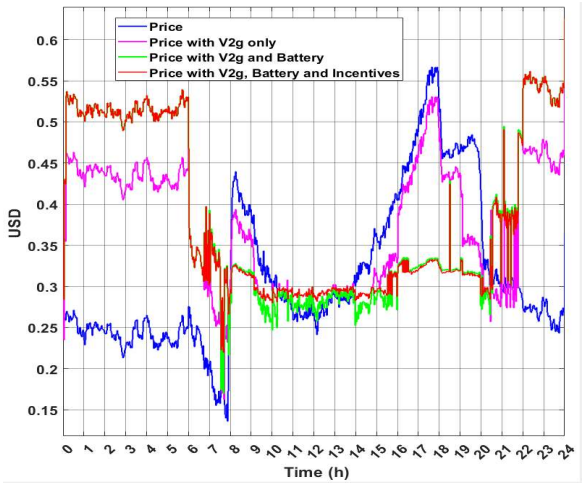


Figure 0.14: Real-Time Pricing Impact – Weekday Scenario 1: Comparison of EMS Strategies (RES Only, +V2G, +BESS, Full EMS.

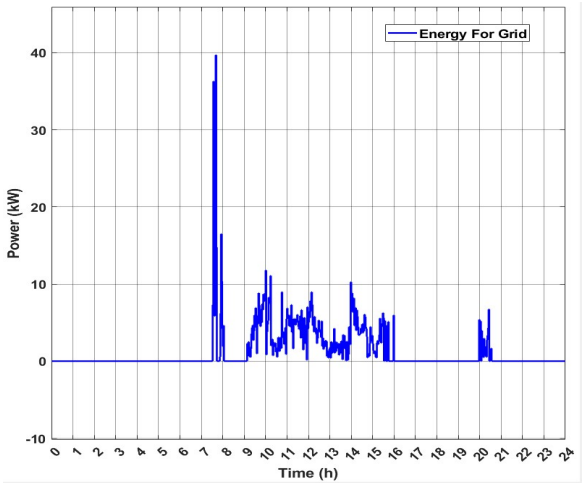


Figure 0.15: Remaining Energy Available for Grid Sale.

The period between 4 p.m. and 8 p.m. illustrates technical and economic optimisation. A peak in consumption occurs when users return home after work, and renewable production struggles to cover it. The EVs plugged in the parks switch to V2G mode to sell energy to the main grid (FigureIV.9), making profits of up to 7 $\$US$ (Figure IV.13), they delivered a power up to 50 kW as shown in figure IV.6. In addition, stored energy from BESS is used, providing nearly 32% of its SOC (Figure IV.8). And the consumers contribute by reducing their usage by a total of 30 kW (Figure IV.11). The rest of the deficit is covered by the grid.

This dynamic illustrates a virtuous cycle directed by the EMS. It uses night-time deficits to 'invest' in low-cost energy, then covers daytime deficits and transforms surpluses into revenue

by selling at the most opportune moments. This management benefits the microgrid by importing less energy from the grid and prioritize renewable energies.

The comparative analysis highlights the significant impact of the energy management system on the performance of the microgrid.

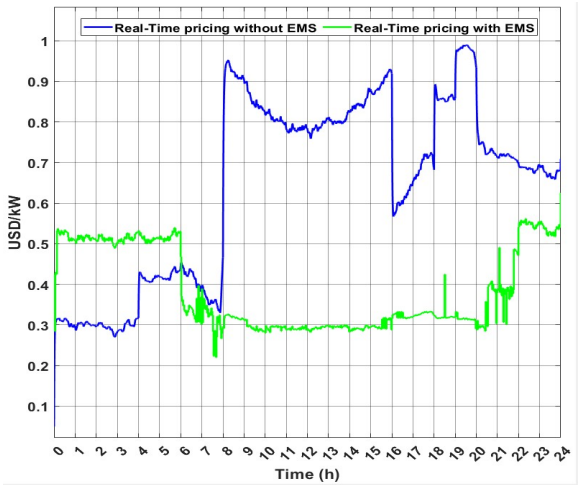


Figure 0.16: Real-time electricity pricing comparison with and without EMS in Scenario 1.

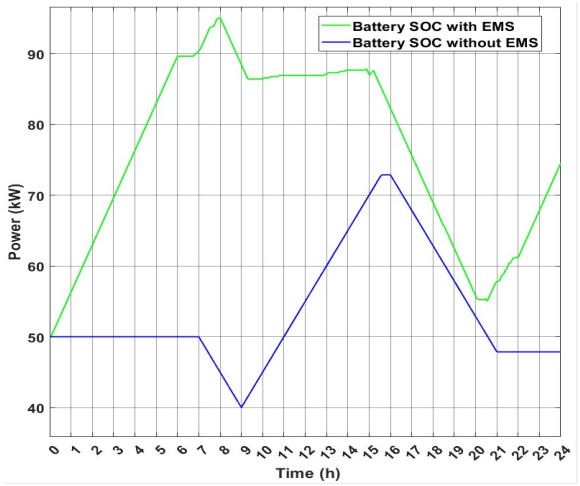


Figure 0.17: Comparison of Battery SOC Dynamics in the Microgrid with and Without EMS.

Without EMS, the operation of the microgrid is technically unpredictable and economically inefficient. The BESSs charging cycle depends on the solar power cycle, not taking in consideration the other conditions (Figure IV.17). This configuration increases the high utilization of the grid, forcing the microgrid prices to reach 0,9 $\$US/kWh$ around 9 a.m. and 4 p.m. (Figure IV.16).

In contrast, the integration of EMS (green curves) completely transforms performance. On the economic side, EMS acts as a financial shield. During periods when the market price without EMS skyrockets to 0.9 $\$US/kWh$, EMS keeps the energy tariff at a controlled level of around 0.30 $\$US/kWh$. This represents a direct reduction in energy costs of more than 65%, equivalent to a saving of 0.60 $ per kWh consumed during the most expensive hours of the day (Figure IV.16).

IV.3.2. Scenario 2

Weekend - Cloudy - Windy - $SOC_i = 20\%$

- **Mode 1:** Without EMS and V2G

This scenario shows the behaviour of the microgrid without an energy management system during a cloudy weekend day with strong winds and an initial SOC of 20% and without the V2G technology.

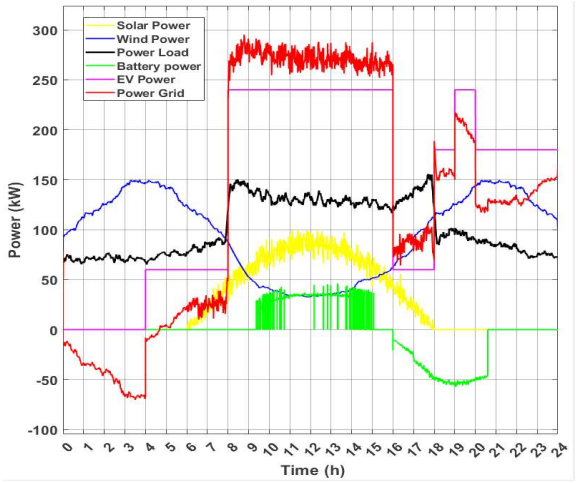


Figure 0.18: Power variation of the microgrid on a cloudy weekend without EMS.

Although the weather and initial conditions are different in this case, the unmanaged operation is the same as in the prior one. Despite strong wind speeds and low grid pricing at that time, figure IV.20 shows that the EVs of park 1 and 2 remain uncharged despite being plugged in, the BESS does not recharge despite its low and inefficient SOC of 20% (Figure IV.19). Consequently, the unused surplus energy is wasted and injected into the grid without taking advantage of it for storage or the residential loads.

When solar production reaches the limit configured in the smart inverter, the BESSs begin charging until they reach an SOC of 43% at 3 p.m.

The simultaneous EV charging of all of the three parks between 8 a.m. and 4 p.m., in addition to the high demand from the loads during these hours (approximately $150\ kW$) illustrated in figure IV.18, results to a significant power deficit, particularly because of the reduced PV generation. The full load is ultimately carried by the grid as its utmost capacity ($300\ kW$).

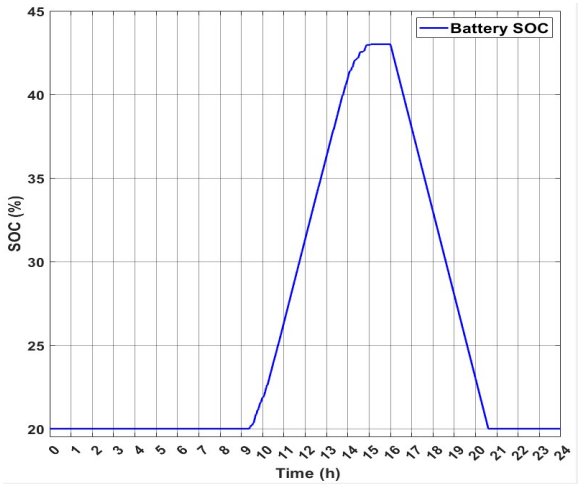


Figure 0.19: Battery state of charge during the day without EMS.

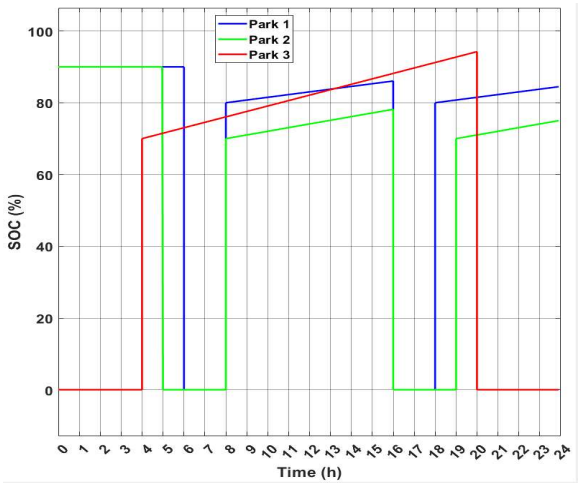


Figure 0.20: EV (V2G) state of charge during the day without EMS.

The period of time between 4 p.m. to 8:30 p.m. is marked by two consumption peaks, appearing one after another at 6 p.m. and 7 p.m. respectively. These elevated consumption peaks are covered by the high wind production during that time (Figure IV.18), the energy stored in the BESS (Figure IV.19), and the remaining power is provided by the grid.

This example supports the same conclusion as the Mode 1 without EMS of the 1st scenario, that in the absence of energy management, the system responds randomly, resulting in missed opportunities to minimize costs and maximize energy efficiency.

- **Mode 2:** With EMS and V2G

This scenario is characterized by the weather conditions of an autumn/winter day. A cloudy day, which means lower solar production, and strong winds, which means higher wind production. The weekend scenario indicates a slight decrease in commercial activity and a higher residential consumption due to increased in-home activities. This scenario aims to test the resilience and behaviour of the energy management system (EMS) in terms of energy optimization under non-optimal weather conditions and with a minimum BESS state of charge of 20%.

Wind energy and the main grid take over the power supply of the loads, the recharging of EVs and BESS due to its relatively low price during the night (Figure IV.27). Figure IV.22 shows that at the end of charging, at 6 a.m., the batteries reach an SOC of 56%. Figure IV.23 shows that the vehicles in parks 1 and 2 were recharged during the night.

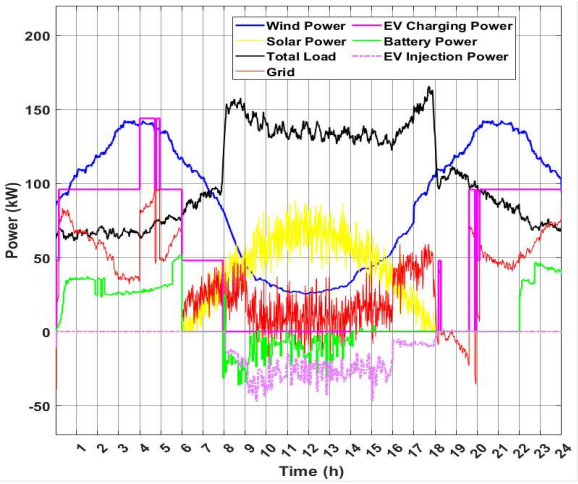


Figure 0.21: Power variation of the microgrid on a cloudy weekend with EMS.

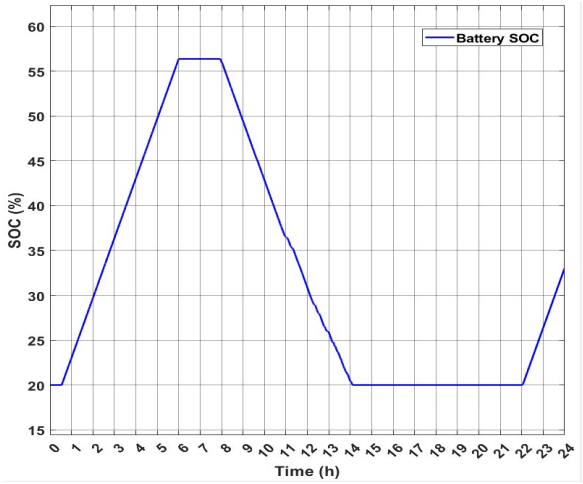


Figure 0.22: Battery state of charge during the day with EMS.

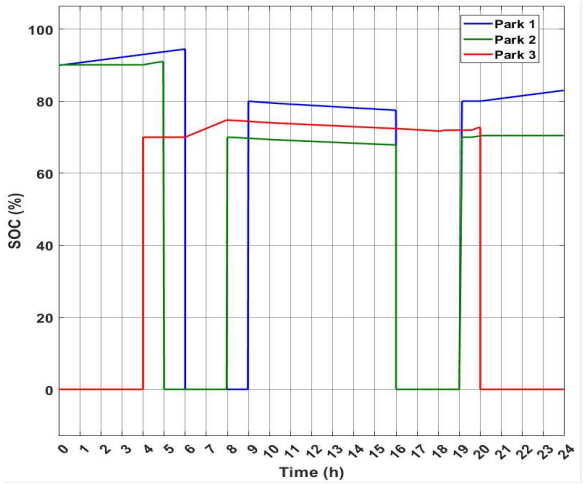


Figure 0.23: EV (V2G) state of charge during the day with EMS.

The period between 8 a.m. and 6 p.m. is the most critical. Weak winds combined with cloudy weather, which limits solar production, create a significant energy deficit. The EMS therefore takes the following measures to remedy this:

- During the weekend days, the average usage of EVs is lower than typical weekday. Figure IV.21 illustrates that the total power injected by the V2G technology is approximately 50 kW, each consumer earning $1 - 2$ $\$US/kWh$ per EV (figure IV.28).
- The storage system supports the grid by discharging from a SOC of 60% to 20% (Figure IV.22).
- The consumers being notified of the energy deficit, they respond by reducing a total power of 10 kW (Figure IV.25). with the incentive, the price of electricity sold to these consumers falls from 0.4 to 0.3 $\$US/kWh$ (Figure IV.27).

- The EMS uses energy from the grid to compensate for the remaining deficit of 40 kW (Figure IV.21).

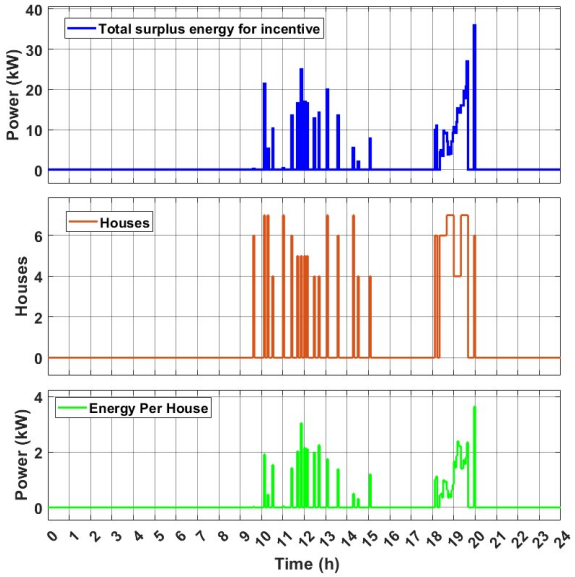


Figure 0.24: Energy Increase Incentives: Total, Household Count, and Average per Time Slot.

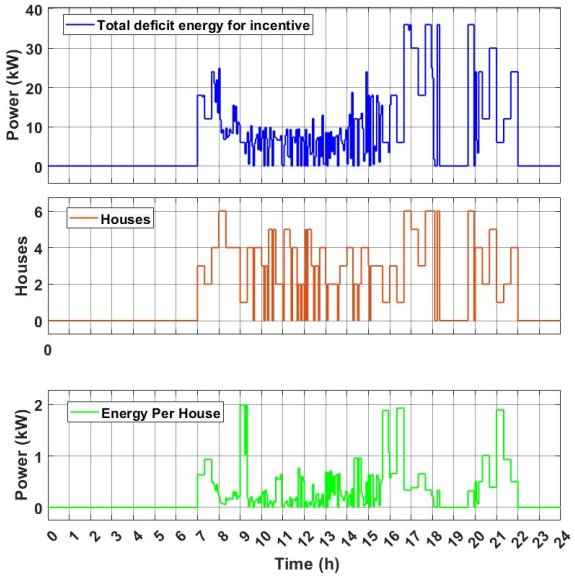


Figure 0.25: Energy Reduction Incentives: Total, Household Count, and Average per Time Slot.

A strong wind recovery coincides with the evening peak in demand between 6 and 8 p.m. WT production covered this demand, even leaving a surplus of energy. The EMS uses part of it for incentives, and approximately six homes responded to the call, each household benefiting between $2 - 4\ kW$ (Figure IV.24).

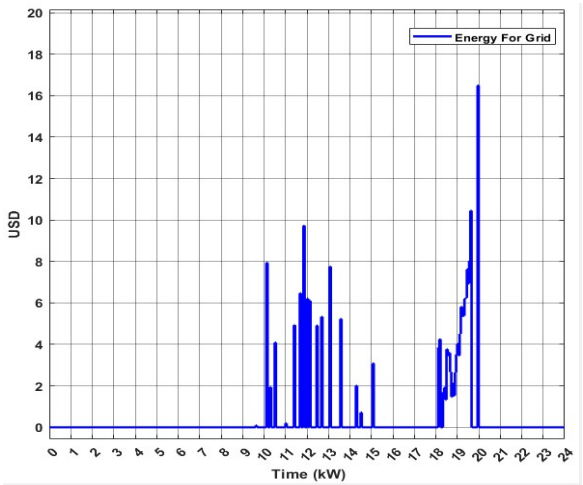


Figure 0.26: Remaining energy available for grid sale on a weekend scenario.

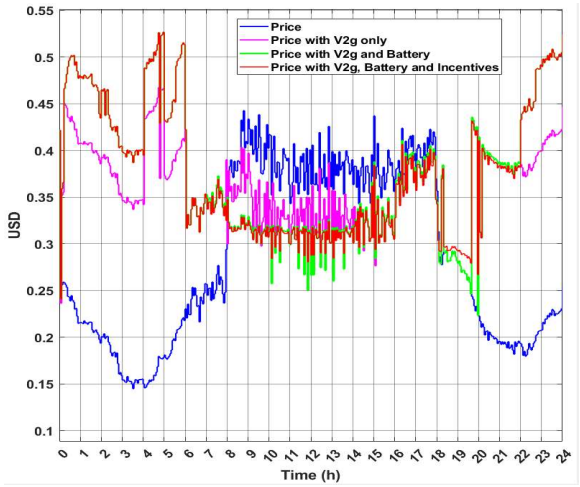


Figure 0.27: Real-time pricing impact on a weekend scenario.

However, optimisation occurs when the EMS sells the remaining surplus to the grid. Figure IV.29 shows a massive peak of surplus production exceeding 16 kW (Figure IV.26). By selling this energy to the grid at that moment, the microgrid generates considerable revenues of more

than 3.5 $\$US$ for each kWh sold (Figure IV.29), and also avoids to consumers purchasing from the grid at the most expensive price of the day (over 0.4 $\$US/kWh$) as shown in figure IV.27.

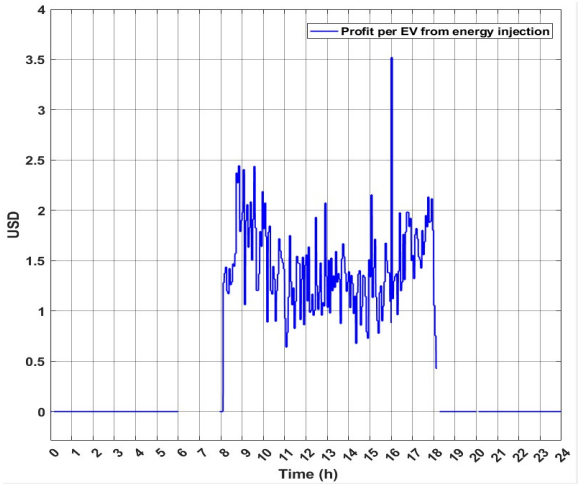


Figure 0.28: Revenue from V2G export to the grid on a weekend scenario.

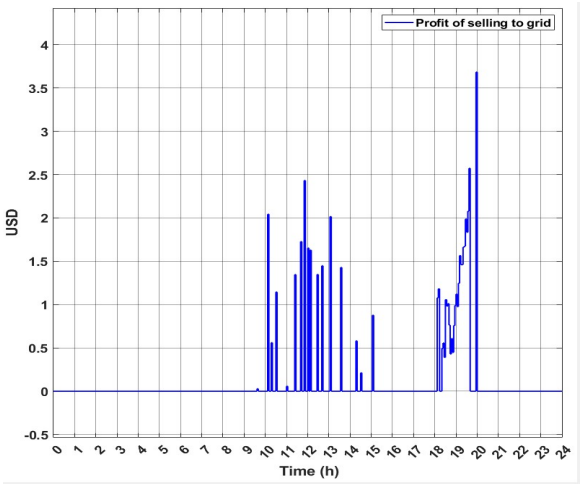


Figure 0.29: Profit from grid energy sales for weekend scenario.

At the end of the evening, after 9 p.m., demand decreases. The EMS returns to BESS and EV charging mode to prepare for the next day.

This scenario demonstrates the EMS's strong adaptability. The system maintained stability and executed an effective financial strategy. Starting from a vulnerable position, it managed the deficit and surplus periods through demand-side management (DSM) techniques. With the objective of maximizing energy efficiency. By dynamically coordinating the distributed resources, the EMS facilitates the integration of renewable energies into the microgrid and guarantees clean power usage even under challenging conditions.

To quantify the EMS impact on the microgrid's performance, we have set a comparative analysis of the two modes of this same scenario.

Without EMS, the microgrid passively suffers from the lack of control. The storage system is an underutilised component, with its SOC remaining inert at 20% during the first 9 hours of the day (Figure IV.30). This inaction is disastrous from both an energy and economic standpoint.

The high rise of electricity prices is visible in Figure IV.31, which peak at around 0.85 $\$US/kWh$ between 9 a.m. and 4 p.m., forcing the microgrid to passively endure high market tariffs for eight consecutive hours.

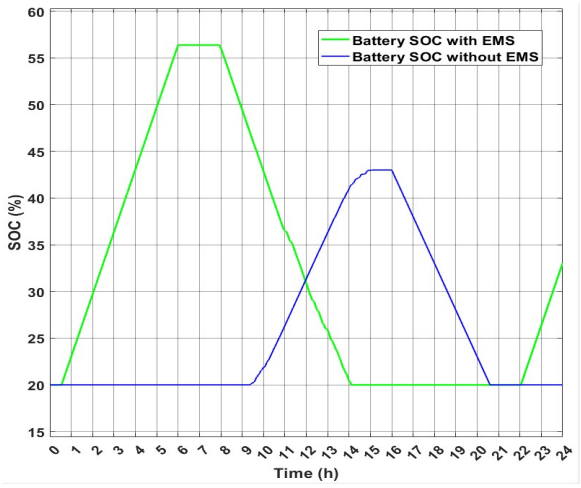


Figure 0.30: Comparison of Battery SOC with and without EMS.

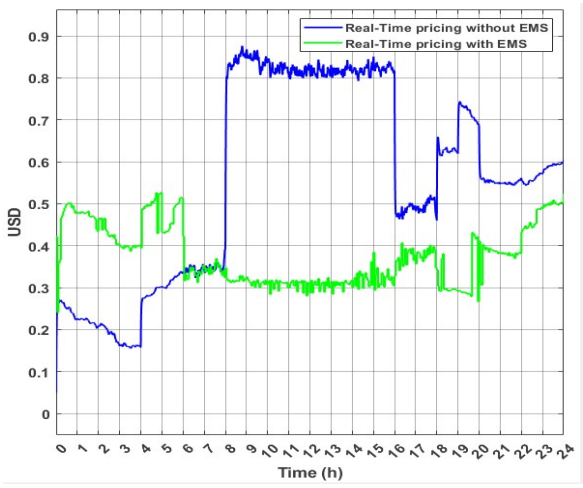


Figure 0.31: Comparison of Real-Time Price with and without EMS.

With the EMS, the logic is reversed for maximum efficiency. The EMS identifies the deficit period and uses the storage system strategically:

- During the first hours (from midnight to 6 a.m.), the EMS charges the batteries to 56% to optimize energy backup.
- Between 6 a.m. and 4 p.m., when in the mode without EMS, BESS is globally less efficient. The batteries in the EMS mode are discharged from 56% to 20% (Figure IV.30). This strategy maintains a stable energy balance and helped to rely less on the grid.

The EMS transforms the dead weight battery into an economic and energy lever. It ensures the transfer of energy so that there is no need to import from the main grid, keeping the price of electricity between $0.3 - 0.5$ $\$US/kWh$ (Figure IV.31).

IV.3.3. Scenario 3

Holiday - Sunny - Normal wind - $SOC_i = 80\%$ - With EMS

This scenario examines the behaviour of the EMS-controlled microgrid operating on a sunny holiday with normal wind speeds. The initial state of charge of the batteries is 80%. The holiday condition represents a load profile that is characterized with low commercial activities, a high residential load as consumers remain at home. The load also consists of critical loads that runs continuously $24/7$, such as the hospital.

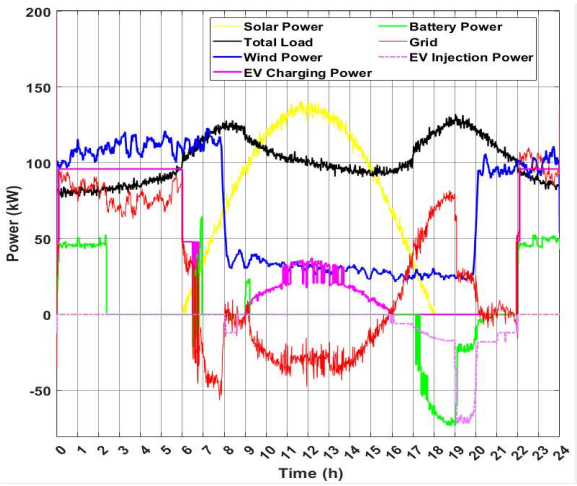


Figure 0.32: Power variation of the microgrid on a sunny holiday with EMS.

During the night, energy demand is low (around 80 kW). Wind power generation alongside the main grid are sufficient to cover this demand and even leaves a small surplus. This surplus is used to recharge the BESS in addition to the first EV park. At dawn, the battery reached the maximum limit of 95% (Figure IV.33), and electric vehicles continued to recharge until around 6 a.m. (Figure IV.34).

Around 8 a.m., we see the appearance of a consumption peak that was easily covered by wind, solar production (Figure IV.32), and a slight contribution from the BESS (Figure IV.33).

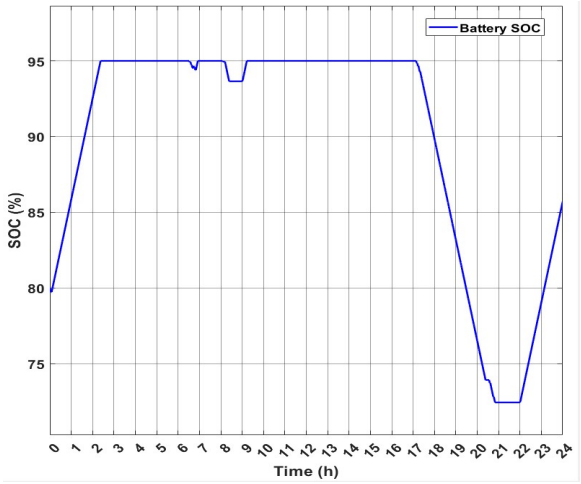


Figure 0.33: Battery state of charge during the holiday with EMS.

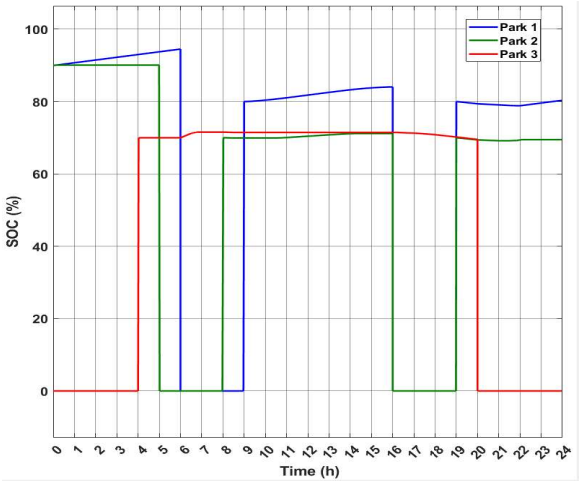


Figure 0.34: EV (V2G) state of charge during the holiday with EMS.

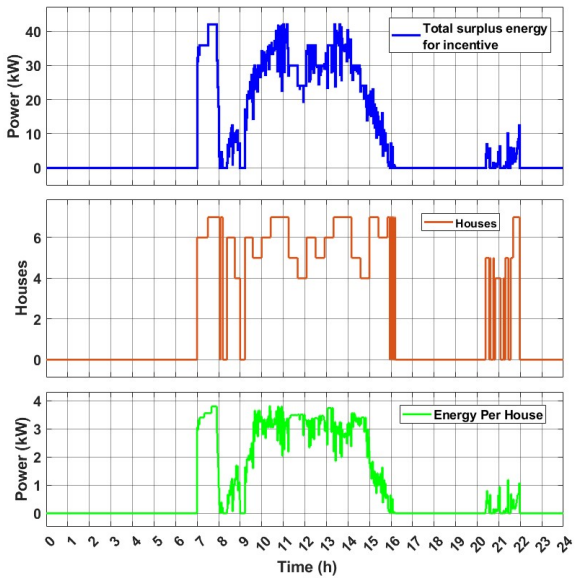


Figure 0.35: Energy Increase Incentives: Total, Household Count, and Average per Time Slot.

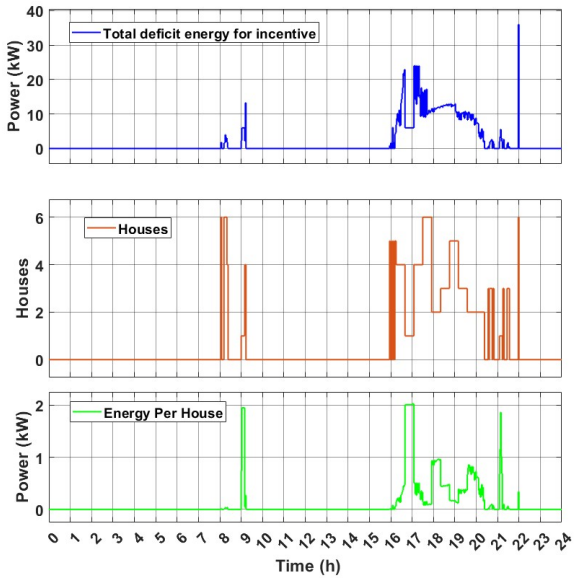


Figure 0.36: Energy Reduction Incentives: Total, Household Count, and Average per Time Slot.

During the day (between 8 a.m. and 4 p.m.), solar and wind energy production powers the microgrid and generates a significant energy surplus. The BESS being already fully charged, the EMS turns to:

- Part of the surplus is offered for charging connected EVs in the three parks (Figure IV.34), as well as to offer incentives to customers. Figure IV.35 shows that around 4 to 6 houses accepted the incentive, and gained around 4 kW each.
- The remaining energy available to sell to the gris is illustrated in figure IV.37 ($15 - 28\ kW$). Figure IV.40 shows a period of continuous profits of selling to the grid that coincides with peaks in solar production (Figure IV.32), reaching profits of $5 - 8\ \$US/kWh$.

During this period ($4 - 8$ p.m.), the EMS apply the steps defined by the algorithm to address the evening peak :

- EVs begin to inject energy at the end of the day, reaching a total power injection of $70\ kW$. Figure IV.39 shows a profit peak of $4.5\ \$US$ for the consumers that inject energy.
- BESS takes over to meet demand by discharging approximately 23% of its energy (Figure IV.33). This effect, combined with V2G and the reducing incentives, impacts final prices. While the grid price was over $0.5\ \$US/kWh$, the EMS algorithm and DSM strategies managed it to reduce around $0.3 - 0.35\ \$US/kWh$ (Figure IV.38).

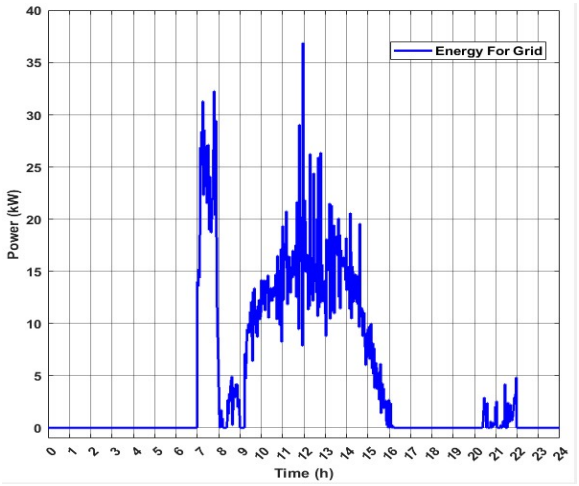


Figure 0.37: Remaining energy available for grid sale on a holiday scenario.

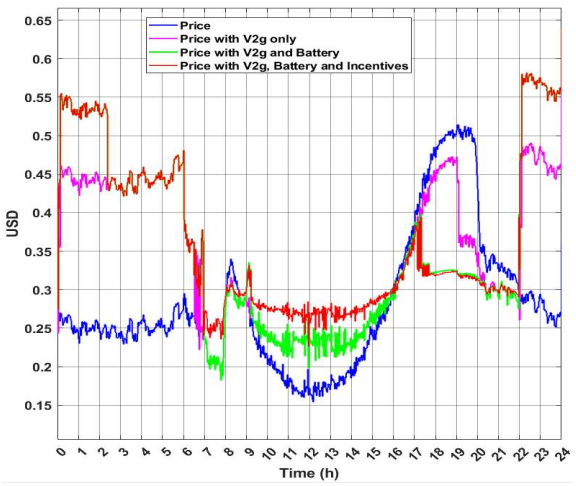


Figure 0.38: Real-Time Pricing impact holiday scenario.

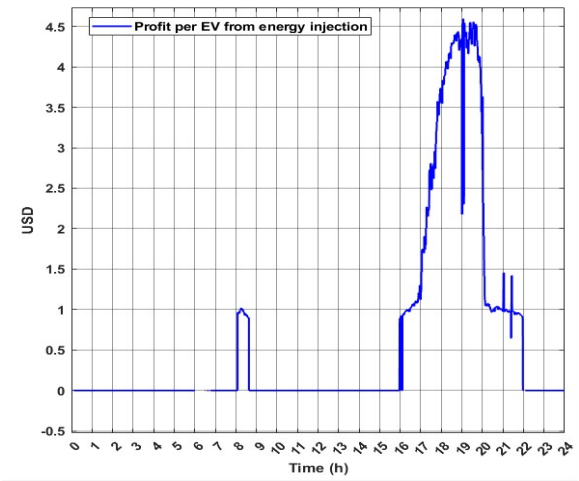


Figure 0.39: Revenue from V2G export to the grid on a holiday scenario.

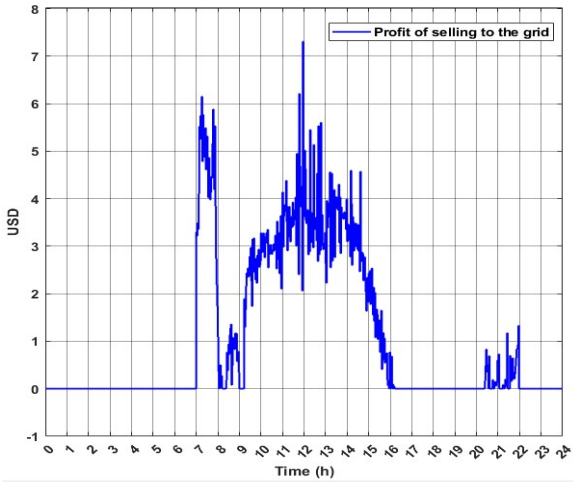


Figure 0.40: Profit from grid energy sales on a holiday scenario.

After this peak has passed, wind power generation and energy from the main grid are used to recharge the BESS and EVs.

This scenario shows how we can make the most of BESS, EVs and incentives to ensure the seamless integration of renewable energy sources, all while maintaining economic profitability for the grid and users.

IV.4. Conclusion

This chapter demonstrated the simulation of the residential microgrid using MATLAB/Simulink. Multiple scenarios were tested to evaluate the behaviour and performance of the proposed energy management system under dynamic conditions. These included changes in generation, storage, and consumption.

The results confirmed that the hybrid EMS combining rule-based algorithm with incentive and market-based demand response effectively adapts to real-time variations, improves energy distribution, and supports consumer engagement through financial incentives. This approach reduces grid dependence and supports the integration of renewables.

This work indicates that microgrids, supported by smart management tactics, can be part of a realistic and scalable energy transition. The developed EMS offers a foundation for future improvements and deployment in real-world applications.

General conclusion

General conclusion

Generation, control, and use of electricity need a drastic change in order to address the energy transition. Electric power systems face additional challenges posed by rising demand, climate targets, the need to balance the proportion of sustainable power, and transportation electrification. This thesis focused primarily on the development of a residential grid-connected including solar and wind energy, a battery energy storage system and EVs that include the V2G technologies. Microgrid Energy Management System (EMS) design and optimization is based on a hybrid approach founded on rule-based algorithm and incentive-based demand response for addressing the operation challenges in modern-day distribution systems.

The outcomes of this thesis are:

We established in the 1st chapter the context by examining the issues and limitations facing the traditional power grids. We explored the modern concept of smart grids and their objectives, presenting them as a flexible solution for supply and demand challenges.

In the 2nd chapter, we presented the microgrid as a promising solution to energy and environmental challenges. We provided a detailed overview of the microgrid's components classifications and control strategies. We reviewed the Energy Management System (EMS) algorithms and demand response programs.

We described in the 3rd chapter the fundamental elements of the designed residential microgrid, outlining the development of a rule-based EMS and a complementary incentive and market-based demand response programs. These components form a hybrid system designed to optimize the energy flow from production to consumption.

In the 4th and final chapter, we validated the system's performance through simulation on MATLAB/Simulink environment. The results confirm that the hybrid EMS effectively adapts to the dynamic changing conditions, improves energy distribution, reduces grid dependence and support the integration of renewable energies. The works concludes that this approach is a realistic and scalable solution for the energy transition, and provides a strong foundation for future real-work development.

The hybrid EMS proved to be very adaptable across various conditions. It facilitated the use of renewable energy resources, reduced energy costs, and directly involved consumers in system optimization. This study gives an example of using well-designed microgrids for smart, green, and robust grid systems.

The research proposes future residential microgrid energy management improvements

- AI can supplement or supplant rule-based EMS. Reinforcement learning, fuzzy logic, implementation with an ANN algorithm for dynamic pricing and electricity load forecasting.
- Simulation with software such as OPAL-RT to study system performance boundaries.
- Safeguarding users' data and protection from cyber-attacks play a pivotal role in the implementation of smart EMS.

These paths can take residential microgrids transition from theory to reality, revolutionizing the world’s energy dynamics

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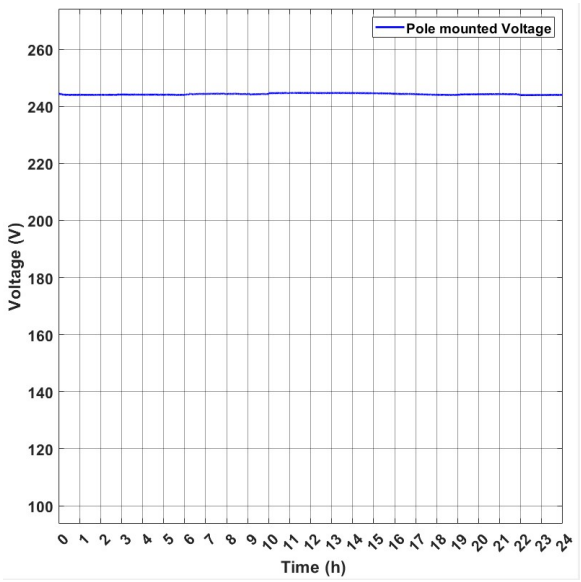
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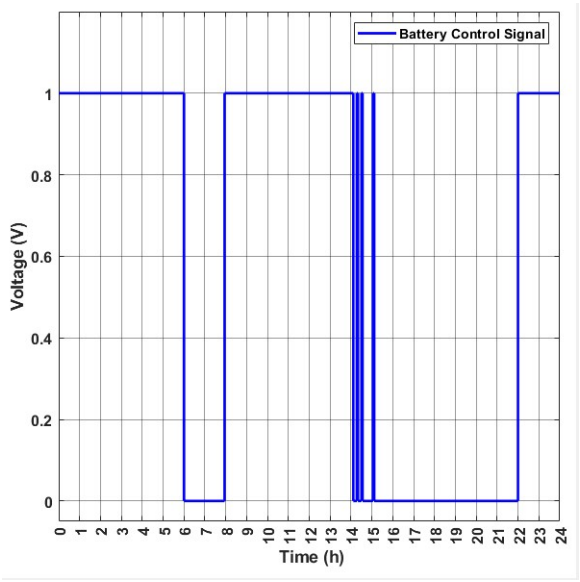
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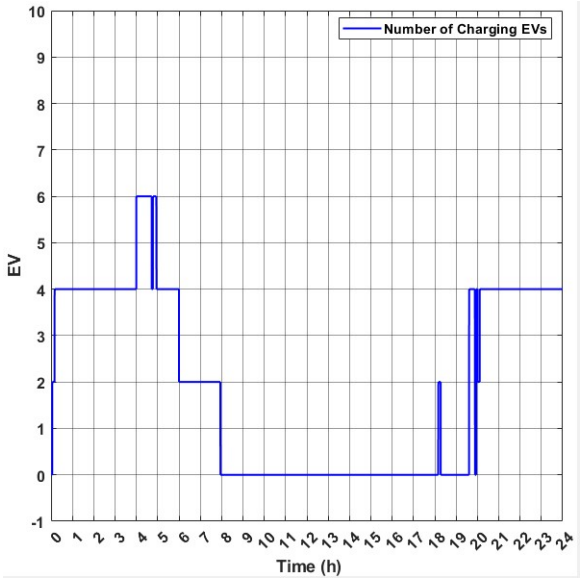
Index A: Scenario 1 additional figures.



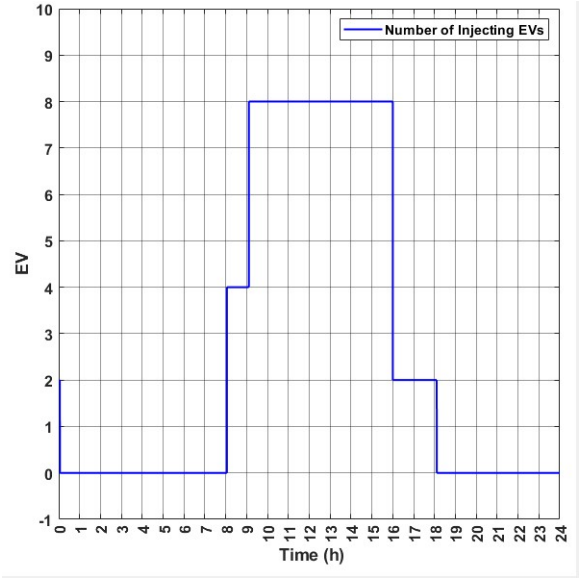
Pole-Mounted transformer voltage



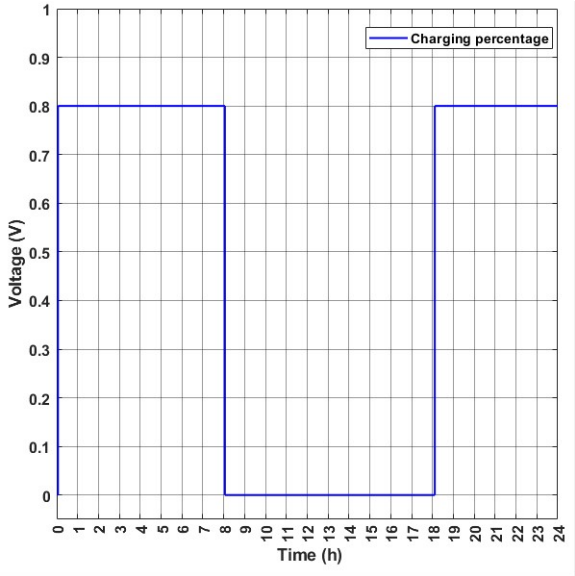
Battery control signal over time



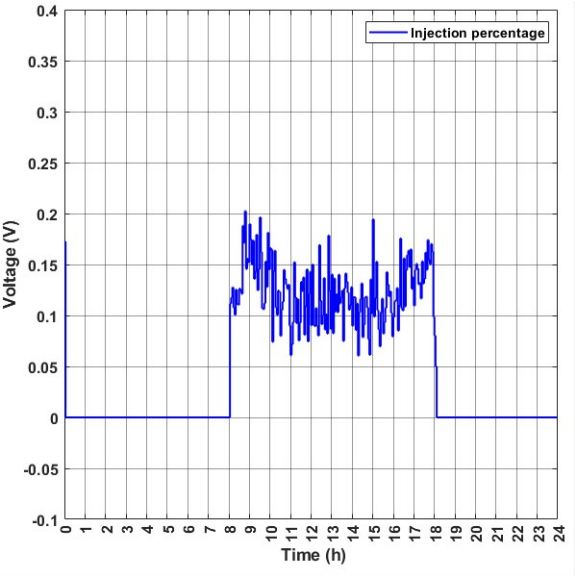
Number of charging EVs



Number of injecting EVs



EV charging percentage over time



EV injection percentage over time